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MASTER THESIS

Implementation of Shrinkage Estimators for Cosmological Precision Matrices

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and
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Declaration of Authorship

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"Any sufficiently advanced technology is indistinguishable from magic."

Arthur C. Clarke

"It's still magic even if you know how it's done."

Terry Pratchett

UNIVERSIDAD DEL NORTE

Abstract

Engineering Division
Department of Software Engineering
and
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Master of Science

Implementation of Shrinkage Estimators for Cosmological Precision Matrices

by Andrés F. Movilla Obregón

Ensemble Kalman filters require the inverse of the background error covariance matrix, but when the ensemble size is small relative to the state dimension, the sample covariance is ill-conditioned and cannot be reliably inverted. We embed the direct precision shrinkage framework of Bodnar et al. into the dual formulation of the Ensemble Kalman Filter, recomputing shrinkage coefficients at each assimilation cycle. Six estimators are tested on Lorenz 96 ($n = 40$, dense precision) and BOSS DR12 cosmological mock data ($d = 18$, near-diagonal precision), alongside two baselines.

Two main findings emerge. First, direct precision shrinkage performance depends critically on target-truth alignment: diagonal targets fail on Lorenz 96 but succeed on cosmological data. Second, the best standalone precision estimator is not the best sequential filter — Ledoit-Wolf produces the worst standalone estimates yet the lowest EnKF RMSE, because its low variance prevents error cascades across assimilation cycles. For sequential data assimilation, variance reduction matters at least as much as bias reduction. All implementations are publicly available in the TEDA framework.

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List of Abbreviations

DA	Data Assimilation
EnKF	Ensemble Kalman Filter
LETKF	Local Ensemble Transform Kalman Filter
LEnKF	Local Ensemble Kalman Filter
3D-Var	Three-Dimensional Variational Assimilation
4D-Var	Four-Dimensional Variational Assimilation
RMSE	Root Mean Square Error
MSE	Mean Squared Error
NERCOME	Nonparametric Eigenvalue-Regularized COvariance Matrix Estimator
LW	Ledoit-Wolf
BOSS	Baryon Oscillation Spectroscopic Survey
DR12	Data Release 12
AR(1)	First-Order Autoregressive Model
TEDA	Toolbox for Ensemble Data Assimilation
OAS	Oracle Approximating Shrinkage
HMM	Hidden Markov Model

To my beloved family, colleagues, and friends

Chapter 1

Introduction

Ensemble Kalman filters require the inverse of the background error covariance matrix, but in most operational settings the ensemble is far too small to estimate that matrix reliably. When the state dimension n exceeds the ensemble size N_e, the sample covariance is rank-deficient and cannot be inverted at all (Houtekamer and Mitchell, 1998); even when $N_e > n$, sampling noise introduces spurious correlations that contaminate the precision matrix (Anderson, 2001). The same problem appears in cosmological data analysis, where the number of mock catalogs available to estimate power spectrum covariances is often comparable to the data dimension (Hamilton et al., 2006).

Covariance matrix estimation is central to both fields. In data assimilation, the background error covariance determines how observations are weighted against the model forecast. In cosmology, the covariance of power spectrum estimates governs the precision of parameter constraints extracted from galaxy surveys. In both cases, the sample covariance computed from a limited number of realizations (ensemble members in data assimilation, mock catalogs in cosmology) is a poor estimator when the sample size is comparable to the data dimension. The eigenvalues of the sample covariance are systematically spread relative to the true eigenvalues: the largest are overestimated, the smallest underestimated, and in the rank-deficient case some eigenvalues are exactly zero (Ledoit and Wolf, 2004).

Shrinkage estimation offers a practical remedy: combine the noisy sample estimate with a structured, low-variance target matrix, accepting some bias in exchange for reduced mean squared error. Most shrinkage work operates in covariance space. (Ledoit and Wolf, 2004) derived an optimal intensity for shrinking toward a scaled identity target, and this approach has become a standard reference in high-dimensional settings. (Pope and Szapudi, 2008) brought shrinkage estimation to cosmology, showing that even a simple diagonal target improves power spectrum covariance estimation when few simulations are available. (Chen et al., 2010) proposed oracle-approximating algorithms that improve on Ledoit-Wolf in high dimensions. (Haff, 1991) derived improved Bayes estimators for the covariance matrix via eigenvalue shrinkage. All of these methods trade bias for variance reduction; they differ in how the target encodes prior assumptions about the covariance structure.

(Bodnar et al., 2016) took a different route, shrinking the precision matrix directly rather than first shrinking the covariance and then inverting. This can be advantageous when the precision matrix has known structure. For instance, when it is approximately diagonal, as is the case for Gaussian random fields observed at independent modes. (Looijmans et al., 2024) applied this direct precision framework to cosmological data, comparing several target choices and showing that domain-informed targets outperform generic ones when the true precision is near-diagonal.

In the data assimilation community, the problem of ill-conditioned covariance estimation has been addressed primarily through localization (Gaspari and Cohn, 1999; Ott et al., 2004; Hunt et al., 2007) and inflation (Anderson and Anderson, 1999; Anderson, 2001). Localization zeroes out long-range correlations that are likely spurious, producing a sparse covariance matrix that is well-conditioned by construction. Inflation artificially increases the ensemble spread to prevent filter divergence. These techniques are complementary to shrinkage estimation: localization imposes sparsity structure, while shrinkage imposes a bias-variance tradeoff. The modified Cholesky decomposition (Nino-Ruiz et al., 2018) provides a third approach, estimating a banded precision matrix directly from local regressions without ever forming or inverting the full covariance matrix.

None of the existing approaches is without practical drawbacks, and a survey of their limitations helps frame the contribution of this work. Covariance-space shrinkage (Ledoit-Wolf, NERCOME) minimizes $\mathbb{E}\|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_F^2$, not the precision-space loss that actually governs the EnKF update; the inverse of a covariance-optimal estimator amplifies errors in the small eigenvalues, which are the very directions where shrinkage was supposed to regularize, an asymmetry already explored by Schäfer and Strimmer (2005) and Touloumis (2015) but not integrated into sequential filtering. Target-free estimators like NERCOME buy robustness against a misspecified structure at the cost of repeated bootstrap eigendecompositions that are prohibitive inside a sequential filter, and random-matrix cleaning schemes (Bun et al., 2017) inherit the same cost profile. Sparse-precision approaches such as the graphical lasso (Friedman et al., 2008), neighborhood selection (Meinshausen and Bühlmann, 2006), and constrained ℓ_1-minimization (Cai et al., 2011) regularize directly in precision space but require a sparsity assumption that is not satisfied by either dense Lorenz-like precision or nearly-diagonal power-spectrum precision. Direct precision shrinkage (Bodnar et al., 2016) removes the inversion bias but is only defined for invertible sample covariances ($N_e > n$) and its target menu in the literature is limited to the identity, eigenvalue-based, and domain-specific (cosmological) options, none of which can be expected to match an arbitrary dense precision structure. Cosmology has explored alternative precision priors (Paz and Sánchez, 2015; Taylor and Joachimi, 2014; Sellentin and Heavens, 2016; Dodelson and Schneider, 2013), but without porting them to sequential filtering either. Localization (Hamill et al., 2001; Furrer et al., 2006; Ménétrier et al., 2015) sidesteps the problem by imposing sparsity, but destroys long-range correlations indiscriminately and invalidates the Wishart assumptions that the shrinkage formulas rely on; adaptive inflation (Anderson, 2009; Zhang et al., 2004) patches over the resulting errors but does not correct their source. A critical discussion of each of these points is given in Sections 2.4.1–2.4.3 and consolidated in Section 2.6.4.

Two gaps remain in the literature. First, (Looijmans et al., 2024) used these estimators for standalone precision matrix estimation, not inside a sequential filter. The Ensemble Kalman Filter recomputes the precision matrix at every assimilation cycle, and errors in the precision estimate feed back through the forecast model into subsequent cycles. It is unclear whether an estimator that performs well in a one-shot setting will also perform well when applied repeatedly across hundreds of assimilation cycles. Second, the direct precision targets in the literature are either generic (identity, eigenvalues) or domain-specific (cosmological), with

no middle ground for practitioners who lack prior knowledge of the precision structure but want something better than the identity.

We address both gaps. We embed the (Bodnar et al., 2016) direct precision shrinkage framework inside the dual formulation of the Ensemble Kalman Filter, recomputing the shrinkage coefficients at each assimilation cycle using the current ensemble. We propose a scaled identity target that uses empirical per-variable variances as a domain-agnostic alternative to the cosmological target. To stress-test the framework under contrasting conditions, we choose two systems with opposite precision structure: the Lorenz 96 model (Lorenz, 1996) ($n = 40$, dense off-diagonal precision from chaotic spatial coupling) and BOSS DR12 cosmological mock data (Looijmans et al., 2024) ($d = 18$, near-diagonal precision from independent modes). This dual-domain design allows us to separate the effect of the shrinkage framework itself from the effect of target-truth alignment, a distinction that single-domain studies cannot make. All implementations are publicly available in the TEDA framework (Nino-Ruiz and Valbuena, 2022).

1.1 Notation

Table 1.1 summarizes the main notation used throughout this thesis.

Table 1.1: Summary of notation.

Symbol	Meaning
n	State dimension
m	Number of observations
N_e	Ensemble size
$\mathbf{x} \in \mathbb{R}^n$	State vector
$\mathbf{y} \in \mathbb{R}^m$	Observation vector
$\mathbf{H} \in \mathbb{R}^{m \times n}$	Observation operator
$\mathbf{R} \in \mathbb{R}^{m \times m}$	Observation error covariance
$\mathbf{P}^b, \mathbf{B} \in \mathbb{R}^{n \times n}$	Background error covariance
$\mathbf{S} \in \mathbb{R}^{n \times n}$	Sample covariance
$\mathbf{\Sigma} \in \mathbb{R}^{n \times n}$	True covariance
$\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$	Sample precision matrix
$\mathbf{\Pi}_0$	Shrinkage target (precision)
$\mathbf{\Pi}_{\mathrm{LS}}$	Shrinkage precision estimator
α^*, β^*	Optimal shrinkage coefficients
λ^*	Covariance shrinkage intensity
$\mathbf{K}$	Kalman gain
$\Delta\mathbf{X}^b$	Ensemble perturbation matrix
$\lVert \cdot \rVert_F$	Frobenius norm
$\lVert \cdot \rVert_{\mathrm{tr}}$	Trace norm
ρ	Inflation factor
r	Cholesky bandwidth / localization radius

1.2 Scope and Contributions

This thesis makes the following contributions:

1. We embed the (Bodnar et al., 2016) framework into the dual formulation of the Ensemble Kalman Filter, recomputing the shrinkage coefficients α^* and β^* at each assimilation cycle using the current ensemble. This is the first application of direct precision shrinkage in a sequential filtering context, as opposed to standalone estimation.

2. We propose a new shrinkage target that uses empirical per-variable variances from the sample covariance, providing a middle ground between the plain identity (which ignores variable scales) and the cosmological target (which requires domain-specific prior knowledge). The factor of 2 mirrors the cosmological $2/N_\ell$ scaling but requires no external parameters.

3. We test six shrinkage-based estimators on two systems with opposite precision structure, Lorenz 96 (dense precision) and BOSS DR12 cosmological mock data (near-diagonal precision), alongside two baselines (standard EnKF, EnKF-Cholesky). This dual-domain design reveals the strong dependence of direct precision shrinkage on target-truth alignment.

4. We show empirically that the best standalone precision estimator (as measured by Frobenius loss) is not the best EnKF filter (as measured by RMSE). We trace this to the sequential accumulation of estimator variance over many assimilation cycles, and argue that variance reduction matters as much as bias reduction in sequential filtering.

5. We provide a systematic study of the shrinkage coefficients, clamping rates, and condition numbers across 12 ensemble sizes, revealing the mechanism behind the failure of diagonal targets on systems with dense precision structure.

1.3 Thesis Organization

The remainder of this thesis is organized as follows.

Chapter 2 reviews the theoretical foundations: data assimilation (sequential and variational), the Ensemble Kalman Filter (stochastic EnKF, dual EnKF, LETKF), filter drawbacks (localization and inflation), the Wishart distribution and sample precision properties, shrinkage estimation for covariance and precision matrices (Ledoit-Wolf, oracle, NERCOME, Bodnar et al.), and the modified Cholesky decomposition.

Chapter 3 presents the proposed method of embedding direct precision shrinkage in the EnKF. It describes the four shrinkage targets (identity, eigenvalue, cosmological, scaled identity), provides a worked numerical example of the coefficient computation, analyzes computational complexity, and describes the TEDA implementation.

Chapter 4 reports the experimental results on Lorenz 96 and BOSS DR12 cosmological data, including RMSE time series, ensemble size sensitivity, shrinkage coefficient diagnostics, standalone precision estimation quality, and computational cost comparisons.

Chapter 5 analyzes the results in depth: the role of target-truth alignment, why Ledoit-Wolf wins in sequential filtering, domain dependence of shrinkage performance, the modified Cholesky alternative, and practical recommendations.

Chapter 6 summarizes the contributions, discusses limitations, and outlines directions for future research including localization interaction, higher-dimensional systems, adaptive target selection, and theoretical analysis of sequential accumulation.

Appendices A–D provide full diagnostic tables and model details.

Chapter 2

Preliminaries

2.1 Data Assimilation

Given a mathematical model $\mathcal{M}$ that describes the evolution of a physical system, the state at time t_{k+1} can be forecast from the state at time t_k:

$$\mathbf{x}_{k+1} = \mathcal{M}(\mathbf{x}_k) + \delta, \tag{2.1}$$

where $\mathbf{x}_k \in \mathbb{R}^{n\times 1}$ is the state vector at time t_k, $\mathcal{M} : \mathbb{R}^n \to \mathbb{R}^n$ is the (generally nonlinear) model operator, and δ represents model error with $\langle \delta \rangle = 0$. The model realization serves as the *background* or *prior* estimate of the system state.

Observations of the system are related to the state through an observation operator:

$$\mathbf{y}_k = \mathcal{H}(\mathbf{x}_k) + \epsilon_o, \quad \epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \tag{2.2}$$

where $\mathbf{y}_k \in \mathbb{R}^{m\times 1}$ are the observations at time t_k, $\mathcal{H} : \mathbb{R}^n \to \mathbb{R}^m$ maps states to observation space, and $\mathbf{R} \in \mathbb{R}^{m\times m}$ is the observation error covariance. In general, not all components of the state are observed ($m \leq n$), and the observation operator may be nonlinear.

Data assimilation combines the model forecast with observations to produce a corrected state estimate (the *analysis*). The correction weights the two sources of information according to their respective uncertainties: the background error covariance $\mathbf{B} \in \mathbb{R}^{n\times n}$ and the observation error covariance $\mathbf{R}$.

Figure 2.1 illustrates this process for a scalar state variable. The prior (orange) represents the model forecast with its uncertainty; the likelihood (blue) represents the information from the observation. The posterior (green) is narrower than either, reflecting the combined information from both sources. The posterior mean is shifted toward the more precise source of information.

Under the assumption that both the prior and likelihood are Gaussian, the posterior distribution is also Gaussian, and the analysis state can be obtained by maximizing the posterior (Lorenc, 1986):

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) \propto \mathcal{P}(\mathbf{x}) \cdot \mathcal{P}(\mathbf{y}|\mathbf{x}), \tag{2.3}$$

where $\mathcal{P}(\mathbf{x}) \propto \exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{x}^b\|^2_{\mathbf{B}^{-1}}\right)$ is the prior and $\mathcal{P}(\mathbf{y}|\mathbf{x}) \propto \exp\left(-\frac{1}{2}\|\mathbf{y} - \mathcal{H}(\mathbf{x})\|^2_{\mathbf{R}^{-1}}\right)$ is the likelihood. The analysis minimizes the cost function:

$$\mathcal{J}(\mathbf{x}) = \frac{1}{2}\|\mathbf{x} - \mathbf{x}^b\|^2_{\mathbf{B}^{-1}} + \frac{1}{2}\|\mathbf{y} - \mathcal{H}(\mathbf{x})\|^2_{\mathbf{R}^{-1}}. \tag{2.4}$$

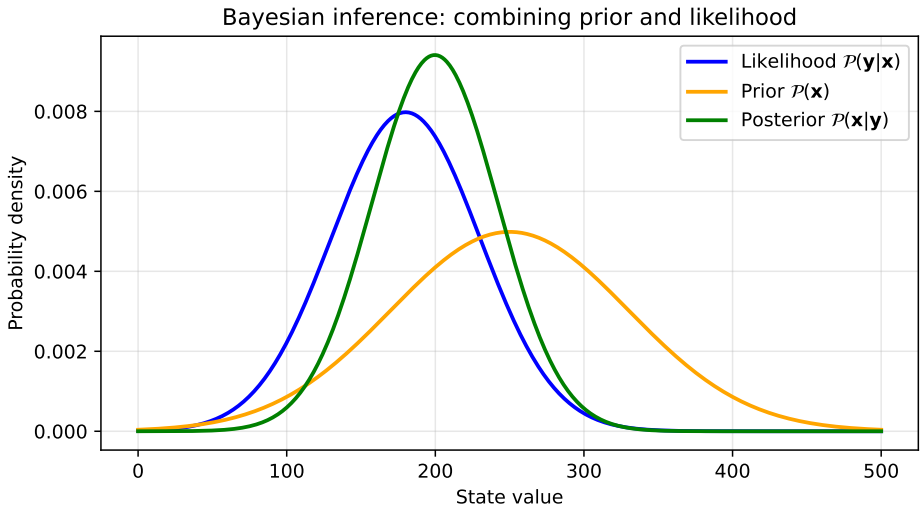

Figure 2.1: Bayesian inference: the posterior (green) combines the prior from the model forecast (orange) with the likelihood from the observation (blue). The posterior is narrower than either source, reflecting the information gain from assimilation.

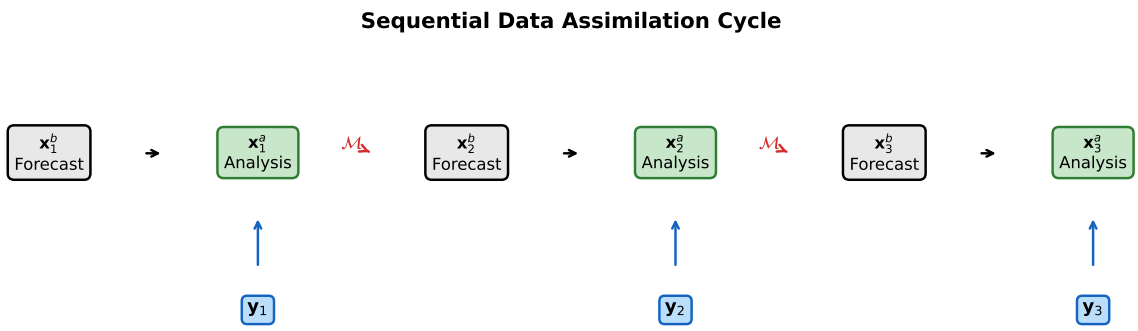

Figure 2.2: The sequential data assimilation cycle. At each step, the forecast (gray) is corrected by observations (blue) to produce the analysis (green), which is propagated by the model (red arrow) to the next forecast.

When $\mathcal{H}$ is linearized as $\mathcal{H}(\mathbf{x}) \approx \mathbf{H}\mathbf{x}$, the minimizer has the closed-form solution:

$$\mathbf{x}^a = \mathbf{x}^b + \mathbf{K}[\mathbf{y} - \mathbf{H}\mathbf{x}^b], \tag{2.5}$$

where $\mathbf{K} = \mathbf{B}\mathbf{H}^T[\mathbf{H}\mathbf{B}\mathbf{H}^T + \mathbf{R}]^{-1}$ is the Kalman gain, weighting the innovation $\mathbf{y} - \mathbf{H}\mathbf{x}^b$ by the relative uncertainties of the model and observations.

Figure 2.2 shows this forecast-analysis cycle schematically. At each time step, the forecast $\mathbf{x}^b$ is corrected using the observation $\mathbf{y}$ to produce the analysis $\mathbf{x}^a$, which is then propagated forward by the model $\mathcal{M}$ to produce the next forecast.

This forecast-analysis cycle is repeated sequentially: after each analysis, the corrected state is propagated forward by the model to produce the next background, and the process repeats when new observations become available. This sequential approach, known as *sequential data assimilation*, processes observations one time step at a time. An alternative

is *variational data assimilation*, which fits the model trajectory to all observations simultaneously.

2.1.1 The Kalman Filter

Before introducing the ensemble approximation, it is useful to review the classical Kalman filter (Lorenc, 1986). For a linear system with Gaussian errors, the Kalman filter provides the exact Bayesian solution. The state evolves as $\mathbf{x}_{k+1} = \mathbf{M}_k \mathbf{x}_k + \boldsymbol{\eta}_k$ with $\boldsymbol{\eta}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_k)$, and observations are $\mathbf{y}_k = \mathbf{H}_k \mathbf{x}_k + \boldsymbol{\epsilon}_k$ with $\boldsymbol{\epsilon}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$.

The filter alternates between two steps:

The forecast step propagates the state and covariance:

$$\mathbf{x}_{k+1}^b = \mathbf{M}_k \mathbf{x}_k^a, \tag{2.6}$$

$$\mathbf{B}_{k+1} = \mathbf{M}_k \mathbf{A}_k \mathbf{M}_k^T + \mathbf{Q}_k. \tag{2.7}$$

The analysis step updates using the observation:

$$\mathbf{K}_k = \mathbf{B}_k \mathbf{H}_k^T (\mathbf{H}_k \mathbf{B}_k \mathbf{H}_k^T + \mathbf{R}_k)^{-1}, \tag{2.8}$$

$$\mathbf{x}_k^a = \mathbf{x}_k^b + \mathbf{K}_k (\mathbf{y}_k - \mathbf{H}_k \mathbf{x}_k^b), \tag{2.9}$$

$$\mathbf{A}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{B}_k. \tag{2.10}$$

The Kalman gain $\mathbf{K}_k$ optimally weights the innovation $\mathbf{y}_k - \mathbf{H}_k \mathbf{x}_k^b$ according to the relative uncertainties of the forecast and observation. When $\mathbf{B}_k$ is large relative to $\mathbf{R}_k$, the gain is large and the analysis trusts the observation; when $\mathbf{B}_k$ is small, the gain is small and the analysis trusts the forecast.

The critical bottleneck is Eq. (2.7): propagating the $n \times n$ covariance matrix through the model requires $\mathcal{O}(n^2)$ storage and $\mathcal{O}(n^3)$ operations (the matrix product $\mathbf{M}_k \mathbf{A}_k \mathbf{M}_k^T$). For $n \sim 10^6$, this is completely infeasible. The Ensemble Kalman Filter avoids this by replacing the explicit covariance propagation with an implicit one: propagating the ensemble members through the model and estimating the covariance from their spread.

2.1.2 Sequential vs. Variational Data Assimilation

In sequential data assimilation, the analysis (2.5) is computed each time observations become available, and the corrected state is propagated forward. The process can be viewed as a Hidden Markov Model: the state $\mathbf{x}_k$ evolves according to the model dynamics, and observations $\mathbf{y}_k$ are noisy projections of the state. The posterior at each step depends only on the current forecast and the current observations (plus the accumulated information from all past observations, encoded in the background).

Variational methods take a different approach. Given observations at multiple times $\{\mathbf{y}_k\}_{k=0}^M$, 4D-Var finds the initial condition $\mathbf{x}_0$ that minimizes the misfit between the model

trajectory and all observations simultaneously:

$$\mathbf{x}_0^a = \arg\min_{\mathbf{x}_0} \left[\frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_0^b\|^2_{\mathbf{B}_0^{-1}} + \sum_{k=0}^{M} \frac{1}{2} \|\mathbf{y}_k - \mathcal{H}_k(\mathcal{M}_{0\to k}(\mathbf{x}_0))\|^2_{\mathbf{R}_k^{-1}} \right], \tag{2.11}$$

where $\mathcal{M}_{0\to k}$ denotes the model integration from time 0 to time k. This requires the adjoint model (the transpose of the linearized model), which is computationally expensive to derive and validate. The 3D-Var simplification drops the time dimension and optimizes at a single time step, equivalent to the analysis (2.5) with a static background covariance.

The Ensemble Kalman Filter, which we describe next, is a sequential method that avoids the need for adjoint models by approximating the background covariance from an ensemble of model runs. This makes it easier to implement and adapt to new models, at the cost of requiring many ensemble members to adequately represent the error statistics.

Table 2.1 summarizes the key differences between the three approaches.

Table 2.1: Comparison of data assimilation approaches. "Model requirement" indicates what information about the dynamical model is needed beyond the forward integration.

Property	Kalman Filter	4D-Var	EnKF
Linearity assumption	Linear only	Nonlinear	Nonlinear
Model requirement	$\mathbf{M}_k$ explicitly	Adjoint model	Forward only
Covariance handling	Explicit ($n \times n$)	Via Hessian	Ensemble
Storage cost	$\mathcal{O}(n^2)$	$\mathcal{O}(n)$	$\mathcal{O}(n \cdot N_e)$
Per-step cost	$\mathcal{O}(n^3)$	$\mathcal{O}(n)$ per iter.	$\mathcal{O}(n^2 N_e)$
Parallelizable	No	Limited	Yes
Gaussian assumption	Exact	Approximate	Approximate
Flow-dependent $\mathbf{B}$	Yes	Limited	Yes

The EnKF's main advantage over 4D-Var is that it provides a flow-dependent background covariance estimate (from the ensemble spread) without requiring the adjoint model. Its main disadvantage is the limited ensemble size, which makes the covariance estimate noisy. This noise in the covariance estimate is the central problem addressed by this thesis.

2.2 The Ensemble Kalman Filter

Computing the background error covariance $\mathbf{B}$ explicitly is infeasible for operational models: with state dimensions on the order of 10^6–10^8, storing and manipulating $\mathbf{B} \in \mathbb{R}^{n\times n}$ requires $\mathcal{O}(n^2)$ storage and $\mathcal{O}(n^3)$ operations for inversion (Evensen, 2003). The Ensemble Kalman Filter (EnKF) avoids this by approximating $\mathbf{B}$ from an ensemble of N_e model realizations.

The EnKF represents the state using an ensemble of N_e state vectors $\{\mathbf{x}_i^b\}_{i=1}^{N_e}$, each $\mathbf{x}_i^b \in \mathbb{R}^{n\times 1}$, obtained by propagating N_e slightly different initial conditions through the model. The ensemble mean

$$\bar{\mathbf{x}}^b = \frac{1}{N_e} \sum_{i=1}^{N_e} \mathbf{x}_i^b \in \mathbb{R}^{n\times 1} \tag{2.12}$$

serves as the background state estimate. The ensemble perturbation matrix

$$\Delta \mathbf{X}^b = \mathbf{X}^b - \bar{\mathbf{x}}^b \mathbf{1}^T \in \mathbb{R}^{n \times N_e}, \tag{2.13}$$

where $\mathbf{X}^b = [\mathbf{x}_1^b, \ldots, \mathbf{x}_{N_e}^b]$, captures the spread of the ensemble around the mean. The sample covariance is

$$\mathbf{P}^b = \frac{1}{N_e - 1} \Delta \mathbf{X}^b (\Delta \mathbf{X}^b)^T \in \mathbb{R}^{n \times n}. \tag{2.14}$$

This estimate has rank at most $N_e - 1$, so when $N_e < n$ it is singular and cannot represent the full covariance structure. Even when $N_e > n$, the sample covariance is a noisy estimate of the true covariance, with eigenvalues that are systematically spread relative to the truth.

The stochastic EnKF (Burgers et al., 1998; Houtekamer and Mitchell, 1998) is one member of a broader family that also includes deterministic square-root filters (Whitaker and Hamill, 2002; Sakov and Oke, 2008) and reduced-rank variants (Verlaan and Heemink, 1997). In what follows we work with the stochastic formulation; the analysis arguments transfer directly to the other variants, and the efficient Sherman–Morrison implementation of Niño-Ruiz et al. (2015) underlies the TEDA code used in Chapter 4. The stochastic EnKF computes analysis updates by solving the linear system

$$[\mathbf{H}\mathbf{P}^b\mathbf{H}^T + \mathbf{R}]\, \Delta \mathbf{X}^a = \mathbf{D}^s \in \mathbb{R}^{m \times N_e}, \tag{2.15}$$

where $\mathbf{H} \in \mathbb{R}^{m \times n}$ is the (linearized) observation operator and $\mathbf{D}^s$ is the innovation matrix of perturbed observation departures. Each ensemble member i is updated as $\mathbf{x}_i^a = \mathbf{x}_i^b + \Delta \mathbf{x}_i^a$, where $\Delta \mathbf{x}_i^a$ is the i-th column of $\Delta \mathbf{X}^a$.

The Kalman gain in ensemble form reads

$$\mathbf{K} = \mathbf{P}^b \mathbf{H}^T [\mathbf{H}\mathbf{P}^b\mathbf{H}^T + \mathbf{R}]^{-1}, \tag{2.16}$$

and the analysis covariance is

$$\mathbf{A} = [\mathbf{B}^{-1} + \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H}]^{-1} \in \mathbb{R}^{n \times n}. \tag{2.17}$$

The *dual EnKF* (Nino-Ruiz et al., 2018) reformulates the analysis in terms of the precision matrix:

$$[(\mathbf{P}^b)^{-1} + \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H}]\, \Delta \mathbf{X}^a = \mathbf{H}^T \mathbf{R}^{-1} \mathbf{D}^s \in \mathbb{R}^{n \times N_e}. \tag{2.18}$$

This formulation requires the precision matrix $(\mathbf{P}^b)^{-1}$, the central object of this work. When $N_e < n$, the sample covariance (2.14) has rank at most $N_e - 1$ and cannot be inverted. Even when $N_e > n$, the sample precision can be poorly conditioned, amplifying noise in the analysis.

A third variant projects the analysis onto the ensemble subspace (Nino-Ruiz and Valbuena, 2022):

$$\mathbf{X}^a = \mathbf{X}^b + \Delta \mathbf{X}^b \mathbf{W}^*, \tag{2.19}$$

where $\mathbf{W}^* \in \mathbb{R}^{N_e \times N_e}$ is obtained from a reduced system, bringing the problem from n di-

mensions down to N_e.

2.2.1 The Local Ensemble Transform Kalman Filter

The LETKF (Hunt et al., 2007) combines the ensemble subspace formulation with domain localization; the serial-observation processing scheme of Nerger (2015) is a representative refinement that illustrates how localization interacts with the order of observation assimilation. For each state variable i, the analysis is computed using only the "local" observations within a radius of influence r. Let $\mathbf{Q}_i = \mathbf{H}_i \Delta \mathbf{X}^b \in \mathbb{R}^{m_i \times N_e}$ be the ensemble perturbations mapped to the local observation space, where $\mathbf{H}_i$ selects the m_i observations within the local box around variable i. The local analysis mean is:

$$\bar{\mathbf{x}}_i^a = \bar{\mathbf{x}}_i^b + \Delta \mathbf{X}_i^b \left[(N_e - 1)\mathbf{I} + \mathbf{Q}_i^T \mathbf{R}_i^{-1} \mathbf{Q}_i \right]^{-1} \mathbf{Q}_i^T \mathbf{R}_i^{-1} [\mathbf{y}_i - \mathbf{H}_i \bar{\mathbf{x}}^b], \tag{2.20}$$

and the local analysis ensemble is:

$$\mathbf{X}_i^a = \bar{\mathbf{x}}_i^a \mathbf{1}^T + \Delta \mathbf{X}_i^b \left[(N_e - 1)\mathbf{I} + \mathbf{Q}_i^T \mathbf{R}_i^{-1} \mathbf{Q}_i \right]^{-1/2}. \tag{2.21}$$

The key computational advantage is that the matrix to be inverted is $N_e \times N_e$ (not $n \times n$), making the cost per local box $\mathcal{O}(N_e^3)$. With n local boxes, the total cost is $\mathcal{O}(n \cdot N_e^3)$, which is much cheaper than the global formulation when $N_e \ll n$. The local boxes can be processed independently, making the LETKF naturally parallelizable.

2.2.2 Filter Drawbacks

In practice, the EnKF suffers from two well-known problems related to the limited ensemble size (Houtekamer and Mitchell, 1998; Anderson, 2001):

The first problem is spurious correlations. The sample covariance (2.14) introduces spurious long-range correlations between physically unrelated state variables. With $N_e \ll n$, two distant grid points may appear correlated simply due to sampling noise. These spurious correlations contaminate the Kalman gain, causing observations to influence state variables they should not affect. *Localization* techniques address this by zeroing out or tapering correlations beyond a specified distance.

The second problem is covariance underestimation. The finite ensemble systematically underestimates the true error variance, leading to overconfidence in the background estimate. Over many assimilation cycles, this causes the ensemble to collapse: the spread shrinks, the filter increasingly ignores observations, and the analysis diverges from the truth. *Inflation* techniques counteract this by artificially increasing the ensemble spread.

2.2.3 Localization

Localization reduces the impact of spurious correlations by exploiting the spatial structure of the system. Given a localization function that assigns a weight based on distance, the localized covariance is obtained via the Schur (element-wise) product (Gaspari and Cohn,

1999):

$$\widetilde{\mathbf{P}}^b = \widehat{\mathbf{\Lambda}} \circ \mathbf{P}^b \in \mathbb{R}^{n \times n}, \tag{2.22}$$

where $\widehat{\mathbf{\Lambda}}$ is a correlation matrix with entries that decay with distance:

$$\left\{\widehat{\mathbf{\Lambda}}\right\}_{i,j} = \exp\left(-\frac{1}{2} \cdot \frac{d(i,j)^2}{r^2}\right), \quad \text{for } 1 \leq i,j \leq n, \tag{2.23}$$

and $r \in \mathbb{R}^+$ is the radius of influence. Entries far from the diagonal (corresponding to distant state variables) are tapered toward zero, enforcing a sparse structure on the estimated covariance. The radius r controls the trade-off: small r aggressively removes spurious correlations but may also discard genuine long-range structure, while large r preserves more information at the cost of retaining more noise.

A widely used alternative to the Gaussian taper (2.23) is the Gaspari-Cohn function (Gaspari and Cohn, 1999), a compactly supported fifth-order piecewise rational function that approximates a Gaussian but is exactly zero beyond a cutoff distance c:

$$\rho_{\mathrm{GC}}(z) = \begin{cases} -\frac{1}{4}z^5 + \frac{1}{2}z^4 + \frac{5}{8}z^3 - \frac{5}{3}z^2 + 1, & 0 \leq z \leq 1, \\ \frac{1}{12}z^5 - \frac{1}{2}z^4 + \frac{5}{8}z^3 + \frac{5}{3}z^2 - 5z + 4 - \frac{2}{3}z^{-1}, & 1 < z \leq 2, \\ 0, & z > 2, \end{cases} \tag{2.24}$$

where $z = d(i,j)/c$. The compact support means the localized covariance is exactly sparse (unlike the Gaussian taper, which is only approximately sparse), which can be exploited for computational savings in the linear solve (2.15).

An alternative approach is *domain localization*, in which the analysis is performed independently within local boxes around each grid point (Ott et al., 2004). The Local Ensemble Kalman Filter (LEnKF) and the Local Ensemble Transform Kalman Filter (LETKF) (Hunt et al., 2007) apply the EnKF equations within each local box using only nearby observations, then assemble the global analysis from the local results. Domain localization has the advantage of naturally parallelizing the analysis step, since each local box can be processed independently.

Figure 2.3 shows the Gaspari-Cohn localization matrix for three different radii on the Lorenz 96 grid ($n = 40$, cyclic boundary conditions). At $r = 2$, the matrix is very sparse, retaining significant weight only for the nearest neighbors. At $r = 10$, most of the domain is included, providing less aggressive filtering.

2.2.4 Inflation

Multiplicative inflation scales the ensemble perturbations by a factor $\rho > 1$ before the analysis step:

$$\mathbf{x}_i^{b,\mathrm{infl}} = \bar{\mathbf{x}}^b + \rho\,(\mathbf{x}_i^b - \bar{\mathbf{x}}^b), \quad i = 1, \ldots, N_e. \tag{2.25}$$

This inflates the background covariance by a factor of ρ^2:

$$\mathbf{P}^{b,\mathrm{infl}} = \rho^2\,\mathbf{P}^b, \tag{2.26}$$

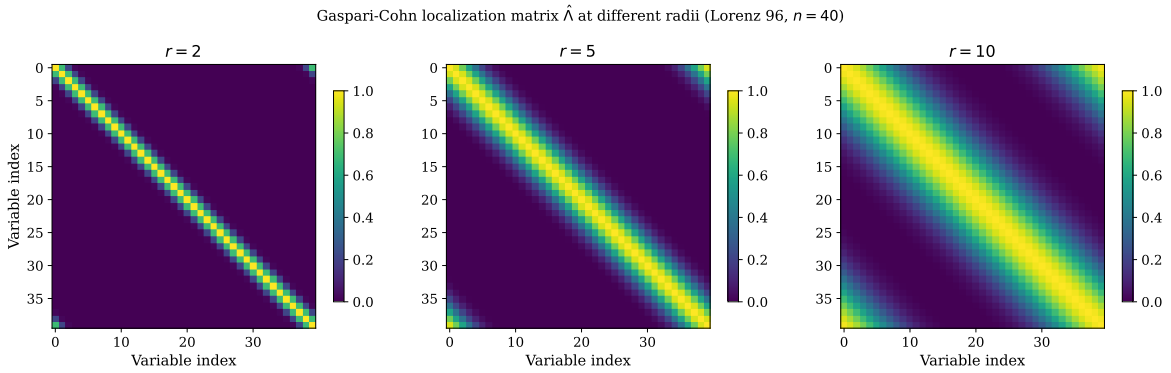

Figure 2.3: Gaspari-Cohn localization matrix $\hat{\mathbf{\Lambda}}$ at different radii of influence on the Lorenz 96 grid ($n = 40$, cyclic). At $r = 2$, the matrix is nearly diagonal; at $r = 10$, most entries are non-zero. The cyclic boundary conditions are visible as the non-zero corners.

counteracting the underestimation due to limited ensemble size. The inflation factor can be set globally (a single ρ for all state variables) or locally (varying ρ across the domain). Typical values range from $\rho = 1.01$ to $\rho = 1.10$, depending on the ensemble size and model dynamics.

Additive inflation takes a different approach, adding random noise drawn from a climatological covariance $\mathbf{Q}_{\text{clim}}$:

$$\mathbf{x}_i^{b,\text{infl}} = \mathbf{x}_i^b + \boldsymbol{\eta}_i, \quad \boldsymbol{\eta}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{\text{clim}}), \tag{2.27}$$

which preserves the ensemble rank and adds variance independently to each member. Multiplicative inflation is simpler to implement and does not require knowledge of the climatological covariance, but it amplifies all features of the ensemble spread equally, including spurious correlations. Additive inflation can be more targeted but requires a reasonable estimate of $\mathbf{Q}_{\text{clim}}$.

The choice of inflation strategy interacts with the covariance estimation method. For the direct precision shrinkage methods studied in this thesis, we use multiplicative inflation because it does not alter the rank or eigenvectors of the sample covariance; only the eigenvalues are scaled uniformly. This preserves the relationship between the sample precision and the shrinkage target, which the Bodnar formulas rely on. With additive inflation, the statistical properties of the sample covariance would change in ways that the Bodnar derivation does not account for.

In our experiments, we use $\rho = 1.05$ for Lorenz 96 and $\rho = 1.02$ for the AR(1) cosmological model. The smaller value for the cosmological model reflects its near-linear dynamics, which produce less covariance underestimation than the chaotic Lorenz 96 system.

2.3 The Wishart Distribution and Sample Precision

The statistical properties of the sample covariance (2.14) are central to understanding why shrinkage is needed and why the Bodnar et al. formulas take the form they do. When the ensemble members $\mathbf{x}_1^b, \ldots, \mathbf{x}_{N_e}^b$ are drawn from a multivariate Gaussian with true covariance

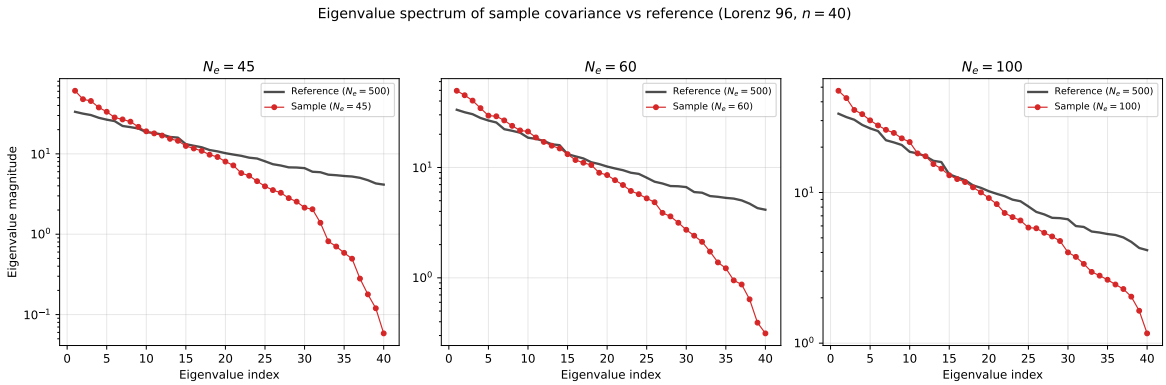

Figure 2.4: Eigenvalue spectrum of the sample covariance $\mathbf{S}$ (red circles) compared to a reference from $N_e = 500$ (black line) on Lorenz 96 ($n = 40$). At small N_e, the eigenvalues are systematically spread: the largest are overestimated and the smallest underestimated.

$\mathbf{\Sigma}$, the matrix $(N_e - 1)\mathbf{S} = \sum_{i=1}^{N_e}(\mathbf{x}_i^b - \bar{\mathbf{x}}^b)(\mathbf{x}_i^b - \bar{\mathbf{x}}^b)^T$ follows a Wishart distribution:

$$(N_e - 1)\mathbf{S} \sim \mathcal{W}_n(N_e - 1, \mathbf{\Sigma}), \tag{2.28}$$

where $\mathcal{W}_n(p, \mathbf{\Sigma})$ denotes the n-dimensional Wishart distribution with p degrees of freedom and scale matrix $\mathbf{\Sigma}$. For this distribution to be non-degenerate (i.e., for $\mathbf{S}$ to be invertible with probability one), we need $p = N_e - 1 \geq n$, or equivalently $N_e > n$.

Two properties of the Wishart distribution are particularly relevant:

The first property is eigenvalue spreading. The eigenvalues of the sample covariance are systematically spread relative to the true eigenvalues. If $\mathbf{\Sigma}$ has eigenvalues $\sigma_1 \geq \cdots \geq \sigma_n$, the eigenvalues of $\mathbf{S}$ are on average more dispersed: the largest sample eigenvalues overestimate the truth, and the smallest underestimate it. This bias increases as n/N_e approaches 1. In the random matrix theory literature, the Marchenko-Pastur law describes the limiting distribution of eigenvalues when n and N_e grow proportionally.

Figure 2.4 illustrates the eigenvalue spreading effect on Lorenz 96. At $N_e = 45$ (barely above the invertibility threshold), the sample eigenvalues are severely spread: the largest are overestimated by a factor of 2–3, and the smallest are underestimated by a similar factor. At $N_e = 100$, the sample eigenvalues are much closer to the reference, but systematic bias remains. This spreading directly affects the sample precision $\mathbf{S}^{-1}$: overestimated small eigenvalues of $\mathbf{S}$ become underestimated large eigenvalues of $\mathbf{S}^{-1}$, and vice versa.

The second property is the bias of the inverse. When $N_e - 1 > n + 1$, the expected value of the inverse Wishart is

$$\mathbb{E}[\mathbf{S}^{-1}] = \frac{N_e - 1}{N_e - n - 2}\mathbf{\Sigma}^{-1}, \tag{2.29}$$

so $\mathbf{S}^{-1}$ is a positively biased estimator of $\mathbf{\Sigma}^{-1}$. The (Hartlap et al., 2007) correction multiplies $\mathbf{S}^{-1}$ by $\gamma = (N_e - n - 2)/(N_e - 1)$ to remove this bias. However, the Bodnar et al. shrinkage formulas are derived for the *uncorrected* inverse $\mathbf{S}^{-1}$, and the $(1 - n/N_e)$ term in Eq. (2.37) already accounts for this bias. Applying the Hartlap correction before the Bodnar formula would double-correct, altering the estimator in a way not supported by the derivation.

2.4 Covariance Matrix Estimation

The quality of the EnKF analysis depends critically on the quality of the background covariance estimate. Shrinkage estimation improves the sample covariance by blending it with a structured target:

$$\hat{\mathbf{P}} = (1-\lambda)\mathbf{S} + \lambda\mathbf{T}, \tag{2.30}$$

where $\mathbf{S}$ is the sample covariance, $\mathbf{T}$ is a structured target matrix, and $\lambda \in [0,1]$ controls the intensity of shrinkage. At $\lambda = 0$ we recover the sample covariance; at $\lambda = 1$ we use only the target. The optimal intensity minimizes the expected Frobenius loss:

$$\lambda^* = \arg\min_{\lambda\in[0,1]} \mathbb{E}\left[\|\hat{\mathbf{P}} - \mathbf{\Sigma}\|_F^2\right], \tag{2.31}$$

where $\mathbf{\Sigma}$ is the true covariance.

2.4.1 Ledoit-Wolf Estimator

(Ledoit and Wolf, 2004) derived a closed-form optimal intensity for the target $\mathbf{T} = \mu\mathbf{I}$, where $\mu = \mathrm{tr}(\mathbf{S})/n$ is the average variance:

$$\lambda^*_{\mathrm{LW}} = \frac{\sum_{i=1}^{N_e} \|\mathbf{z}_i\mathbf{z}_i^T - \mathbf{S}\|_F^2 / N_e^2}{\|\mathbf{S} - \mu\mathbf{I}\|_F^2}, \tag{2.32}$$

where $\mathbf{z}_i$ are standardized observations. The numerator measures the variability of sample covariance estimates across observations (the higher the variability, the more shrinkage is needed), while the denominator measures how far the sample covariance is from the target (the closer it already is, the less shrinkage helps). This estimator is asymptotically optimal, achieving the lowest possible Frobenius loss in the class of linear combinations of $\mathbf{S}$ and $\mu\mathbf{I}$ as $N_e \to \infty$, and has become a standard reference in high-dimensional covariance estimation.

Limitations. The guarantees of Ledoit-Wolf apply to the covariance matrix, not to its inverse. When the estimator is used inside an EnKF or anywhere the precision $\mathbf{\Sigma}^{-1}$ is required, shrinking toward $\mu\mathbf{I}$ distorts the eigenvalue spectrum in a direction that is favorable for estimating $\mathbf{\Sigma}$ but unfavorable for estimating $\mathbf{\Sigma}^{-1}$: small eigenvalues of $\hat{\mathbf{\Sigma}}$ (directions of low variance) become large eigenvalues of $\hat{\mathbf{\Sigma}}^{-1}$ (directions of high precision), and Ledoit-Wolf's biasing of those small eigenvalues toward μ translates into precision errors in the most informative directions. The target $\mu\mathbf{I}$ also does not encode any structural prior beyond uniform scaling, so any non-trivial correlation pattern in $\mathbf{\Sigma}$ (banded, block-diagonal, power-law decay) must be recovered entirely from the sample, which is noisy in the very regime where shrinkage is supposed to help. Subsequent refinements—nonlinear shrinkage that applies an eigenvalue-dependent intensity (Ledoit and Wolf, 2012, 2020), Stein-type shrinkage in the high-dimensional setting (Touloumis, 2015), and biology-motivated adaptive variants (Schäfer and Strimmer, 2005)—partially address the structural rigidity of $\mu\mathbf{I}$, but all remain formulated in covariance space. Its optimality is additionally asymptotic in N_e: the closed-form λ^*_{LW} plugs in sample estimates of the population moments, and at mod-

erate ensemble sizes these plug-in estimates can themselves be noisy enough to produce a suboptimal intensity (Warton, 2008). Alternative regularizers that bound the condition number explicitly (Won et al., 2013) or penalize the log-likelihood directly (Bickel and Levina, 2008) offer different trade-offs but still operate on $\mathbf{\Sigma}$. We revisit these points in Chapter 5 once the empirical behavior on Lorenz 96 and BOSS data is in view.

2.4.2 Oracle Estimator

When the true covariance $\mathbf{\Sigma}$ is known (as in simulation studies), the oracle intensity

$$\lambda^*_{\text{oracle}} = \frac{\text{tr}[(\mathbf{S} - \mathbf{\Sigma})(\mathbf{T} - \mathbf{\Sigma})]}{\|\mathbf{T} - \mathbf{S}\|_F^2} \tag{2.33}$$

gives a best-case benchmark. It is not available in practice but is useful for judging how close a data-driven estimator gets to the theoretical optimum.

Limitations. The oracle is informative only as a reference ceiling: its dependence on $\mathbf{\Sigma}$ forbids operational use, and any gap between a deployed estimator and this ceiling reflects not only method design but also the quality of the sample estimates of $\text{tr}[(\mathbf{S} - \mathbf{\Sigma})(\mathbf{T} - \mathbf{\Sigma})]$ and $\|\mathbf{T} - \mathbf{S}\|_F^2$. The oracle also assumes a fixed target $\mathbf{T}$; it cannot say anything about the choice of target itself, which is often the dominant decision in practice.

2.4.3 NERCOME Estimator

NERCOME (Nonparametric Eigenvalue-Regularized COvariance Matrix Estimator) (Lam, 2016) sidesteps the choice of target entirely. The algorithm proceeds as follows:

1. Split the N_e samples randomly into two disjoint subsets of sizes N_1 and $N_2 = N_e - N_1$.
2. Compute the sample covariance $\mathbf{S}_1$ from the first subset and eigendecompose it: $\mathbf{S}_1 = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$.
3. Compute the sample covariance $\mathbf{S}_2$ from the second subset, rotate it into the eigenbasis of $\mathbf{S}_1$: $\tilde{\mathbf{S}}_2 = \mathbf{U}^T\mathbf{S}_2\mathbf{U}$, and keep only the diagonal.
4. Average over many random splits to reduce the variance of the estimate.

The split ratio N_1/N_e that minimizes cross-validated loss is selected. By using one subset to determine the eigenbasis and the other to estimate eigenvalues, NERCOME avoids the systematic bias that affects the eigenvalues of a single sample covariance, producing a better-conditioned estimate without structural assumptions.

Estimating eigenvectors and eigenvalues from the same data introduces a positive bias in the eigenvalues (the eigenvectors are chosen to maximize the sample variance along each direction, so the corresponding sample variances are necessarily overestimates). By splitting the data, NERCOME decouples the eigenvector estimation from the eigenvalue estimation, removing this bias at the cost of increased variance from using smaller subsets.

In practice, the algorithm performs B random splits (typically $B = 100$–200) and averages the results:

$$\hat{\mathbf{\Sigma}}_{\text{NERCOME}} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{U}_b \, \text{diag}(\tilde{\mathbf{S}}_{2,b}) \, \mathbf{U}_b^T, \tag{2.34}$$

where $\mathbf{U}_b$ is the eigenvector matrix from the first subset in split b, and $\text{diag}(\tilde{\mathbf{S}}_{2,b})$ is the diagonal of the rotated second-subset covariance. The averaging reduces the variance introduced by the random splitting.

NERCOME was developed by Lam (2016) as a general statistical method and later applied to cosmological large-scale structure covariance estimation by Joachimi (2017), where the number of mock catalogs (N_e) is often only 2–3 times the data dimension (n). In this regime, the eigenvalue bias of the standard sample covariance is substantial, and NERCOME's debiasing provides a clear improvement. We include it in our comparison as a representative of target-free covariance estimators.

Limitations. Avoiding a target has a price. The cost per estimate is roughly B times that of a single-pass shrinkage estimator (for $B \approx 100$–200 random splits), each pass carrying an $\mathcal{O}(n^3)$ eigendecomposition; this makes NERCOME attractive when one pays the estimation cost once, but prohibitive inside a sequential filter that recomputes the precision matrix at every cycle. The random splitting also injects an additional source of variance (between splits, in addition to between ensembles), which we show in Chapter 4 propagates through the EnKF update and produces less stable filtering behavior than much cheaper linear-shrinkage alternatives. This behavior is consistent with the more general lesson from random matrix theory that eigenvalue cleaning is a delicate operation whose variance can dominate its bias reduction when the cleaning itself is sample-dependent (Bun et al., 2017). Finally, NERCOME regularizes the eigenvalues of $\mathbf{S}$ but does not act in precision space directly; when the downstream quantity of interest is $\mathbf{S}^{-1}$, the residual noise in small eigenvalues is still amplified by inversion, and NERCOME's debiasing in covariance space does not translate into a proportional improvement in precision space—an issue that motivated sparse-precision approaches such as the graphical lasso (Friedman et al., 2008), neighborhood selection (Meinshausen and Bühlmann, 2006), and constrained ℓ_1-minimization (Cai et al., 2011), none of which however assume the Wishart structure that Bodnar et al. exploit.

2.5 Why Shrink? The Bias-Variance Tradeoff

Before describing precision shrinkage, we examine why shrinkage works at all. The sample covariance $\mathbf{S}$ is an unbiased estimator of $\mathbf{\Sigma}$: $\mathbb{E}[\mathbf{S}] = \mathbf{\Sigma}$. However, unbiasedness does not imply low error. The mean squared error (MSE) of an estimator $\hat{\mathbf{\Sigma}}$ can be decomposed as:

$$\mathbb{E}[\|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_F^2] = \|\text{Bias}(\hat{\mathbf{\Sigma}})\|_F^2 + \text{Var}(\hat{\mathbf{\Sigma}}), \tag{2.35}$$

where $\text{Bias}(\hat{\mathbf{\Sigma}}) = \mathbb{E}[\hat{\mathbf{\Sigma}}] - \mathbf{\Sigma}$ and $\text{Var}(\hat{\mathbf{\Sigma}}) = \mathbb{E}[\|\hat{\mathbf{\Sigma}} - \mathbb{E}[\hat{\mathbf{\Sigma}}]\|_F^2]$.

For the sample covariance, the bias is zero but the variance is $\mathcal{O}(n^2/N_e)$. When n is comparable to N_e, this variance dominates the MSE. The shrinkage estimator $\hat{\mathbf{P}} = (1 -$

$\lambda)\mathbf{S} + \lambda\mathbf{T}$ introduces bias $\lambda(\mathbf{T} - \mathbf{\Sigma})$ but reduces variance by a factor of $(1-\lambda)^2$. The optimal λ trades off these two effects:

- When n/N_e is small (many samples relative to dimensions), the variance is already low and shrinkage provides little benefit. $\lambda^* \to 0$.
- When n/N_e is close to 1 (few samples relative to dimensions), the variance is large and shrinkage provides substantial benefit. λ^* can approach 1 if the target is a good match.
- When the target $\mathbf{T}$ is close to $\mathbf{\Sigma}$, the bias from shrinkage is small regardless of λ, so the optimal λ is larger. When $\mathbf{T}$ is far from $\mathbf{\Sigma}$, the bias penalty limits how much shrinkage is beneficial.

This tradeoff applies equally to covariance shrinkage and precision shrinkage. The (Bodnar et al., 2016) framework minimizes the MSE of the precision estimator, with the same bias-variance structure but different mathematical details because the inverse Wishart distribution replaces the Wishart.

2.6 Inverse Covariance Matrix Estimation

Rather than shrinking the covariance and then inverting, (Bodnar et al., 2016) proposed shrinking the precision matrix directly. The estimator combines the sample precision $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$ with a structured target $\mathbf{\Pi}_0$:

$$\mathbf{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\mathbf{\Pi}}_S + \beta^* \mathbf{\Pi}_0, \tag{2.36}$$

where α^* and β^* are derived analytically by minimizing the expected quadratic loss $\mathbb{E}[\|\mathbf{\Pi}_{\mathrm{LS}} - \mathbf{\Sigma}^{-1}\|_F^2]$ under the assumption that $\mathbf{S}$ follows a Wishart distribution:

$$\alpha^* = 1 - \frac{n}{N_e} - \frac{N_e^{-1}\|\hat{\mathbf{\Pi}}_S\|_{\mathrm{tr}}^2 \, \|\mathbf{\Pi}_0\|_F^2}{\|\hat{\mathbf{\Pi}}_S\|_F^2 \, \|\mathbf{\Pi}_0\|_F^2 - \left(\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0)\right)^2}, \tag{2.37}$$

$$\beta^* = \left(1 - \frac{n}{N_e} - \alpha^*\right) \frac{\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0)}{\|\mathbf{\Pi}_0\|_F^2}, \tag{2.38}$$

where $\|\cdot\|_{\mathrm{tr}}$ denotes the trace norm (sum of eigenvalues) and $\|\cdot\|_F$ the Frobenius norm. The term $(1 - n/N_e)$ accounts for the known positive bias of the Wishart inverse: the expected value of $\mathbf{S}^{-1}$ is not $\mathbf{\Sigma}^{-1}$ but rather $\frac{N_e-1}{N_e-n-2}\mathbf{\Sigma}^{-1}$, so the correction factor ensures that the coefficients are calibrated for the raw inverse rather than the Hartlap-corrected one (Hartlap et al., 2007).

The denominator of the correction term in α^* involves the quantity $\|\hat{\mathbf{\Pi}}_S\|_F^2 \|\mathbf{\Pi}_0\|_F^2 - (\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0))^2$, which by the Cauchy-Schwarz inequality is non-negative and equals zero only when $\hat{\mathbf{\Pi}}_S$ is proportional to $\mathbf{\Pi}_0$. When the target is a good match for the sample precision, this quantity is small, and the correction drives α^* down (more weight to the target). When the target is a poor match, the correction is small relative to the denominator, and α^* stays near $1 - n/N_e$.

2.6.1 Cosmological Precision Matrix Structure

In galaxy surveys, the data vector consists of power spectrum measurements $\hat{C}_\ell$ at different multipole moments ℓ and, for tomographic analyses, different redshift bins. For a Gaussian random field observed over a sky fraction $f_{\rm sky}$, the covariance of power spectrum estimates is (Pope and Szapudi, 2008):

$$\mathrm{Cov}(\hat{C}_{\ell_i}, \hat{C}_{\ell_j}) = \frac{2}{(2\ell_i + 1) f_{\rm sky}} \left(C_{\ell_i} + \frac{1}{n_g} \right)^2 \delta_{\ell_i \ell_j}, \tag{2.39}$$

where C_ℓ is the true power spectrum, n_g the galaxy number density, and $(2\ell + 1) f_{\rm sky}$ counts the independent modes N_ℓ at multipole ℓ. The $1/n_g$ term is shot noise. The diagonal structure follows from mode independence under the Gaussian assumption: power spectrum estimates at different multipoles are uncorrelated.

The precision matrix is then also diagonal:

$$\mathbf{P}_{ij}^{-1} = \frac{(2\ell_i + 1)}{2} f_{\rm sky} \left(C_{\ell_i} + \frac{1}{n_g} \right)^{-2} \delta_{\ell_i \ell_j} \delta_{z_i z_j}. \tag{2.40}$$

This diagonal structure motivates a cosmological shrinkage target. Following (Pope and Szapudi, 2008), as adopted by (Looijmans et al., 2024), the target covariance is

$$T_{ii}^{(2)} = \frac{2}{N_{\ell_i}} C_{\ell_i}^2, \tag{2.41}$$

where $N_{\ell_i} = (2\ell_i + 1) f_{\rm sky}$. This captures the leading-order cosmic variance contribution and, when the assumed power spectrum model is close to the truth, provides a much more informative shrinkage direction than a generic identity target.

2.6.2 Related Shrinkage Approaches

(Pope and Szapudi, 2008) introduced shrinkage estimation to cosmology, showing that even a simple diagonal target improves power spectrum covariance estimation over the raw sample covariance when few simulations are available. (Looijmans et al., 2024) compared covariance shrinkage (with empirical and analytical targets) against precision shrinkage (with eigenvalue-based and cosmological targets, following Bodnar et al. 2016) and the NERCOME estimator. They found that incorporating cosmological prior information helps, especially at larger sample sizes, with the benefit depending on N_e/n and the accuracy of the assumed cosmological model.

(Haff, 1991) derived improved Bayes estimators for the covariance matrix via eigenvalue shrinkage, and (Chen et al., 2010) proposed oracle-approximating algorithms that improve on Ledoit-Wolf in high dimensions. A parallel Bayesian tradition places priors directly on the covariance or precision matrix (Kaufman and Sain, 2010), with performance driven by how well the prior captures the dominant structure of the true matrix. All of these methods trade bias for variance reduction; they differ in how the target encodes prior assumptions about the covariance structure.

2.6.3 Comparison of Covariance Estimators

Table 2.2 summarizes the properties of the covariance estimation methods described above.

Table 2.2: Comparison of covariance matrix estimators with their dominant failure modes. "Bias" refers to the bias of the covariance estimate; "Variance" to the variance across different realizations. "Fails when" lists the regime in which the estimator is known to degrade or be unusable.

Estimator	Bias	Variance	Target	Key assumption	Fails when
Sample $\mathbf{S}$	None	High	None	$N_e > n$ for invertibility	$N_e \leq n$ (singular); n/N_e near 1 (ill-conditioned)
Ledoit-Wolf	Low	Low	$\mu\mathbf{I}$	Asymptotic optimality	$\mathbf{\Sigma}$ far from $\mu\mathbf{I}$; precision is the target quantity
Oracle	Minimal	Low	Any	True $\mathbf{\Sigma}$ known	Operationally (requires $\mathbf{\Sigma}$)
NERCOME	Low	Medium	None	Gaussian, sufficient N_e	Sequential use ($B\times$ cost per cycle); precision is the target quantity

The sample covariance is the starting point: unbiased but high variance, singular when $N_e \leq n$. Ledoit-Wolf adds a small bias toward a scaled identity (which may not match the true covariance structure) but dramatically reduces variance, especially when n/N_e is large. The oracle estimator represents the best achievable performance for any linear combination of $\mathbf{S}$ and a given target, but requires knowledge of the true covariance. NERCOME avoids the target choice entirely through cross-validated eigenvalue estimation but is computationally expensive.

All estimators in this comparison trade some accuracy (bias) for stability (reduced variance). The question is where to apply this tradeoff: in covariance space (Ledoit-Wolf, NERCOME) or in precision space (Bodnar et al.). As we will see in Chapter 3, the answer depends on the structure of the true precision matrix and on whether the estimator is used once (standalone estimation) or repeatedly (sequential filtering).

2.6.4 Critical Comparison of Existing Approaches

The estimators reviewed above address the same problem (ill-conditioned sample covariance) from complementary angles, but none is uniformly preferable, and the discussion in the preceding subsections allows us to consolidate their practical limitations.

Shrinkage in covariance space is not equivalent to shrinkage in precision space. Given a covariance estimator $\hat{\mathbf{\Sigma}}$, the induced precision estimator $\hat{\mathbf{\Sigma}}^{-1}$ inherits its bias and variance, but through the matrix inverse, which amplifies errors in small eigenvalues. Ledoit-Wolf minimizes $\mathbb{E}\|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_F^2$, not $\mathbb{E}\|\hat{\mathbf{\Sigma}}^{-1} - \mathbf{\Sigma}^{-1}\|_F^2$; the two objectives are genuinely different, and an optimal estimator for one can be suboptimal for the other. This distinction mirrors the classical observation from ridge regression that regularizing a matrix and regularizing its inverse are different problems (Hoerl and Kennard, 1970; Friedman, 1989). It motivates the direct precision framework of (Bodnar et al., 2016), but also explains why Ledoit-Wolf,

despite operating in the "wrong" space, can still give competitive precision estimates in practice: its aggressive variance reduction produces estimates whose small eigenvalues are rarely close to zero, which in turn prevents the inversion from amplifying them catastrophically. Similar considerations in cosmology have prompted the use of surrogate priors on the precision matrix itself (Paz and Sánchez, 2015; Taylor and Joachimi, 2014), with downstream consequences for parameter inference well documented by Dodelson and Schneider (2013) and Sellentin and Heavens (2016).

Target-free methods pay for their generality. NERCOME removes the need to specify a target at the cost of $\mathcal{O}(B)$ repeated eigendecompositions and additional between-split variance. If the target is known to approximately match $\mathbf{\Sigma}$ (as for power spectrum data), a single-pass linear shrinkage with that target is both cheaper and lower-variance than NERCOME. If no such target is available, NERCOME buys robustness at a computational cost that becomes prohibitive inside the EnKF assimilation loop, where every cycle recomputes the precision matrix. Section 4.6 quantifies this gap at two orders of magnitude.

Diagonal targets assume the problem away. All of the precision targets we review (identity, eigenvalue, cosmological) are diagonal. This is appropriate for data where the underlying covariance is near-diagonal by construction (Gaussian random fields at independent Fourier modes), but for dynamical systems with spatial coupling, the true precision is dense and off-diagonal. Data assimilation has historically confronted this mismatch with localization—tapering of long-range correlations—rather than shrinkage (Gaspari and Cohn, 1999; Hamill et al., 2001; Furrer et al., 2006; Ménétrier et al., 2015), with the side effect of breaking the Wishart assumption the Bodnar formulas rely on. Shrinking toward a diagonal target in such a case is not merely suboptimal: the (Bodnar et al., 2016) formulas produce coefficients that drive the target component to near-zero (to reduce bias) while simultaneously shrinking the sample component through the $(1 - n/N_e)$ factor, yielding a compromise estimate dominated by neither ingredient. The resulting behavior in sequential filtering is erratic, and Chapter 4 reports clamping rates in excess of 80% and unclamped β^* values in the 10^8–10^9 range on Lorenz 96 for exactly this reason. Alternative regularizers that impose a penalty rather than a target—ridge on covariance (Warton, 2008), covariance-regularized regression (Witten and Tibshirani, 2009), condition-number constraints (Won et al., 2013)—avoid the diagonal-target commitment but do not produce the closed-form shrinkage coefficients that make the Bodnar estimator tractable inside the EnKF loop.

Standalone optimality is not sequential optimality. Every estimator in Table 2.2 is optimized for a single-shot loss: given one sample, minimize $\mathbb{E}\|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_F^2$ or its precision analogue. Inside an EnKF, the precision is recomputed at every cycle, and error accumulates over hundreds of updates. Even the Stein-type shrinkage literature from which Ledoit-Wolf descends (Stein, 1956; James and Stein, 1961; Efron and Morris, 1975) is built around a single-sample decision problem, and its extensions to dynamical settings are recent. A low-bias, high-variance estimator that performs best in the single-shot sense can accumulate enough between-cycle variance to be outperformed, over 100 cycles, by a higher-bias,

lower-variance alternative. Review articles of the EnKF literature (Houtekamer and Zhang, 2016; Reich and Cotter, 2015) describe this sequential structure but do not connect it to the shrinkage-estimation choice, and the consequences are typically absorbed into inflation tuning (Anderson, 2009; Zhang et al., 2004) rather than estimator selection. This observation is the central empirical finding of this thesis (Chapter 4–5) and points to a gap in the existing literature, which has almost exclusively evaluated precision estimators in a standalone setting.

These four limitations jointly define the gap that motivates the proposed method: a precision-space estimator whose target can be chosen to match the problem at hand, whose cost is $\mathcal{O}(n^3)$ per cycle (comparable to the inversion that the EnKF already requires), and whose variance properties can be compared against Ledoit-Wolf in the same sequential benchmark.

2.7 Modified Cholesky Decomposition

The modified Cholesky decomposition (Nino-Ruiz et al., 2018) factors the precision matrix as

$$(\mathbf{P}^b)^{-1} \approx \mathbf{L}^{-T}\mathbf{D}^{-1}\mathbf{L}^{-1}, \tag{2.42}$$

where $\mathbf{L}$ is unit lower triangular and $\mathbf{D}$ is diagonal. The key idea is that each row of $\mathbf{L}$ can be estimated independently by regressing a state variable on its spatial neighbors within a bandwidth r. Specifically, for each state variable i, we define the set of *predecessors* $\Pi(i, r)$ as those variables whose indices fall within r positions of i (in some ordering of the state variables). The regression coefficients for variable i depend only on the ensemble values of variable i and its predecessors, yielding a banded lower-triangular matrix.

The regression for variable i uses the ensemble values of variables in $\Pi(i, r)$ as predictors and the ensemble values of variable i as the response. Let $\mathbf{x}_i \in \mathbb{R}^{N_e}$ be the vector of ensemble values for variable i, and let $\mathbf{X}_{\Pi(i,r)} \in \mathbb{R}^{N_e \times |\Pi(i,r)|}$ be the matrix of ensemble values for the predecessors. The regression coefficients are:

$$\boldsymbol{\ell}_i = (\mathbf{X}_{\Pi(i,r)}^T \mathbf{X}_{\Pi(i,r)})^{-1} \mathbf{X}_{\Pi(i,r)}^T \mathbf{x}_i, \tag{2.43}$$

and the residual variance is:

$$d_i = \frac{1}{N_e - |\Pi(i, r)| - 1} \|\mathbf{x}_i - \mathbf{X}_{\Pi(i,r)} \boldsymbol{\ell}_i\|_2^2. \tag{2.44}$$

The coefficients $\boldsymbol{\ell}_i$ fill the i-th row of $\mathbf{L}$, and d_i is the i-th diagonal entry of $\mathbf{D}$. Because each regression involves at most r predictors, the cost per variable is $\mathcal{O}(r^2 N_e)$, and the total cost is $\mathcal{O}(n \cdot r^2 \cdot N_e)$, which is independent of the full state dimension n aside from the linear factor.

This produces a sparse, banded precision matrix by construction, avoiding the need to invert $\mathbf{P}^b$ entirely. The bandwidth r controls the sparsity: small r enforces locality (reducing noise but potentially discarding real correlations), while large r retains more structure at the cost of higher variance. For the Lorenz 96 model with nearest-neighbor coupling, a

bandwidth of $r = 2$–5 captures the dominant covariance structure while filtering out noise from distant grid points.

2.8 EnKF via Modified Cholesky Decomposition

Substituting (2.42) into the dual EnKF formulation (2.18) gives an analysis that uses the banded precision estimate directly:

$$[\mathbf{L}^{-T}\mathbf{D}^{-1}\mathbf{L}^{-1} + \mathbf{H}^T\mathbf{R}^{-1}\mathbf{H}]\,\Delta\mathbf{X}^a = \mathbf{H}^T\mathbf{R}^{-1}\mathbf{D}^s. \tag{2.45}$$

The computational advantage is twofold. First, the regression step for each variable involves only r predictors, so building the Cholesky factors costs $\mathcal{O}(n \cdot r \cdot N_e)$ instead of the $\mathcal{O}(n^2 N_e + n^3)$ required for full inversion. Second, the banded structure provides implicit localization: long-range correlations are zeroed out by construction, without requiring a separate localization matrix. (Nino-Ruiz et al., 2018) showed that even modest bandwidths ($r = 2$–5) substantially improve the EnKF on spatially extended systems like the Lorenz 96 model, achieving comparable performance to explicit localization methods at lower computational cost.

Chapter 3

Proposed Method

3.1 Direct Precision Shrinkage for the EnKF

The (Bodnar et al., 2016) framework was developed for standalone precision matrix estimation and applied to cosmological data by (Looijmans et al., 2024). Neither work embedded the estimator inside a sequential filter. We do so here: at each assimilation cycle of the dual EnKF (2.18), we replace the raw sample precision $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$ with the shrinkage estimator

$$\mathbf{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\mathbf{\Pi}}_S + \beta^* \mathbf{\Pi}_0, \tag{3.1}$$

where $\mathbf{\Pi}_0$ is a structured target and the coefficients α^*, β^* are computed from Eqs. (2.37)–(2.38) at every cycle using the current ensemble.

The weight α^* on the sample precision is bounded above by $1 - n/N_e$, reflecting the known positive bias of the Wishart inverse. When N_e is only slightly larger than n (as is typical in data assimilation), this bound is tight: for $N_e = 50$ and $n = 40$, the maximum weight on the sample precision is $1 - 40/50 = 0.2$, meaning at least 80% of the estimator comes from the target. This makes the choice of target critically important in the data assimilation setting.

The remaining weight β^* on the target depends on how well $\mathbf{\Pi}_0$ aligns with $\hat{\mathbf{\Pi}}_S$: when $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0)$ is large relative to $\|\mathbf{\Pi}_0\|_F^2$, more weight goes to the target. Both coefficients are clamped to $[0, 1]$ with $\alpha^* + \beta^* \leq 1$ as a finite-sample safeguard, since the asymptotic guarantees of (Bodnar et al., 2016) do not hold at the ensemble sizes typical of data assimilation ($N_e = 20$–100).

3.1.1 Why Shrink the Precision Directly?

An alternative to direct precision shrinkage is to shrink the covariance matrix (e.g., using Ledoit-Wolf) and then invert. This "shrink-then-invert" approach has the advantage of working with well-established covariance shrinkage estimators, but it has a fundamental limitation when the goal is precision estimation.

Consider the Ledoit-Wolf estimator $\hat{\mathbf{P}} = (1 - \lambda)\mathbf{S} + \lambda\mu\mathbf{I}$. The shrinkage pulls the eigenvalues of $\mathbf{S}$ toward the grand mean μ, reducing their spread. When inverted, the small eigenvalues of $\hat{\mathbf{P}}$ (which correspond to directions of low variance) become large eigenvalues of $\hat{\mathbf{P}}^{-1}$ (directions of high precision). The shrinkage makes these small eigenvalues larger (pulling them toward μ), which makes their reciprocals smaller, *reducing* the precision in directions where it should be high.

In contrast, direct precision shrinkage operates on $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$ and pulls its eigenvalues

toward those of the target $\mathbf{\Pi}_0$. If $\mathbf{\Pi}_0$ captures the dominant structure of the true precision (e.g., diagonal with appropriate magnitudes), the shrinkage improves the precision estimate in the right coordinate system. This is why direct precision shrinkage outperforms shrink-then-invert on the cosmological data, where the precision is near-diagonal: the target is well-aligned with the true precision structure.

The cost of direct precision shrinkage is that it requires $\mathbf{S}$ to be invertible ($N_e > n$), which rules out the rank-deficient regime. Covariance shrinkage has no such requirement (it works for any N_e), making it applicable in a wider range of settings.

To illustrate the difference concretely, consider a 2×2 example. Suppose the true covariance is $\mathbf{\Sigma} = \mathrm{diag}(1, 0.01)$, so one direction has 100 times more variance than the other. The true precision is $\mathbf{\Sigma}^{-1} = \mathrm{diag}(1, 100)$.

With Ledoit-Wolf shrinkage toward $\mu\mathbf{I}$ where $\mu = (1 + 0.01)/2 = 0.505$, the shrunk covariance at $\lambda = 0.5$ is:

$$\hat{\mathbf{P}} = 0.5 \begin{pmatrix} 1 & 0 \\ 0 & 0.01 \end{pmatrix} + 0.5 \begin{pmatrix} 0.505 & 0 \\ 0 & 0.505 \end{pmatrix} = \begin{pmatrix} 0.753 & 0 \\ 0 & 0.258 \end{pmatrix}.$$

Inverting: $\hat{\mathbf{P}}^{-1} = \mathrm{diag}(1.33, 3.88)$. The precision in the high-precision direction ($j = 2$) is 3.88 instead of the true 100, a factor of 25 too small. The covariance shrinkage has "filled in" the small eigenvalue of $\mathbf{\Sigma}$, which is correct for covariance estimation, but the resulting precision in that direction is destroyed.

With direct precision shrinkage toward $\mathbf{I}$ at $\alpha = 0.5$, $\beta = 0.5$:[1]

$$\mathbf{\Pi}_{\mathrm{LS}} = 0.5 \begin{pmatrix} 1 & 0 \\ 0 & 100 \end{pmatrix} + 0.5 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 50.5 \end{pmatrix}.$$

The precision in the high-precision direction is 50.5, still biased downward, but only by a factor of 2 rather than 25. Direct precision shrinkage preserves more of the precision structure because it operates in the right space.

Embedding this estimator in the EnKF introduces a consideration absent from standalone estimation: the precision matrix is recomputed at every assimilation cycle, and errors in the analysis state propagate through the forecast model into subsequent cycles. An estimator that is accurate on average but variable from cycle to cycle can accumulate larger errors over a long assimilation window than a slightly biased but stable one. This sequential accumulation effect is not captured by one-shot metrics like Frobenius loss and must be evaluated through filtering experiments (Chapter 4).

3.2 Shrinkage Targets

The four target variants below differ only in their choice of $\mathbf{\Pi}_0$. The first three follow (Looijmans et al., 2024); the fourth is a domain-agnostic adaptation we propose.

[1] The values $\alpha = 0.5$, $\beta = 0.5$ are chosen for illustration; the Bodnar formula (2.37)–(2.38) would compute different optimal coefficients for this particular $\mathbf{S}$. The point of the example is to compare the effect of operating in covariance space versus precision space at the same shrinkage intensity, not to show the optimal coefficients.

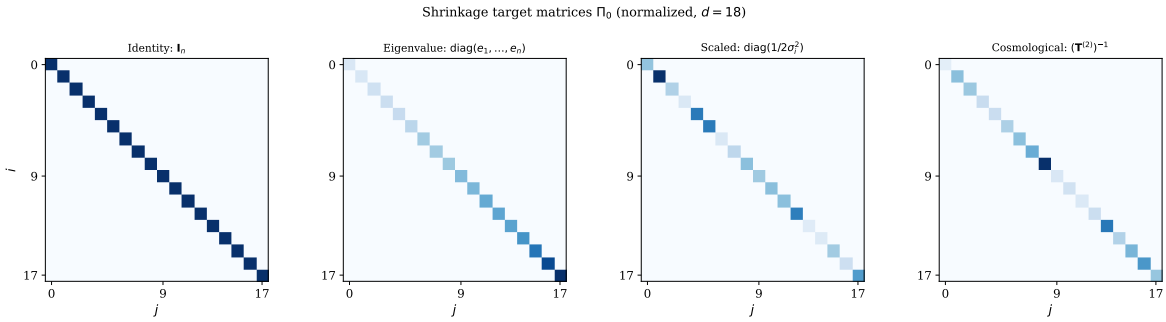

Figure 3.1: The four shrinkage target matrices $\mathbf{\Pi}_0$ (normalized, $d = 18$). All are diagonal by construction. The identity target assigns equal weight to all variables; the eigenvalue target uses the sample precision's spectral information; the scaled and cosmological targets incorporate variance structure from the data or domain knowledge.

Algorithm 1 Direct Precision Shrinkage (Identity Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$

1: Compute sample covariance: $\mathbf{S} = \frac{1}{N_e - 1}\mathbf{DX}\,\mathbf{DX}^T$
2: Compute sample precision: $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$
3: Set target: $\mathbf{\Pi}_0 = \mathbf{I}_n$
4: Compute α^*, β^* via Eqs. (2.37)–(2.38)
5: Clamp: $\alpha^* \leftarrow \max(0, \min(1, \alpha^*))$, $\beta^* \leftarrow \max(0, \min(1 - \alpha^*, \beta^*))$
6: **return** $\mathbf{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\mathbf{\Pi}}_S + \beta^* \mathbf{\Pi}_0$

Figure 3.1 provides a visual comparison of the four target matrices. The identity and eigenvalue targets are domain-agnostic; the scaled identity and cosmological targets incorporate information about the variable-specific variance structure.

3.2.1 Identity Target

The simplest choice: $\mathbf{\Pi}_0 = \mathbf{I}_n$, assuming independent unit-variance priors. This carries no domain-specific information, but the direct precision formulation still produces a valid precision matrix. Inverting a shrunk covariance, by contrast, can amplify estimation error.

For the identity target, the terms in the Bodnar formula simplify: $\|\mathbf{\Pi}_0\|_F^2 = n$, $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0) = \mathrm{tr}(\hat{\mathbf{\Pi}}_S)$, and $\beta^* = (1 - n/N_e - \alpha^*) \cdot \mathrm{tr}(\hat{\mathbf{\Pi}}_S)/n$. When the true precision has large diagonal entries relative to n (as happens in Lorenz 96), the factor $\mathrm{tr}(\hat{\mathbf{\Pi}}_S)/n$ becomes very large, driving β^* to extreme values that require clamping.

3.2.2 Eigenvalue Target

A diagonal matrix of the sorted eigenvalues of $\hat{\mathbf{\Pi}}_S$: $\mathbf{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$ with $e_1 \leq \cdots \leq e_n$. This retains the spectral information of the sample precision while zeroing out off-diagonal correlations, and requires no external domain knowledge. It corresponds to the first precision target ($\mathbf{\Pi}_0^{(1)}$) in (Looijmans et al., 2024).

Because the eigenvalue target is derived from the same sample as the precision matrix it is shrinking toward, the Cauchy-Schwarz term $\|\hat{\mathbf{\Pi}}_S\|_F^2 \|\mathbf{\Pi}_0^{(1)}\|_F^2 - (\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0^{(1)}))^2$ tends to be small, which drives α^* down and β^* toward zero. In our experiments, β^* is effectively zero

Algorithm 2 Direct Precision Shrinkage (Eigenvalue Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$
1: Compute $\mathbf{S}$, $\hat{\mathbf{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: Compute eigenvalues $e_1 \leq \cdots \leq e_n$ of $\hat{\mathbf{\Pi}}_S$
3: Set target: $\mathbf{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$
4: Compute α^*, β^* via Eqs. (2.37)–(2.38), clamp to $[0, 1]$
5: **return** $\mathbf{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\mathbf{\Pi}}_S + \beta^* \mathbf{\Pi}_0^{(1)}$

Algorithm 3 Direct Precision Shrinkage (Cosmological Target)

Require: Deviation matrix $\mathbf{DX}$, power spectrum C_ℓ, mode counts N_ℓ
1: Compute $\mathbf{S}$, $\hat{\mathbf{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: **for** each data element i mapped to multipole ℓ_i **do**
3: Compute diagonal covariance target: $T_{ii}^{(2)} = \frac{2}{N_{\ell_i}} C_{\ell_i}^2$
4: **end for**
5: Set target precision: $\mathbf{\Pi}_0^{(2)} = (\mathbf{T}^{(2)})^{-1}$
6: Compute α^*, β^* via Eqs. (2.37)–(2.38), clamp to $[0, 1]$
7: **return** $\mathbf{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\mathbf{\Pi}}_S + \beta^* \mathbf{\Pi}_0^{(2)}$

($\sim 10^{-4}$) for the eigenvalue target, so the estimator reduces to $\alpha^* \hat{\mathbf{\Pi}}_S$, a simple rescaling of the sample precision without meaningful shrinkage toward the target.

3.2.3 Cosmological Target

The analytical target from (Looijmans et al., 2024), derived from the Gaussian power spectrum covariance (2.39). Each diagonal entry of the target covariance is $T_{ii}^{(2)} = (2/N_{\ell_i}) C_{\ell_i}^2$, encoding cosmic variance at multipole ℓ_i (Hamilton et al., 2006), and the target precision is its inverse: $\mathbf{\Pi}_0^{(2)} = (\mathbf{T}^{(2)})^{-1}$. When the assumed power spectrum model is close to the truth, this provides a stronger shrinkage direction than the generic alternatives.

3.2.4 Scaled Identity Target (Novel Adaptation)

The cosmological target requires power spectrum data C_ℓ and mode counts N_ℓ that are unavailable outside cosmology. We propose a domain-agnostic alternative that uses empirical per-variable variances from the sample covariance:

$$\mathbf{\Pi}_0 = \mathrm{diag}\left(\frac{1}{2\sigma_1^2}, \ldots, \frac{1}{2\sigma_n^2} \right), \tag{3.2}$$

where σ_i^2 are the diagonal entries of $\mathbf{S}$. The factor of 2 mirrors the $2/N_\ell$ scaling of the cosmological target (2.41): in the cosmological case, the target precision element for multipole ℓ is $N_\ell/(2C_\ell^2)$, which can be written as $1/(2 \cdot C_\ell^2/N_\ell) = 1/(2 \cdot \sigma_\ell^2)$ where $\sigma_\ell^2 = C_\ell^2/N_\ell$ is the cosmic variance. Our scaled target replaces the theoretical cosmic variance with the empirical variance from the ensemble.

This variant serves two purposes. First, it provides a middle ground between the plain identity (which ignores variable-specific scales) and the cosmological target (which requires

domain knowledge). Second, by comparing it against the other targets, we can quantify how much of the direct precision shrinkage performance depends on domain-specific target knowledge versus the general framework itself.

3.3 Worked Example

To make the shrinkage coefficient computation concrete, consider a small example with $n = 3$ and $N_e = 5$. Suppose the sample covariance is

$$\mathbf{S} = \begin{pmatrix} 2.1 & 0.3 & 0.1 \\ 0.3 & 1.8 & 0.2 \\ 0.1 & 0.2 & 2.5 \end{pmatrix},$$

so the sample precision is

$$\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1} = \begin{pmatrix} 0.490 & -0.077 & -0.013 \\ -0.077 & 0.575 & -0.042 \\ -0.013 & -0.042 & 0.408 \end{pmatrix}.$$

For the identity target $\mathbf{\Pi}_0 = \mathbf{I}_3$, we compute:

$$\begin{aligned} \|\hat{\mathbf{\Pi}}_S\|_{\text{tr}} &= 0.490 + 0.575 + 0.408 = 1.473, \\ \|\hat{\mathbf{\Pi}}_S\|_F^2 &= 0.490^2 + 0.575^2 + 0.408^2 + 2(0.077^2 + 0.013^2 + 0.042^2) = 0.751, \\ \|\mathbf{\Pi}_0\|_F^2 &= 3, \\ \text{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0) &= \text{tr}(\hat{\mathbf{\Pi}}_S) = 1.473. \end{aligned}$$

The Cauchy-Schwarz denominator is $\|\hat{\mathbf{\Pi}}_S\|_F^2 \|\mathbf{\Pi}_0\|_F^2 - (\text{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0))^2 = 0.751 \times 3 - 1.473^2 = 2.253 - 2.170 = 0.083$. Then:

$$\alpha^* = 1 - \frac{3}{5} - \frac{(1/5) \times 1.473^2 \times 3}{0.083} = 0.4 - \frac{1.302}{0.083} = 0.4 - 15.7 = -15.3.$$

After clamping to $[0, 1]$: $\alpha^* = 0$. The correction term is much larger than $1 - n/N_e = 0.4$, driving α^* negative. This happens because the identity target is not well-aligned with the sample precision (the Cauchy-Schwarz denominator is small, so the correction is large).

$$\beta^* = (1 - 3/5 - 0) \times \frac{1.473}{3} = 0.4 \times 0.491 = 0.196.$$

After clamping: $\beta^* = 0.196$. The final estimator is $\mathbf{\Pi}_{\text{LS}} = 0 \cdot \hat{\mathbf{\Pi}}_S + 0.196 \cdot \mathbf{I}_3 = 0.196\, \mathbf{I}_3$. All weight goes to the target, and the sample precision is discarded entirely. This is the correct behavior when the target does not match the sample: the formula detects the mismatch and assigns zero weight to the sample precision rather than producing a compromised blend.

In practice, when $n = 40$ and $N_e = 50$ (as in our Lorenz 96 experiments), the correction term in α^* is even more extreme because the Cauchy-Schwarz denominator becomes very

Table 3.1: Computational complexity per assimilation cycle. n: state dimension, N_e: ensemble size, m: observation dimension, r: Cholesky bandwidth, B: number of NERCOME bootstrap draws.

Method	Cost per cycle
EnKF (standard)	$\mathcal{O}(n^2 N_e + m^2 N_e)$
EnKF-Cholesky	$\mathcal{O}(n \cdot r \cdot N_e)$
Identity Shrinkage	$\mathcal{O}(n^2 N_e + n^3)$
Scaled Shrinkage	$\mathcal{O}(n^2 N_e + n^3)$
Cosmological Shrinkage	$\mathcal{O}(n^2 N_e + n^3)$
Eigenvalue Shrinkage	$\mathcal{O}(n^2 N_e + 2n^3)$
NERCOME	$\mathcal{O}(B \cdot (n^2 N_e + n^3))$
Ledoit-Wolf	$\mathcal{O}(n^2 N_e)$

small relative to $\|\hat{\mathbf{\Pi}}_S\|_{\mathrm{tr}}^2 \|\mathbf{\Pi}_0\|_F^2$, and the unclamped β^* values reach 10^8 to 10^9.

3.4 Computational Complexity

All four direct precision shrinkage variants share a common computational core: forming the sample covariance $\mathbf{S}$ costs $\mathcal{O}(n^2 N_e)$, and inverting it to obtain $\hat{\mathbf{\Pi}}_S$ costs $\mathcal{O}(n^3)$. Computing α^* and β^* requires the Frobenius norm $\|\hat{\mathbf{\Pi}}_S\|_F^2$, the trace norm $\|\hat{\mathbf{\Pi}}_S\|_{\mathrm{tr}}$, and the product $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0)$. For a dense target, each of these costs $\mathcal{O}(n^2)$; for a diagonal target (identity, scaled identity, cosmological), the cost is $\mathcal{O}(n)$ since $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0) = \sum_i (\hat{\mathbf{\Pi}}_S)_{ii} (\mathbf{\Pi}_0)_{ii}$. The final linear combination (3.1) costs $\mathcal{O}(n^2)$. Since the $\mathcal{O}(n^3)$ inversion dominates, all four variants have the same asymptotic cost per assimilation cycle.

They differ in target construction:

- **Identity**: No computation needed ($\mathbf{\Pi}_0 = \mathbf{I}$).
- **Scaled identity**: Reads n diagonal entries of $\mathbf{S}$ and computes n reciprocals. Cost: $\mathcal{O}(n)$.
- **Cosmological**: Computes n diagonal entries from precomputed C_ℓ and N_ℓ values. Cost: $\mathcal{O}(n)$.
- **Eigenvalue**: Requires a symmetric eigendecomposition of the $n \times n$ matrix $\hat{\mathbf{\Pi}}_S$. Cost: $\mathcal{O}(n^3)$, which doubles the effective per-cycle cost.

Table 3.1 summarizes the per-cycle cost of each method tested in this work, including the baselines.

The modified Cholesky approach avoids the $\mathcal{O}(n^3)$ inversion entirely by exploiting banded structure, making it the cheapest option on spatially extended systems where $r \ll n$. Ledoit-Wolf shrinks the covariance directly and does not require inversion ($\mathcal{O}(n^2 N_e)$), though the subsequent EnKF solve still involves an $\mathcal{O}(m^2 N_e)$ linear system. NERCOME is the most expensive: its B bootstrap resamples (typically $B = 200$) each require a covariance computation and eigendecomposition, resulting in a cost roughly B times that of the other shrinkage methods.

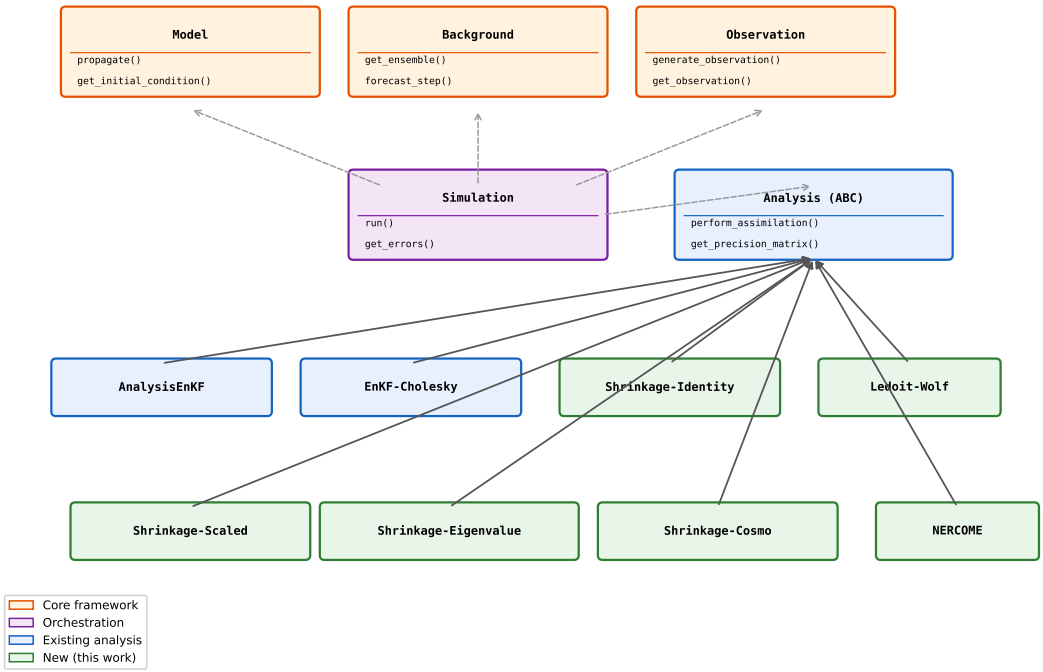

Figure 3.2: Class diagram of the TEDA framework, highlighting the new shrinkage estimator classes (shaded). All analysis methods inherit from the abstract `Analysis` base class and are registered via the `AnalysisFactory`, enabling interchangeability.

In our experiments, the practical timing differences are substantial. On the AR(1) cosmological model ($n = 18$, $N_e = 30$), NERCOME takes approximately 27 seconds per 100 assimilation cycles versus approximately 0.2 seconds for the other shrinkage methods. On Lorenz 96 ($n = 40$, $N_e = 80$), the gap widens to approximately 215 seconds versus approximately 26 seconds.

3.5 Implementation in TEDA

All estimators are implemented in TEDA (Toolbox for Ensemble Data Assimilation) (Nino-Ruiz and Valbuena, 2022), a Python-based educational framework that implements ensemble methods alongside toy dynamical models such as Lorenz 96. TEDA uses an object-oriented design where dynamical models, ensemble management, observation handling, and analysis methods are separate modular classes. Figure 3.2 shows the relevant class hierarchy.

3.5.1 Architecture

The main classes in TEDA are:

- **Model**: Abstract base for dynamical systems (e.g., Lorenz 96, AR(1)), defining the `propagate()` interface that advances the state by one time step.

- **Background**: Manages the ensemble of state vectors, computes ensemble statistics (mean, perturbation matrix, sample covariance), and applies inflation.
- **Observation**: Generates synthetic observations by applying the observation operator $\mathbf{H}$ to the true state and adding noise, and stores $\mathbf{H}$ and $\mathbf{R}$.
- **Analysis**: Abstract base for assimilation methods. Subclasses override `perform_assimilation()` to implement specific analysis schemes. The six shrinkage estimators introduced in this work are all subclasses of `Analysis`.
- **Simulation**: Orchestrates the forecast-analysis cycle, calling `Model.propagate()` and `Analysis.perform_assimilation()` in sequence, and storing results for post-processing.

New analysis methods are registered via a factory pattern. The `@register_analysis` decorator allows users to select methods by name at runtime:

```
from pyteda.analysis import AnalysisFactory
method = AnalysisFactory.create("EnKF-Shrinkage-Identity",
                                params=config)
```

A typical comparative experiment requires only changing the method name:

```
methods = ["EnKF", "EnKF-Cholesky",
           "EnKF-Shrinkage-Identity",
           "EnKF-Shrinkage-Scaled",
           "EnKF-Shrinkage-Eigenvalues",
           "EnKF-Shrinkage-Cosmo",
           "EnKF-NERCOME",
           "EnKF-Ledoit-Wolf"]
for name in methods:
    sim = Simulation(model, background,
                     AnalysisFactory.create(name),
                     observation)
    sim.run()
```

3.5.2 Estimator Classes

Each of the four direct precision shrinkage variants is implemented as a separate class, differing only in the `get_target()` method:

- `AnalysisEnKFDirectPrecisionShrinkageIdentity`: Returns $\mathbf{I}_n$.
- `AnalysisEnKFDirectPrecisionShrinkageIdentityScaled`: Computes $\mathrm{diag}(1/(2\sigma_1^2), \ldots, 1/(2\sigma_n^2))$ from the diagonal of $\mathbf{S}$.
- `AnalysisEnKFDirectPrecisionShrinkageEigenvalues`: Eigendecomposes $\hat{\mathbf{\Pi}}_S$ and returns the sorted eigenvalue diagonal.

Table 3.2: Comparison of all methods tested in this thesis. "Space" indicates whether the method operates on the covariance matrix ($\mathbf{P}$) or the precision matrix ($\mathbf{\Pi}$). "Target" indicates whether domain-specific prior knowledge is required. "Min N_e" is the minimum ensemble size for the method to be applicable.

Method	Space	Target	Min N_e	Key property
EnKF (standard)	$\mathbf{P}$	None	2	Baseline, no regularization
EnKF-Cholesky	$\mathbf{\Pi}$	None	$r+2$	Banded, implicit localization
Identity Shrinkage	$\mathbf{\Pi}$	$\mathbf{I}_n$	$n+2$	Simplest direct precision
Scaled Shrinkage	$\mathbf{\Pi}$	Empirical	$n+2$	Domain-agnostic (novel)
Eigenvalue Shrinkage	$\mathbf{\Pi}$	Data-driven	$n+2$	Spectral information
Cosmo. Shrinkage	$\mathbf{\Pi}$	Domain	$n+2$	Best for diagonal $\mathbf{\Pi}$
NERCOME	$\mathbf{P}$	None	3	Target-free, bootstrap
Ledoit-Wolf	$\mathbf{P}$	$\mu\mathbf{I}$	2	Low variance, standard ref.

- `AnalysisEnKFDirectPrecisionShrinkageCosmo`: Reads BOSS power spectrum parameters and constructs the cosmological target (2.41).

All four share a common `compute_shrinkage_coefficients()` method that implements Eqs. (2.37)–(2.38) with clamping. The implementation follows the Bodnar formula exactly, operating on the raw sample inverse $\mathbf{S}^{-1}$ without the Hartlap correction. This matches the derivation in (Bodnar et al., 2016), which accounts for the estimation bias through the $(1 - n/N_e)$ term.

NERCOME is implemented as `AnalysisEnKFNercomeShrinkage`, which performs 200 bootstrap resamples per assimilation cycle. Ledoit-Wolf is implemented as `AnalysisEnKFLedoitWolfShrinl` wrapping the `sklearn.covariance.LedoitWolf` estimator and inverting the result.

All code is publicly available.[2]

3.6 Summary of Methods

Table 3.2 provides a systematic comparison of all eight methods tested in this thesis, including their estimation space (covariance vs. precision), target requirements, minimum ensemble size, and key properties.

The methods differ in the prior assumptions they bring to the estimation problem. The standard EnKF uses the raw sample covariance with no regularization, which works well at large ensemble sizes but degrades at small ones. The cosmological shrinkage target occupies the opposite end: it encodes detailed domain knowledge about the precision structure and achieves the best standalone estimation quality when that knowledge is correct. The remaining methods fall between these two points, each trading some generality for some robustness.

A practical user choosing among these methods faces two key questions: (1) Is the ensemble large enough for direct precision methods ($N_e > n+2$)? If not, the covariance-space methods (EnKF, Ledoit-Wolf, NERCOME) or EnKF-Cholesky are the only options.

[2]TEDA source code: https://github.com/enino84/TEDA.git

(2) Does domain knowledge about the precision structure exist? If so, incorporating it through a domain-specific target can improve estimation quality. If not, Ledoit-Wolf provides the safest default for sequential filtering, and the scaled identity target provides a middle ground for standalone precision estimation.

Chapter 4

Benchmark

4.1 Experimental Setup

We test on two systems with different precision matrix structures, chosen to bracket the range of conditions the direct precision shrinkage framework is likely to encounter.

4.1.1 Lorenz 96

The Lorenz 96 model (Lorenz, 1996) is a system of n coupled ordinary differential equations:

$$\frac{dx_i}{dt} = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F, \quad i = 1, \ldots, n, \tag{4.1}$$

with cyclic boundary conditions ($x_{-1} = x_{n-1}$, $x_0 = x_n$, $x_{n+1} = x_1$). With $n = 40$ and forcing $F = 8$, the system exhibits chaotic dynamics with a Lyapunov time of approximately 0.3 time units. The spatial coupling through the quadratic advection terms produces a dense background covariance, and consequently a dense precision matrix with significant off-diagonal entries. This makes Lorenz 96 a demanding test for estimators that assume diagonal or near-diagonal precision structure.

We use the following configuration:

- **State dimension**: $n = 40$.
- **Observations**: $m = 32$ randomly placed, Gaussian error $\sigma_o = 0.01$.
- **Ensemble sizes**: $N_e \in \{50, 60, 80, 100\}$, all above the $N_e > n + 2 = 42$ threshold required for the sample covariance to be invertible (a prerequisite for the direct precision methods).
- **Assimilation window**: 100 cycles, observation frequency $\Delta t_{\text{obs}} = 0.1$ (total time $T = 10$).
- **Inflation**: Multiplicative, $\rho = 1.05$.
- **Repetitions**: 30 runs per configuration with different random seeds.

Seven methods are compared: standard EnKF, EnKF with modified Cholesky decomposition (bandwidth $r = 2$), three direct precision shrinkage variants (identity, scaled identity, eigenvalue targets), NERCOME, and Ledoit-Wolf covariance shrinkage. The cosmological target is excluded from the Lorenz 96 experiments since this model does not produce power spectrum data.

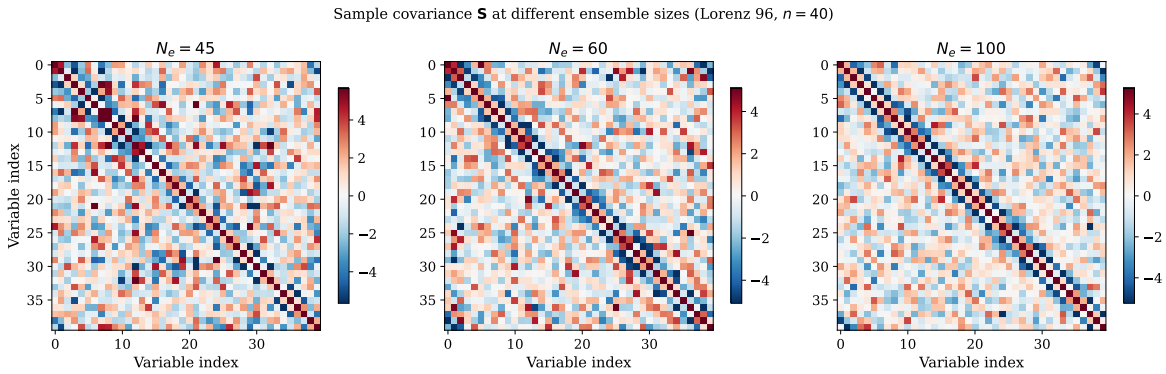

Figure 4.1: Sample covariance matrices $\mathbf{S}$ on Lorenz 96 ($n = 40$) at three ensemble sizes. At $N_e = 45$, the matrix is dominated by sampling noise; at $N_e = 100$, the banded structure from spatial coupling is clearly visible.

Figure 4.1 illustrates the quality of the sample covariance estimate at different ensemble sizes on Lorenz 96. At $N_e = 45$ (barely above the invertibility threshold), the sample covariance is noisy and the banded structure of the true covariance is obscured. At $N_e = 100$, the tridiagonal pattern from Lorenz 96's nearest-neighbor coupling is clearly visible, and the off-diagonal entries are well resolved. The direct precision shrinkage methods must invert these sample covariance matrices, so the noise at small N_e directly translates into ill-conditioned precision estimates.

4.1.2 AR(1) Cosmological Model

To test on data with near-diagonal precision structure, we build an AR(1) dynamical model calibrated to BOSS DR12 Patchy mock catalog statistics from (Looijmans et al., 2024):

$$\mathbf{x}_{t+1} = \phi\, \mathbf{x}_t + (1-\phi)\, \boldsymbol{\mu} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{4.2}$$

with $\boldsymbol{\mu}$ the mean power spectrum from the mocks, $\phi = 0.95$, and $\mathbf{Q} = (1-\phi^2)\boldsymbol{\Sigma}_{\text{mock}}$ so the stationary distribution has covariance $\boldsymbol{\Sigma}_{\text{mock}}$. The AR(1) structure provides temporal dynamics (necessary for sequential filtering) while preserving the near-diagonal covariance structure of the mock catalogs. This is admittedly a much simpler dynamical system than a realistic cosmological simulation, but it serves the purpose of testing whether the estimators behave as expected when the precision matrix matches their assumed structure.

Configuration:

- **State dimension**: $d = 18$ (9 k-bins $\times$ 2 multipoles).
- **Observations**: $m = 14$ ($\approx 78\%$ coverage), observation error $\sigma_o = 100$.
- **Ensemble sizes**: $N_e \in \{24, 30, 40, 50\}$.
- **Inflation**: Multiplicative, $\rho = 1.02$.
- **Repetitions**: 30 runs per configuration.

All eight methods are tested, including the cosmological target with BOSS power spectrum parameters.

4.1.3 BOSS DR12 Mock Catalog Data

The BOSS (Baryon Oscillation Spectroscopic Survey) DR12 data used in our experiments comes from the Patchy mock catalog suite (Looijmans et al., 2024). These mocks underpin the clustering covariances reported by Percival et al. (2014) and related BOSS analyses; they simulate the galaxy power spectrum of the BOSS Northern Galactic Cap (NGC) at redshift z_1, producing 2048 independent realizations of the $d = 18$ dimensional data vector (9 k-bins $\times$ 2 multipole moments, $\ell = 0$ and $\ell = 2$).

The "true" precision matrix is computed from all 2048 mocks using the Hartlap correction (Hartlap et al., 2007):

$$\mathbf{\Pi}_{\text{true}} = \frac{N_{\text{mocks}} - d - 2}{N_{\text{mocks}} - 1} \left(\frac{1}{N_{\text{mocks}} - 1} \sum_{i=1}^{N_{\text{mocks}}} (\mathbf{p}_i - \bar{\mathbf{p}})(\mathbf{p}_i - \bar{\mathbf{p}})^T \right)^{-1}, \tag{4.3}$$

where $\mathbf{p}_i$ is the power spectrum vector from mock i and $\bar{\mathbf{p}}$ is the mean. With $N_{\text{mocks}} = 2048$ and $d = 18$, the Hartlap factor is $\gamma = (2048 - 18 - 2)/(2048 - 1) = 0.990$, so the correction is minimal. This "true" precision serves as the ground truth against which all estimators are evaluated.

The cosmological shrinkage target $\mathbf{T}^{(2)}$ is constructed from the theoretical power spectrum model:

$$T_{ii}^{(2)} = \frac{2}{N_{\ell_i}} C_{\ell_i}^2, \tag{4.4}$$

using the power spectrum values C_ℓ and mode counts N_ℓ from the BOSS survey parameters. The target matrix is pre-computed and stored as a 18×18 file (`T_18_18.matrix`).

4.1.4 Choice of Ensemble Size

The ensemble sizes listed in Sections 4.1.1 and 4.1.2 ($N_e \in \{50, 60, 80, 100\}$ for Lorenz 96 and $N_e \in \{24, 30, 40, 50\}$ for the cosmological model) are not arbitrary. They are chosen to cover the regime in which shrinkage estimation of the precision matrix is both *necessary* and *non-trivial*, bounded below by a hard invertibility constraint and above by the point where shrinkage stops providing any benefit over the sample estimate.

Theoretical lower bound. The direct precision methods of (Bodnar et al., 2016) operate on the sample precision $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$, which exists only when the sample covariance $\mathbf{S}$ is full-rank. For a Gaussian ensemble, $\mathbf{S}$ follows a Wishart distribution with $N_e - 1$ degrees of freedom, which is non-singular almost surely iff $N_e \geq n + 1$. The corresponding inverse Wishart has finite first two moments only for $N_e \geq n + 4$, so the Bodnar closed-form coefficients are rigorously defined only in this regime. For Lorenz 96 ($n = 40$), this translates into $N_e \geq 44$; our lowest tested value $N_e = 50$ exceeds this by six members, enough to keep the Hartlap factor $\gamma = (N_e - n - 2)/(N_e - 1) = 8/49 \approx 0.163$ away from zero while still occupying the

small-N_e end of the practical range. For the cosmological model ($n = 18$), $N_e = 24$ gives $\gamma = 4/23 \approx 0.174$, quantitatively comparable.

Upper bound where shrinkage is uninteresting. From the bias-variance decomposition reviewed in Section 2.5, the variance of the sample covariance scales as $\mathcal{O}(n^2/N_e)$. When $N_e \gg n$, this variance is small, the sample precision is close to the true precision, and the optimal shrinkage intensity $\lambda^* \to 0$. In the specific case of the (Bodnar et al., 2016) α^*, the leading term $1 - n/N_e$ (Eq. 2.37) approaches 1 as N_e grows, meaning almost all weight is placed on the sample precision and the target becomes irrelevant. Testing at N_e much larger than n therefore adds little beyond confirming convergence to the sample estimator; the regime where the choice of target, clamping behavior, and coefficient stability actually matters is $n/N_e \in [0.3, 0.8]$, which bounds our top tested values $N_e = 100$ (Lorenz 96, $n/N_e = 0.4$) and $N_e = 50$ (cosmo, $n/N_e = 0.36$) from above.

Regime map. Figure 5.1 (analyzed in Chapter 5) places both experiments on a two-dimensional plane spanned by the concentration ratio n/N_e and the structural distance from the shrinkage target. The four N_e values per domain span the right half of that plane: rank-deficiency is avoided, but N_e is never so large that the sample estimator becomes optimal on its own. This is the regime that is both operationally relevant (ensemble sizes in weather prediction are in the tens, not thousands) and statistically interesting (the shrinkage intensity is neither 0 nor 1 but varies with the target).

Empirical validation. Beyond the theoretical bounds, we verified empirically that the four chosen values of N_e per domain sample the interior of the regime of interest rather than crowd at one end. Figures 4.10, 4.11, 4.12, and 4.13 (discussed in Section 4.2) show that the shrinkage coefficients α^* and β^*, the condition number of $\hat{\mathbf{\Pi}}_S$, and the resulting RMSE all change by orders of magnitude across $N_e \in \{45, 50, \ldots, 100\}$, so this range cannot be condensed to a single representative value. The four reported sizes were selected as a regularly spaced subsample: $N_e = 50$ stresses the small-ensemble end where the Bodnar coefficients are most sensitive; $N_e = 100$ represents the well-sampled regime where RMSE has stabilized; $N_e = 60$ and 80 interpolate. For the cosmological domain, the analogous rationale applies: $N_e = 24$ matches the small-sample regime typical of mock-catalog availability in observational cosmology, $N_e = 50$ gives $N_e/n \approx 2.8$ (close to the convergence threshold), and $N_e = 30$ and 40 interpolate.

Summary. Table 4.1 consolidates the rationale for each N_e value tested. The span has been chosen so that every reported value lies strictly between the invertibility bound ($N_e > n + 2$) and the variance-negligible bound ($N_e \gtrsim 3n$), with a diagnostic sweep in the appendix confirming that no further insight is obtained from finer sampling.

4.1.5 Standalone Precision Estimation

We also replicate the standalone precision estimation experiment of (Looijmans et al., 2024) using the same BOSS DR12 mock catalogs (2048 power spectrum realizations with $d = 18$

Table 4.1: Ensemble sizes tested, with per-value motivation. n/N_e is the concentration ratio; the regime labels are: RD (rank-deficient, $N_e \leq n$, not tested), ILL (ill-conditioned, N_e slightly above n), MOD (moderately ill-conditioned, shrinkage most useful), WS (well-sampled, shrinkage intensity small).

Domain	n	N_e	n/N_e	Regime	Role in the study
Lorenz 96	40	50	0.80	ILL	Stress test of Bodnar coefficients near the invertibility boundary
	40	60	0.67	MOD	Interior of the moderately ill-conditioned band
	40	80	0.50	MOD	Upper interior; RMSE transition point
	40	100	0.40	MOD–WS	Largest tested; shrinkage intensity small but non-zero
Cosmological	18	24	0.75	ILL	Matches operational mock-catalog availability (Looijmans et al., 2024)
	18	30	0.60	MOD	Standard working point in Looijmans et al.
	18	40	0.45	MOD	Mid-range; coefficients stabilize
	18	50	0.36	MOD–WS	Approaches $N_e/n \approx 2.8$, convergence threshold

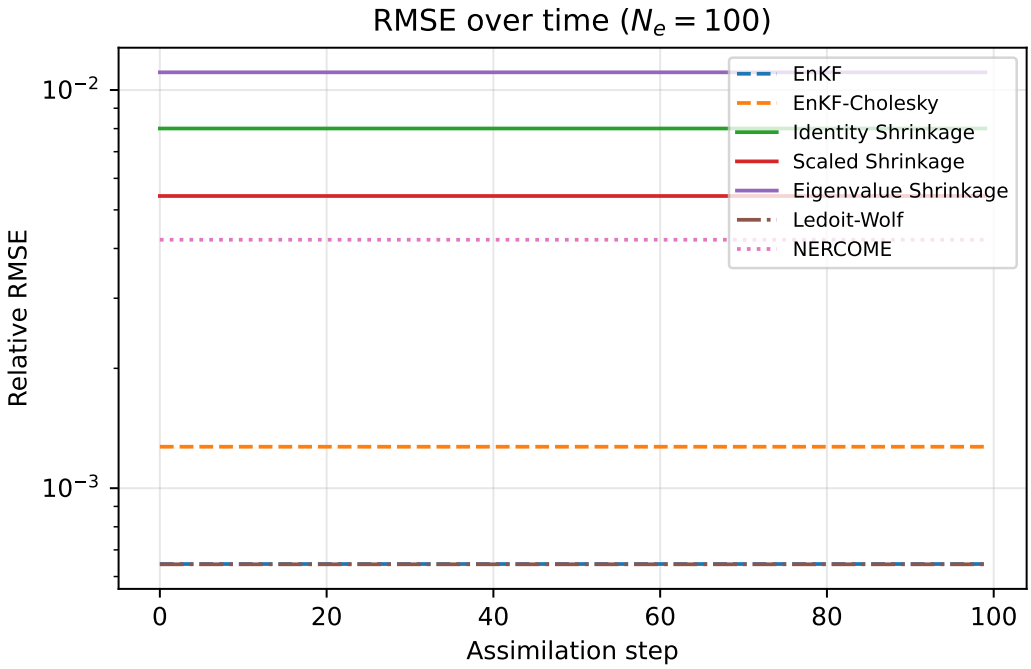

Figure 4.2: Evolution of relative RMSE over 100 assimilation cycles for different analysis methods with $N_e = 100$. Shrinkage-based methods (solid lines) and Ledoit-Wolf (dash-dot) are compared against the standard EnKF and EnKF-Cholesky (dashed).

bins). For each subsample size $n \in \{24, 30, 40, 50\}$, we draw $M = 100$ random subsets of n mocks, compute each estimator's precision matrix, and measure the relative Frobenius loss against the "true" precision (computed from all 2048 mocks with Hartlap correction (Hartlap et al., 2007)):

$$\mathcal{L}(\hat{\mathbf{\Pi}}, \mathbf{\Pi}_{\text{true}}) = \frac{\|\hat{\mathbf{\Pi}} - \mathbf{\Pi}_{\text{true}}\|_F}{\|\mathbf{\Pi}_{\text{true}}\|_F}. \tag{4.5}$$

This experiment provides a direct comparison with (Looijmans et al., 2024) and establishes a baseline for standalone estimation quality, against which the filtering results can be compared.

4.1.6 Error Metrics

For filtering experiments, we report the relative Root Mean Square Error (RMSE) of the analysis state:

$$\text{RMSE}(t) = \frac{\|\bar{\mathbf{x}}^a(t) - \mathbf{x}^{\text{true}}(t)\|_2}{\|\mathbf{x}^{\text{true}}(t)\|_2}, \tag{4.6}$$

time-averaged over the last 50 cycles to discard the initial transient. For standalone estimation, we report the relative Frobenius loss (4.5).

4.2 Lorenz 96 Results

Figure 4.2 shows RMSE over 100 assimilation cycles for six methods at $N_e = 100$.

At smaller ensemble sizes, the instability of the direct precision methods is more pro-

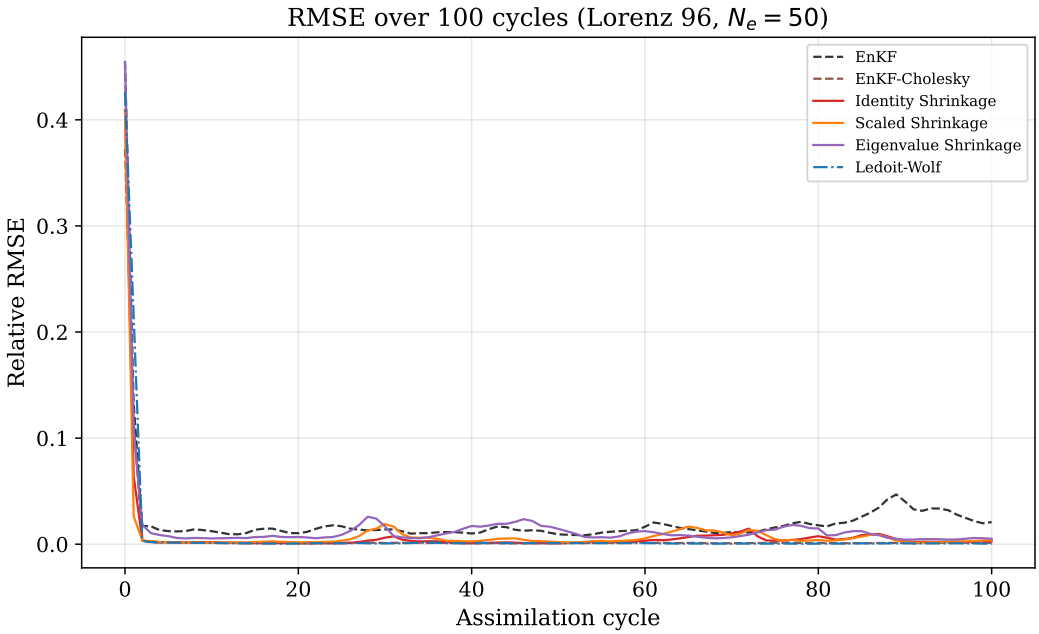

Figure 4.3: RMSE over 100 assimilation cycles on Lorenz 96 with $N_e = 50$. The direct precision shrinkage methods (red, orange, purple) show large RMSE spikes, while Ledoit-Wolf (blue) and EnKF-Cholesky (brown) remain stable.

nounced. Figures 4.3 and 4.4 show the RMSE time series at $N_e = 50$ and $N_e = 60$, respectively. At $N_e = 50$ (only 8 members above the invertibility threshold), the eigenvalue and identity shrinkage methods exhibit large spikes that persist for many cycles. At $N_e = 60$, the behavior is similar though slightly less extreme. In contrast, Ledoit-Wolf and EnKF-Cholesky maintain stable, low RMSE throughout the assimilation window at all ensemble sizes.

Figures 4.3–4.2 show several consistent patterns. The standard EnKF at $N_e = 100$ converges quickly after an initial transient of approximately 10 cycles and maintains low RMSE thereafter. The direct precision shrinkage methods, by contrast, exhibit intermittent spikes even at $N_e = 100$: episodes where the RMSE jumps by an order of magnitude before recovering. These spikes are more frequent and severe at smaller ensemble sizes: at $N_e = 50$, the identity and eigenvalue shrinkage methods spend a significant fraction of the assimilation window in a high-error regime. Ledoit-Wolf and EnKF-Cholesky are remarkably stable across all ensemble sizes, with no visible spikes.

The spikes correspond to assimilation cycles where the shrinkage precision estimate is particularly ill-conditioned, producing an analysis update that moves the ensemble mean away from the truth rather than toward it. Because the analysis becomes the next forecast, this error propagates forward and can persist for several cycles before the filter recovers. This "error cascade" effect is a direct consequence of using the precision matrix inside a sequential filter, a consideration that does not arise in standalone estimation.

Table 4.2 summarizes performance across ensemble sizes. Ledoit-Wolf achieves the lowest RMSE at three of four ensemble sizes, with near-zero variance across runs. The direct precision shrinkage methods show erratic non-monotonic behavior (RMSE sometimes jumps at

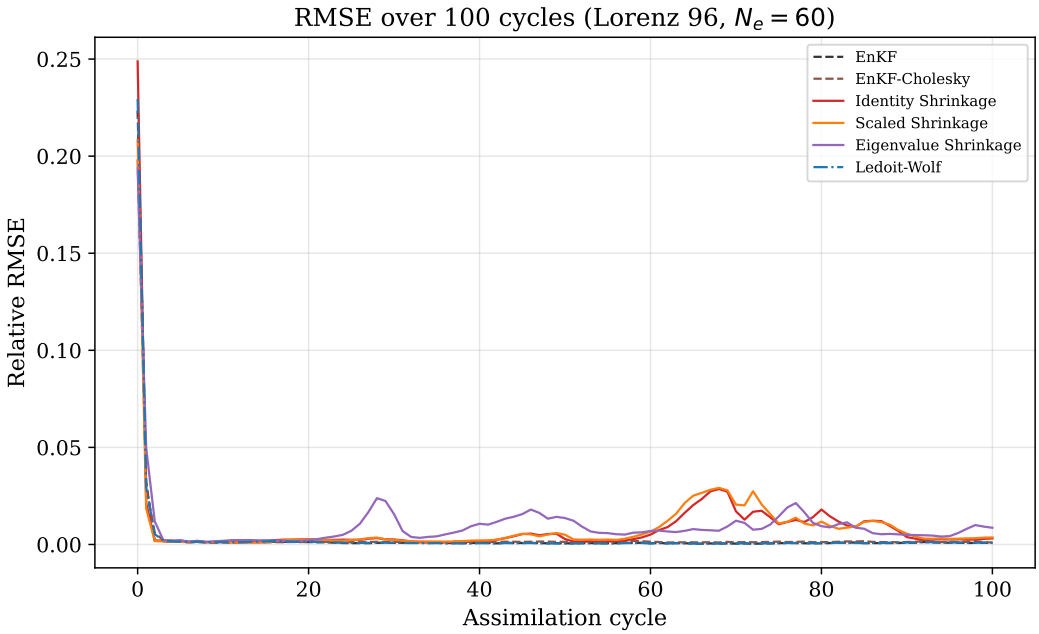

Figure 4.4: RMSE over 100 assimilation cycles on Lorenz 96 with $N_e = 60$. Similar pattern to $N_e = 50$ but with slightly reduced spike magnitudes.

intermediate ensemble sizes before dropping again) and high standard deviations reflecting seed-dependent instability.

Looking at the table entries in detail: the standard EnKF exhibits catastrophic behavior at small ensemble sizes, with RMSE of 3.06×10^{-2} at $N_e = 50$ and 1.15×10^{-2} at $N_e = 60$, accompanied by enormous standard deviations (± 11.45 and ± 4.33). These large values reflect occasional filter divergence: a few seeds produce ill-conditioned sample covariances that cause the analysis to diverge, inflating both the mean and variance across 30 runs. At $N_e = 80$ and $N_e = 100$, the standard EnKF stabilizes sharply and achieves the best RMSE of any method (0.06×10^{-2} at $N_e = 100$). This reflects the fact that at sufficiently large ensemble sizes, the sample covariance is a good estimator and shrinkage provides no benefit, only added risk.

Figures 4.5 and 4.6 show box plots at $N_e = 50$ and $N_e = 100$, respectively, complementing the $N_e = 80$ box plot in Figure 4.8. At $N_e = 50$, the direct precision methods show the widest interquartile ranges, with several runs producing RMSE values 3–5 times the median. At $N_e = 100$, the distributions tighten but the direct precision methods still show more outliers than Ledoit-Wolf. The standard EnKF has negligible variance at $N_e = 100$ (± 0.01) but extreme variance at $N_e = 50$ (± 11.45), reflecting occasional filter divergence when the sample covariance is ill-conditioned.

EnKF-Cholesky is the most consistent method: its RMSE stays at 0.13–0.14×10^{-2} with standard deviation ± 0.01 regardless of ensemble size. The banded precision structure imposed by the modified Cholesky decomposition acts as implicit localization, preventing the ill-conditioning that affects the other precision-based methods. The price is a ceiling on accuracy: even at $N_e = 100$, EnKF-Cholesky cannot match the standard EnKF because the bandwidth $r = 2$ discards real correlations beyond two grid points.

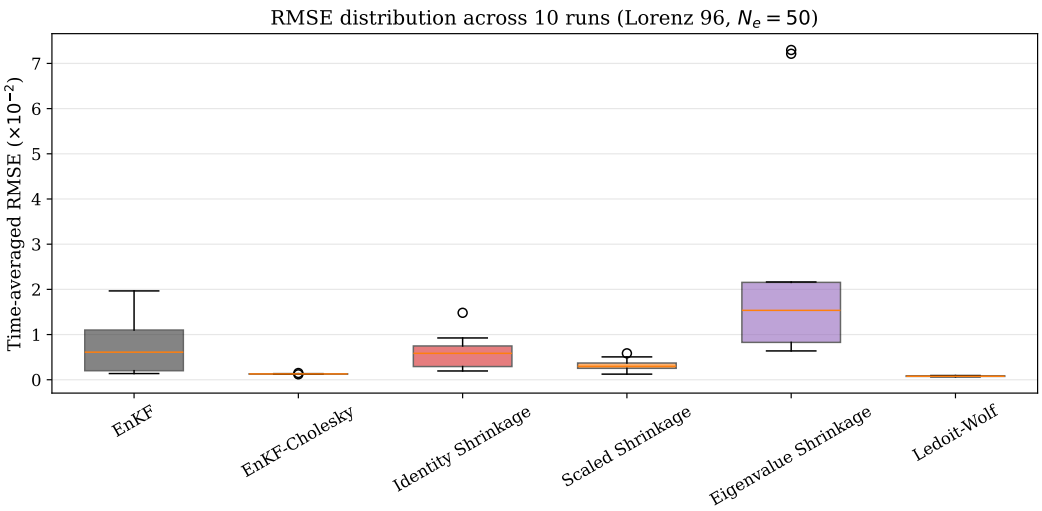

Figure 4.5: RMSE distribution across 30 runs on Lorenz 96 with $N_e = 50$. The direct precision methods show the widest distributions, reflecting seed-dependent instability.

Table 4.2: Time-averaged relative RMSE ($\times 10^{-2}$) for different methods and ensemble sizes on Lorenz 96. Best result per column in **bold**. †Scaled Shrinkage is a novel domain-agnostic adaptation proposed in this work.

Method	$N_e = 50$	$N_e = 60$	$N_e = 80$	$N_e = 100$
EnKF (standard)	3.06 ± 11.45	1.15 ± 4.33	0.08 ± 0.02	$\mathbf{0.06 \pm 0.01}$
EnKF-Cholesky	0.13 ± 0.01	0.14 ± 0.01	0.13 ± 0.01	0.13 ± 0.01
Identity Shrinkage	0.84 ± 0.57	1.13 ± 1.19	0.85 ± 0.46	0.80 ± 0.65
Scaled Shrinkage†	0.51 ± 0.28	0.73 ± 0.62	0.61 ± 0.33	0.54 ± 0.36
Eigenvalue Shrinkage	1.72 ± 0.83	1.81 ± 1.58	1.43 ± 1.01	1.11 ± 0.89
NERCOME	0.23 ± 0.15	0.28 ± 0.15	0.34 ± 0.17	0.42 ± 0.18
Ledoit-Wolf	$\mathbf{0.09 \pm 0.01}$	$\mathbf{0.08 \pm 0.01}$	$\mathbf{0.07 \pm 0.01}$	0.06 ± 0.01

Figure 4.7 shows the RMSE time series at $N_e = 80$, an intermediate ensemble size. The direct precision methods still exhibit occasional spikes, though less severe than at $N_e = 50$ or $N_e = 60$. The eigenvalue shrinkage method shows the most persistent instability, consistent with its high RMSE in Table 4.2. Ledoit-Wolf and EnKF-Cholesky maintain their characteristic stability.

To quantify the run-to-run variability, Figure 4.8 shows box plots of the time-averaged RMSE across 30 independent runs at $N_e = 80$. Ledoit-Wolf has the tightest distribution (interquartile range $< 0.01 \times 10^{-2}$), confirming its consistency. The direct precision methods have wide distributions with several outlier runs where the RMSE is 2–3 times the median, reflecting the seed-dependent instability seen in the time series plots.

Figure 4.9 provides an alternative view, showing each method's RMSE time series across all four ensemble sizes in a single panel. This makes it easy to compare how each method responds to increasing N_e. For the standard EnKF, larger ensembles consistently reduce RMSE. For the direct precision methods, the relationship is non-monotonic: some ensemble sizes produce smooth traces while others produce persistent spikes. Ledoit-Wolf and EnKF-Cholesky show nearly identical traces across all N_e, confirming their stability.

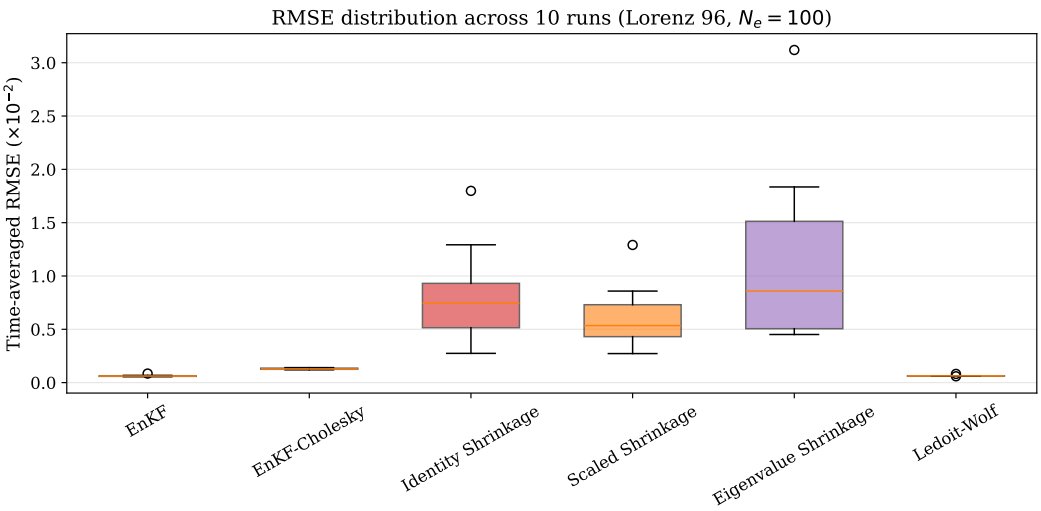

Figure 4.6: RMSE distribution across 30 runs on Lorenz 96 with $N_e = 100$. Distributions are tighter than at $N_e = 50$, but the direct precision methods still show more variability than Ledoit-Wolf.

4.2.1 Shrinkage Coefficient Analysis

To understand the instability of the direct precision methods, we ran a systematic diagnostic study across 12 ensemble sizes from $N_e = 45$ to $N_e = 100$ in steps of 5, recording the shrinkage coefficients α^* and β^* (both clamped and unclamped), the Cauchy-Schwarz denominator, and the condition number of the resulting precision estimator at each assimilation step. The results are averaged over all 100 assimilation cycles per run.

Behavior of α^*.

Figure 4.10 shows α^* as a function of ensemble size for the three direct precision targets. All three methods produce α^* values that track the theoretical bound $1 - n/N_e$ closely. At $N_e = 45$ (barely above the invertibility threshold $n + 2 = 42$), $\alpha^* \approx 0.09$, meaning only 9% of the weight goes to the sample precision and 91% must come from the target. At $N_e = 100$, α^* reaches approximately 0.6, giving the sample precision a majority weight.

The identity and cosmological targets produce nearly identical α^* values (differing only in the fourth decimal place), because the correction term in Eq. (2.37) depends on the target only through the ratio of norms, which is similar for these diagonal targets. The eigenvalue target gives slightly higher α^* at all ensemble sizes, because its target is derived from the sample precision itself, making the Cauchy-Schwarz term in the denominator closer to zero.

Behavior of β^* (unclamped).

Figure 4.11 reveals the core problem. The unclamped β^* values differ by many orders of magnitude across targets:

- **Identity target**: $|\beta^*| \sim 10^7$–10^9, requiring clamping to 1 at every assimilation step (100% clamping rate). The formula assigns enormous weight to the identity target

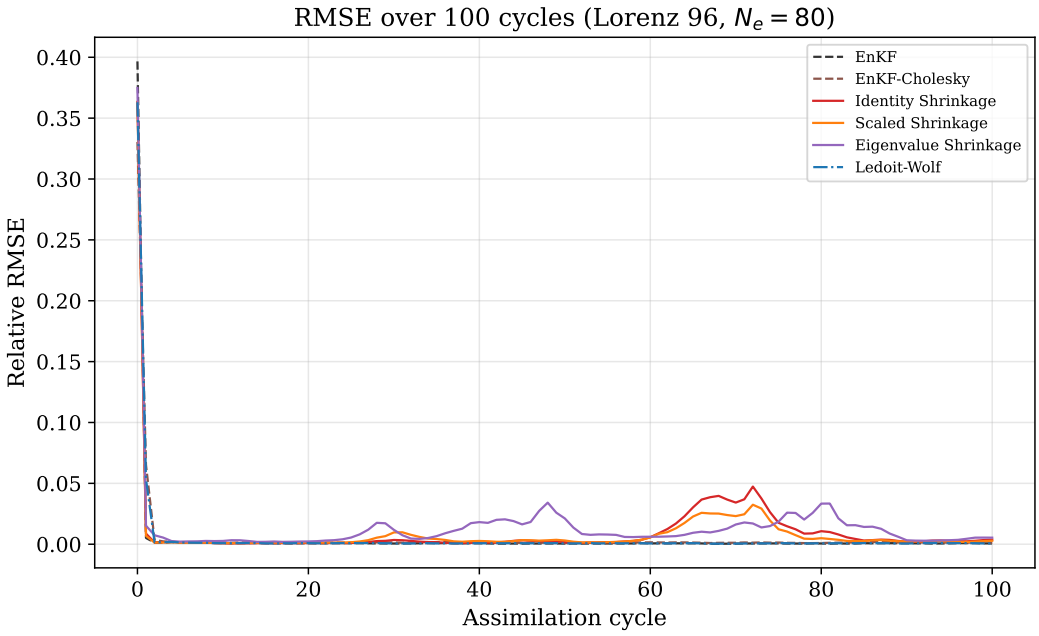

Figure 4.7: RMSE over 100 assimilation cycles on Lorenz 96 with $N_e = 80$. The direct precision methods show reduced but still present instability compared to $N_e = 50$ and $N_e = 60$.

because $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{I}) = \mathrm{tr}(\hat{\mathbf{\Pi}}_S)$ is very large for the Lorenz 96 precision matrix, while $\|\mathbf{I}\|_F^2 = n = 40$ is modest.

- **Scaled target**: $|\beta^*| \sim 10^3$–10^5, also requiring 100% clamping but with values two to four orders of magnitude smaller than the identity target. Scaling by empirical variances reduces the mismatch but does not eliminate it.

- **Eigenvalue target**: $|\beta^*| \sim 10^{-4}$, well below the clamping threshold. Because the eigenvalue target is derived from $\hat{\mathbf{\Pi}}_S$ itself, the Cauchy-Schwarz denominator $\|\hat{\mathbf{\Pi}}_S\|_F^2 \|\mathbf{\Pi}_0\|_F^2 - (\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0))^2$ is very large (close to its maximum), driving β^* toward zero. Effectively, no shrinkage toward the eigenvalue target occurs.

Condition number of the estimator.

Figure 4.12 shows the condition number of the resulting precision estimator $\mathbf{\Pi}_{\mathrm{LS}}$. All three targets produce estimators with condition numbers of 10^7–10^{11}, with the identity target being the worst (mean condition number $\sim 10^9$ at $N_e = 80$) and the eigenvalue target slightly better ($\sim 10^8$). The shaded regions show the range from mean to maximum condition number across assimilation steps, illustrating the substantial step-to-step variability.

For comparison, Ledoit-Wolf produces precision matrices with condition numbers of 10^2–10^3 on the same system. The five to eight orders of magnitude difference in conditioning accounts for the numerical instability of the direct precision methods as EnKF precision estimators on Lorenz 96.

The root cause is target-truth mismatch. The identity target $\mathbf{\Pi}_0 = \mathbf{I}$ is a poor match for Lorenz 96, whose precision matrix has significant off-diagonal entries from spatial coupling.

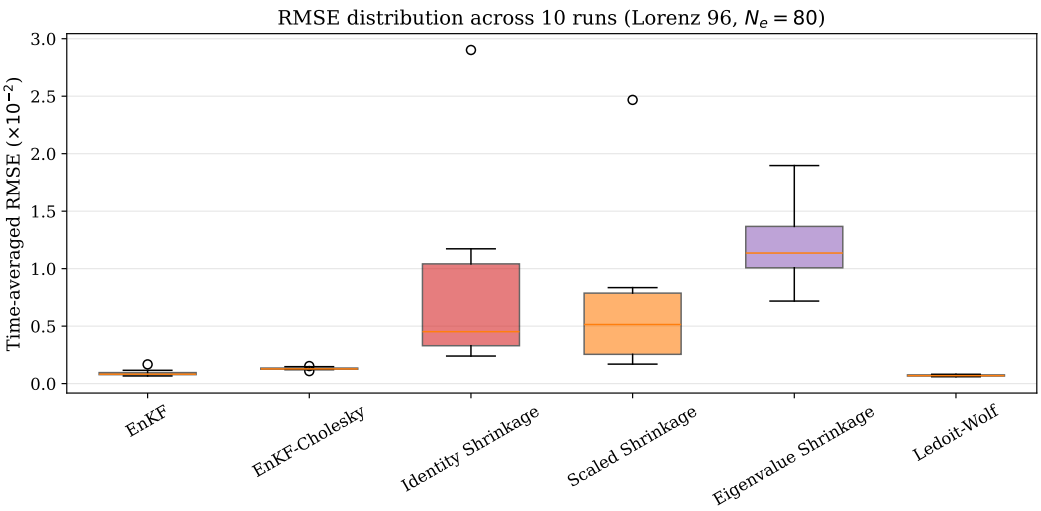

Figure 4.8: Distribution of time-averaged RMSE ($\times 10^{-2}$) across 30 runs on Lorenz 96 with $N_e = 80$. Ledoit-Wolf shows the tightest distribution; the direct precision methods show wide spread with outlier runs.

In the formula for β^* (2.38), the factor $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0) / \|\mathbf{\Pi}_0\|_F^2$ becomes very large when the sample precision has large diagonal entries relative to the target norm, driving β^* to extreme values. After clamping, the mismatch between the intended and actual coefficient values produces ill-conditioned precision matrices.

4.2.2 Ensemble Size Sensitivity

Figure 4.13 plots the time-averaged RMSE against N_e. The standard EnKF and Ledoit-Wolf improve monotonically with larger ensembles. Modified Cholesky gives stable though not optimal performance across all sizes.

NERCOME shows a counterintuitive trend: RMSE *increases* with ensemble size (0.23 at $N_e = 50$ to 0.42 at $N_e = 100$). At small N_e, the sample covariance is noisy and bootstrap resampling provides genuine regularization. At larger N_e, the sample covariance is already reasonable, and the variance introduced by 200 bootstrap draws per cycle compounds over 100 sequential updates. This is another manifestation of the sequential accumulation effect discussed in Section 4.8: estimator variance matters more in sequential filtering than in one-shot estimation.

4.2.3 Precision Matrix Quality

Figure 4.14 compares the precision matrices produced by each method at a representative assimilation step.

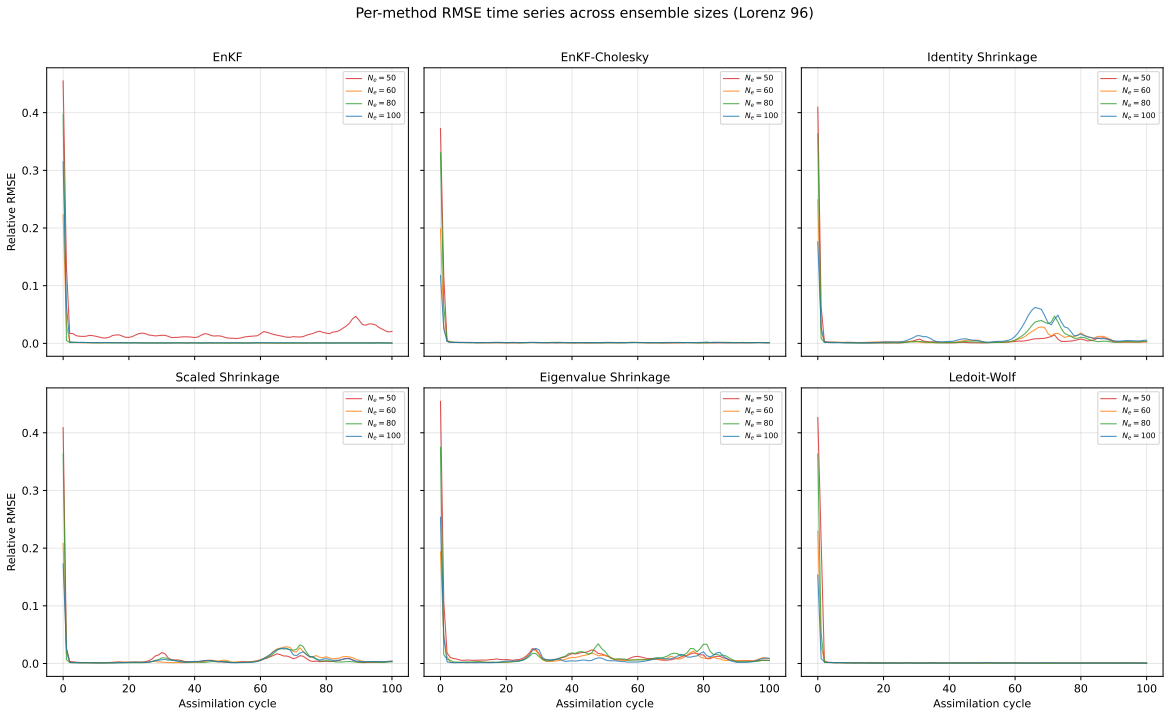

Figure 4.9: Per-method RMSE time series across all ensemble sizes on Lorenz 96. Each panel shows one method with $N_e = 50$ (red), 60 (orange), 80 (green), and 100 (blue). Stable methods (Ledoit-Wolf, Cholesky) show consistent improvement with N_e; unstable methods (Identity, Eigenvalue) show erratic behavior.

4.3 Cosmological Results

4.3.1 Standalone Precision Estimation

Table 4.3 shows the relative Frobenius loss for standalone precision matrix estimation on BOSS DR12 mock data. The results reverse compared to Lorenz 96: the direct precision shrinkage methods now outperform Ledoit-Wolf by a wide margin. At $n = 50$, the identity, eigenvalue, and cosmological targets all achieve losses of ≈ 0.41 versus 0.93 for Ledoit-Wolf. This confirms the findings of (Looijmans et al., 2024) that direct precision shrinkage works well when the true precision matrix is near-diagonal.

Figure 4.15 visualizes these trends; the following paragraphs examine each in turn.

Ledoit-Wolf's loss stays nearly flat at $\approx$ 0.93–0.96 regardless of subsample size. This is because Ledoit-Wolf shrinks the *covariance* toward a diagonal target and then inverts, a sub-optimal route when the goal is to estimate a near-diagonal *precision* matrix. The shrinkage is applied in the wrong space: improving the covariance estimate does not necessarily improve the precision estimate when the two have different structure. Even at $n = 50$ (nearly three times the data dimension), the Frobenius loss barely changes from $n = 24$. This is not a failure of the Ledoit-Wolf estimator as such; it is well-suited for covariance estimation. The flat loss is rather a consequence of applying it in a setting where precision estimation is the goal.

NERCOME achieves the lowest loss at $n = 24$ (0.82 ± 0.06) despite having higher mean

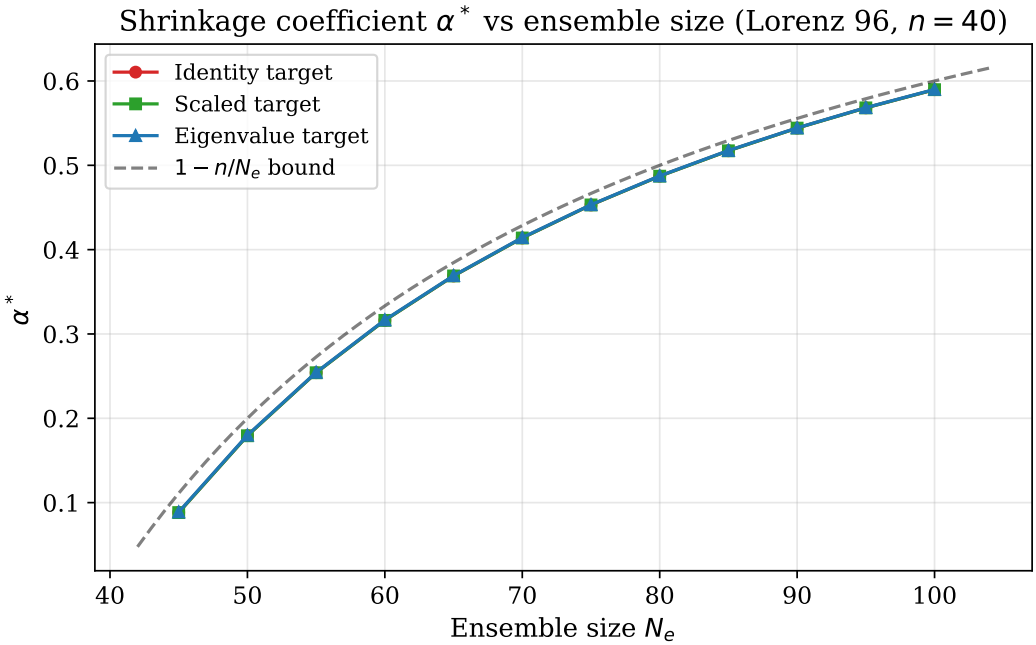

Figure 4.10: Shrinkage coefficient α^* as a function of ensemble size N_e on Lorenz 96 ($n = 40$). All three targets produce values close to the theoretical bound $1 - n/N_e$ (dashed line). The weight on the sample precision increases with N_e, but remains below 0.6 even at $N_e = 100$.

Table 4.3: Relative Frobenius loss of precision matrix estimates on BOSS DR12 mock data ($d = 18$). Lower is better. Best result per column in **bold**.

Method	$n = 24$	$n = 30$	$n = 40$	$n = 50$
Sample (Hartlap)	1.15 ± 1.01	0.74 ± 0.44	0.47 ± 0.15	0.43 ± 0.16
Identity Shrinkage	1.18 ± 1.13	0.68 ± 0.43	$\mathbf{0.45 \pm 0.12}$	$\mathbf{0.41 \pm 0.14}$
Scaled Shrinkage†	0.90 ± 0.96	$\mathbf{0.60 \pm 0.36}$	0.51 ± 0.11	0.46 ± 0.13
Eigenvalue Shrinkage	1.20 ± 1.13	0.70 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
Cosmological Shrinkage	1.19 ± 1.13	0.69 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
NERCOME	$\mathbf{0.82 \pm 0.06}$	0.73 ± 0.08	0.63 ± 0.08	0.53 ± 0.08
Ledoit-Wolf	0.96 ± 0.01	0.95 ± 0.01	0.94 ± 0.01	0.93 ± 0.01

loss than the Bodnar methods at larger n. The distinguishing feature is its extremely low variance: ± 0.06 versus ± 1.13 for the direct precision methods. At small sample sizes, the sample precision $\mathbf{S}^{-1}$ is so noisy that the Bodnar methods occasionally produce very poor estimates, inflating both their mean loss and standard deviation. NERCOME avoids this by never computing $\mathbf{S}^{-1}$ directly; instead, it builds a regularized covariance through cross-validated eigenvalue estimation.

At $n = 40$ and $n = 50$, the identity, eigenvalue, and cosmological targets converge to nearly identical loss values (0.41–0.46). This convergence occurs because at large enough sample sizes, the sample precision is already close to the truth and the shrinkage coefficients allocate most of the weight to $\hat{\mathbf{\Pi}}_S$ rather than the target. The target choice becomes less important as α^* approaches $1 - n/N_e \approx 0.6$–0.7.

The scaled identity target achieves the lowest mean loss at $n = 30$ (0.60 ± 0.36), outperforming both the other Bodnar variants and NERCOME. At $n = 30$, α^* is relatively low, so a

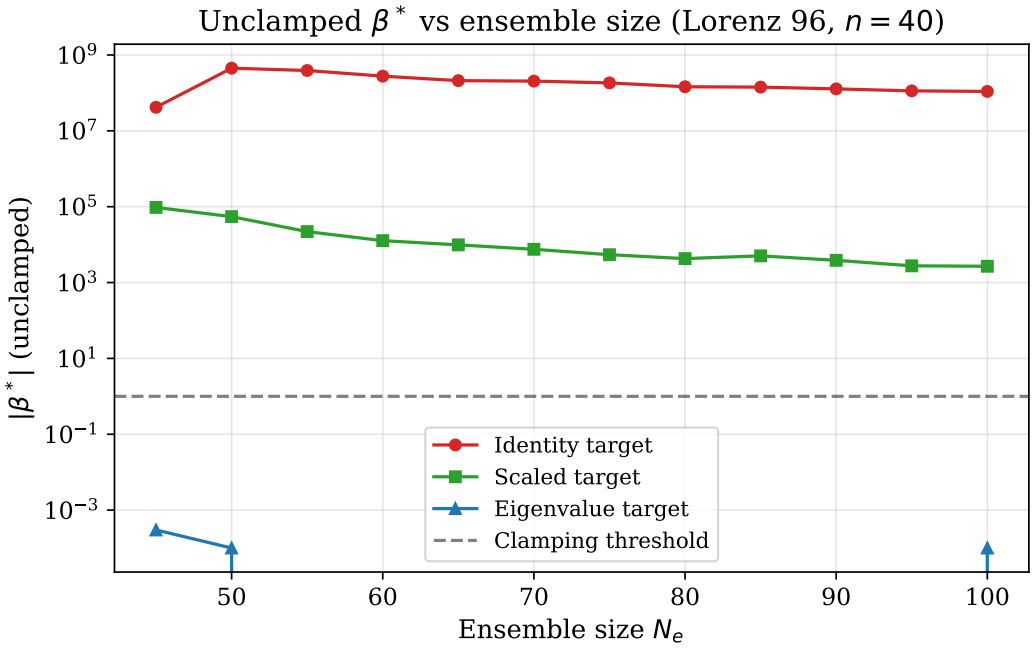

Figure 4.11: Absolute value of unclamped β^* as a function of ensemble size N_e on Lorenz 96 ($n = 40$), log scale. The dashed line shows the clamping threshold ($\beta^* = 1$). The identity target produces β^* values 10^7–10^9 times too large; the scaled target is better but still requires clamping; the eigenvalue target produces negligible β^*.

meaningful fraction of the estimator comes from $\mathbf{\Pi}_0$, and the scaled identity turns out to be a reasonable match for the near-diagonal precision structure. At $n = 24$, the sample precision is too noisy for any Bodnar method to compete with NERCOME; at $n = 40$–50, the sample is good enough that the target barely matters. The advantage of the scaled identity target is most apparent at this intermediate sample size.

4.3.2 EnKF Filtering

Before examining the summary statistics, Figures 4.16–4.19 show the RMSE time series for the AR(1) cosmological model at each ensemble size. In contrast to the Lorenz 96 time series (Figures 4.3–4.2), all methods, including the direct precision shrinkage variants, exhibit smooth, stable behavior without spikes. This confirms that the instability observed on Lorenz 96 is caused by target-truth mismatch, not by the shrinkage framework itself.

At $N_e = 24$, the methods are most spread apart, with the standard EnKF starting at the highest RMSE and the shrinkage methods providing clear regularization benefit. By $N_e = 50$, all methods have largely converged. The cosmological target does not provide a visible advantage over the identity target in any of these time series, consistent with the convergence behavior noted in the standalone estimation experiment.

Table 4.4 reports the time-averaged RMSE for the AR(1) cosmological model. Ledoit-Wolf achieves the lowest RMSE at all four ensemble sizes, despite having the worst standalone precision estimates. Unlike on Lorenz 96, the direct precision shrinkage methods now improve smoothly and monotonically with ensemble size, confirming that the erratic Lorenz 96 behavior was caused by target-truth mismatch, not a fundamental flaw in the

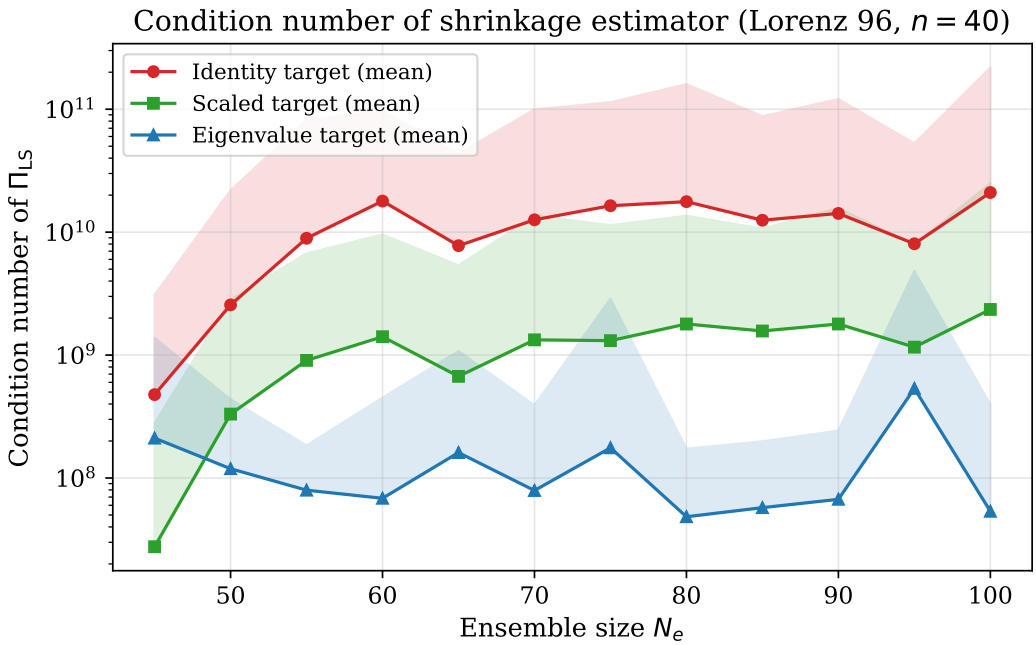

Figure 4.12: Condition number of the shrinkage precision estimator $\mathbf{\Pi}_{\mathrm{LS}}$ as a function of N_e on Lorenz 96 ($n = 40$), log scale. All targets produce condition numbers of 10^7–10^{11}, compared to 10^2–10^3 for Ledoit-Wolf. Shaded regions show the range from mean to maximum condition number across assimilation steps.

framework.

Figure 4.21 plots RMSE against ensemble size. All methods converge as N_e grows, forming a much tighter cluster than in the Lorenz 96 case. The spread between the best method (Ledoit-Wolf at 1.76×10^{-2}) and the worst (EnKF at 2.32×10^{-2}) at $N_e = 30$ is only 0.56×10^{-2}, compared to a spread of 1.73×10^{-2} on Lorenz 96 at $N_e = 60$.

Pairwise Welch's t-tests ($n = 30$ runs) show that on Lorenz 96, Ledoit-Wolf's advantage over the direct precision methods is statistically significant ($p < 0.05$) at most ensemble sizes. On the cosmological data, most shrinkage methods do not differ significantly from Ledoit-Wolf ($p > 0.05$) at $N_e \geq 40$. The identity and scaled identity targets match Ledoit-Wolf to within one standard deviation at all ensemble sizes: at $N_e = 30$, for instance, the identity target achieves 1.80×10^{-2} and the scaled target 1.81×10^{-2}, compared to 1.76×10^{-2} for Ledoit-Wolf. These differences are not statistically significant.

The most notable difference from the Lorenz 96 results is the behavior of the eigenvalue and cosmological targets. On Lorenz 96, the eigenvalue target performs worst among all methods. On cosmological data, it performs comparably to the other Bodnar variants, confirming that the Lorenz 96 failure is target-specific rather than method-specific. Similarly, the cosmological target (which uses the theoretically optimal prior for this data) performs comparably to but not significantly better than the simpler identity target. This is consistent with the convergence behavior observed in the standalone estimation experiment: at $N_e/d \geq 1.5$, the sample precision is good enough that the target barely matters.

Figure 4.20 shows box plots of the time-averaged RMSE across 30 runs at $N_e = 30$. Unlike on Lorenz 96, the distributions are tight for all methods, with the interquartile range below

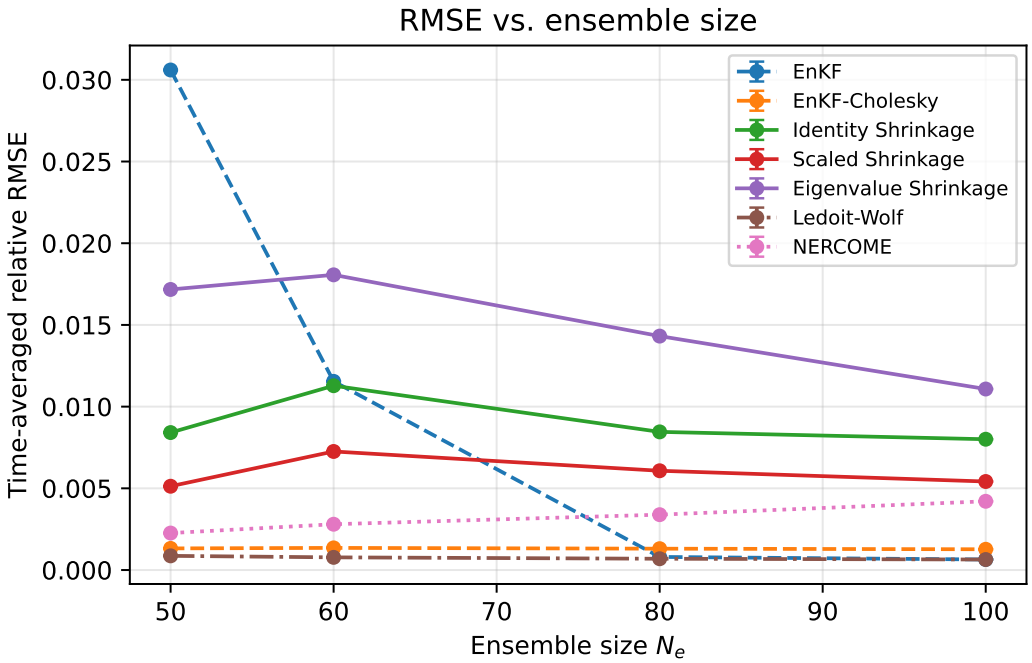

Figure 4.13: Time-averaged relative RMSE (over last 50 cycles) as a function of ensemble size N_e on Lorenz 96. Error bars show ± 1 standard deviation over 30 runs.

Table 4.4: Time-averaged relative RMSE ($\times 10^{-2}$) for the AR(1) cosmological model. Best result per column in **bold**. †Scaled Shrinkage is a novel adaptation proposed in this work.

Method	$N_e = 24$	$N_e = 30$	$N_e = 40$	$N_e = 50$
EnKF (standard)	2.75 ± 0.46	2.32 ± 0.39	2.17 ± 0.20	2.11 ± 0.34
EnKF-Cholesky	1.99 ± 0.29	1.81 ± 0.26	1.87 ± 0.25	1.88 ± 0.31
Identity Shrinkage	1.95 ± 0.25	1.80 ± 0.26	1.88 ± 0.21	1.86 ± 0.30
Scaled Shrinkage†	1.99 ± 0.29	1.81 ± 0.26	1.89 ± 0.23	1.87 ± 0.31
Eigenvalue Shrinkage	2.35 ± 0.33	2.05 ± 0.30	2.00 ± 0.20	1.97 ± 0.32
Cosmological Shrinkage	2.21 ± 0.30	2.06 ± 0.31	2.04 ± 0.21	2.02 ± 0.33
NERCOME	1.88 ± 0.20	1.82 ± 0.19	1.91 ± 0.29	1.85 ± 0.27
Ledoit-Wolf	$\mathbf{1.88 \pm 0.26}$	$\mathbf{1.76 \pm 0.26}$	$\mathbf{1.86 \pm 0.21}$	$\mathbf{1.84 \pm 0.30}$

0.3×10^{-2} for every method. This confirms that the cosmological domain produces stable, reproducible results regardless of the random seed, consistent with the smooth RMSE time series in Figures 4.16–4.19.

The standard EnKF improves the most with ensemble size, dropping from 2.75×10^{-2} at $N_e = 24$ to 2.11×10^{-2} at $N_e = 50$, a 23% reduction. All shrinkage methods show flatter curves, reflecting their regularization effect: they provide the most benefit at small ensemble sizes (where the sample covariance is noisy) and diminishing returns at larger sizes. NERCOME shows essentially flat RMSE from $N_e = 30$ onward, suggesting its 200 bootstrap resamples introduce enough variance to offset the gains from larger ensembles, the same phenomenon observed on Lorenz 96, albeit less pronounced on this simpler system.

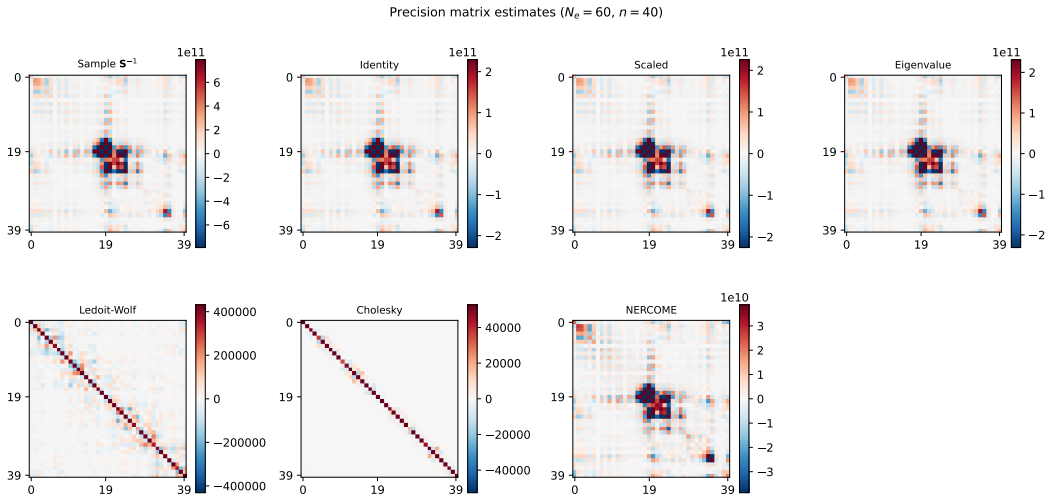

Figure 4.14: Precision matrix estimates (40×40) from different methods with $N_e = 60$: sample precision $\mathbf{S}^{-1}$, identity shrinkage, scaled shrinkage, eigenvalue shrinkage, Ledoit-Wolf, and modified Cholesky. The direct precision shrinkage estimators produce matrices with much larger magnitude than Ledoit-Wolf and Cholesky, reflecting the ill-conditioning discussed above.

4.3.3 The Scaled Identity Target

The scaled identity target occupies a middle ground. On Lorenz 96, it outperforms the other direct precision variants (Table 4.2): its RMSE ranges from 0.51 to 0.73×10^{-2} versus 0.80–1.81×10^{-2} for the identity and eigenvalue targets. Scaling by empirical variances gives a better-conditioned target than the plain identity, reducing (though not eliminating) the coefficient miscalibration.

On cosmological data, the scaled target achieves the lowest mean Frobenius loss at $n = 30$ (Table 4.3) but falls slightly behind the Bodnar methods at larger sample sizes. In the EnKF experiments, it matches the identity target almost exactly (Table 4.4).

The scaled target serves its intended purpose: modest gains over the identity target without requiring external information, but no substitute for a domain-specific target when one is available. Its ad-hoc factor of 2, borrowed by analogy from the cosmological $2/N_\ell$ term, could potentially be improved through a data-driven selection criterion.

To understand *why* the scaled target outperforms the identity on Lorenz 96, consider the shrinkage coefficients. The identity target has $\|\mathbf{\Pi}_0\|_F^2 = n = 40$ and $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0) = \mathrm{tr}(\hat{\mathbf{\Pi}}_S)$, which is large because the Lorenz 96 precision matrix has diagonal entries much greater than 1. The factor $\mathrm{tr}(\hat{\mathbf{\Pi}}_S)/n$ that multiplies $(1 - n/N_e - \alpha^*)$ in the β^* formula is therefore very large, driving β^* to extreme values.

The scaled identity target has $\|\mathbf{\Pi}_0\|_F^2 = \sum_i 1/(4\sigma_i^4)$, which is calibrated to the actual scale of the state variables. The factor $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0)/\|\mathbf{\Pi}_0\|_F^2$ is closer to 1 than for the identity target, producing more moderate β^* values (though still requiring clamping in most cases). The net effect is that the clamped β^* is less wrong: the mismatch between the intended and actual coefficient values is smaller, leading to better-conditioned precision estimates.

On cosmological data, the difference between scaled and identity targets is less pro-

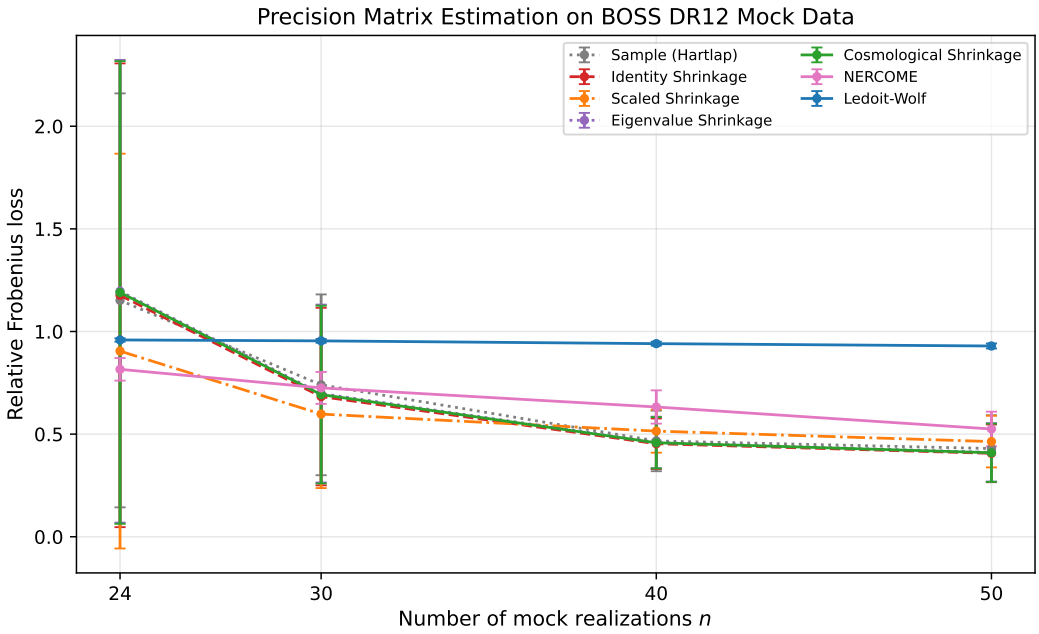

Figure 4.15: Relative Frobenius loss of precision matrix estimates vs. subsample size on BOSS DR12 mock data ($d = 18$). Direct precision shrinkage methods (solid lines) achieve the lowest loss at $n \geq 40$, while Ledoit-Wolf (blue) remains flat near 0.93. Error bars show ± 1 standard deviation over 100 random draws.

nounced because the true precision is close to diagonal with entries of order 1 relative to the target norms, so both targets produce reasonable β^* values without clamping.

4.4 Effect of Inflation on Shrinkage Performance

All experiments in this thesis use a fixed inflation factor ($\rho = 1.05$ for Lorenz 96, $\rho = 1.02$ for the cosmological model). The choice of inflation interacts with the precision estimation in two ways.

First, inflation increases the eigenvalues of the sample covariance by a factor of ρ^2, making the sample covariance better conditioned. This reduces the condition number of $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$ and should in principle help the direct precision methods. However, the improvement is modest: $\rho^2 = 1.1025$ for $\rho = 1.05$, which changes the condition number by at most a factor of $\rho^2 \approx 1.1$. Since the condition numbers in our experiments are 10^7–10^{11}, a 10% change is negligible.

Second, inflation increases the forecast error covariance, which means the analysis trusts the observations more (the Kalman gain $\mathbf{K}$ increases). For the direct precision methods, the analysis update, which is computed using the potentially ill-conditioned shrinkage precision, then has a larger magnitude, amplifying any errors in the precision estimate. This is a subtle but important effect: more inflation helps the sample covariance quality but hurts the analysis quality when the precision estimate is poor.

The interaction between inflation and precision estimation is complex and configuration-dependent. A systematic study varying ρ from 1.01 to 1.10 for each method would quantify this tradeoff but is beyond the scope of the present work.

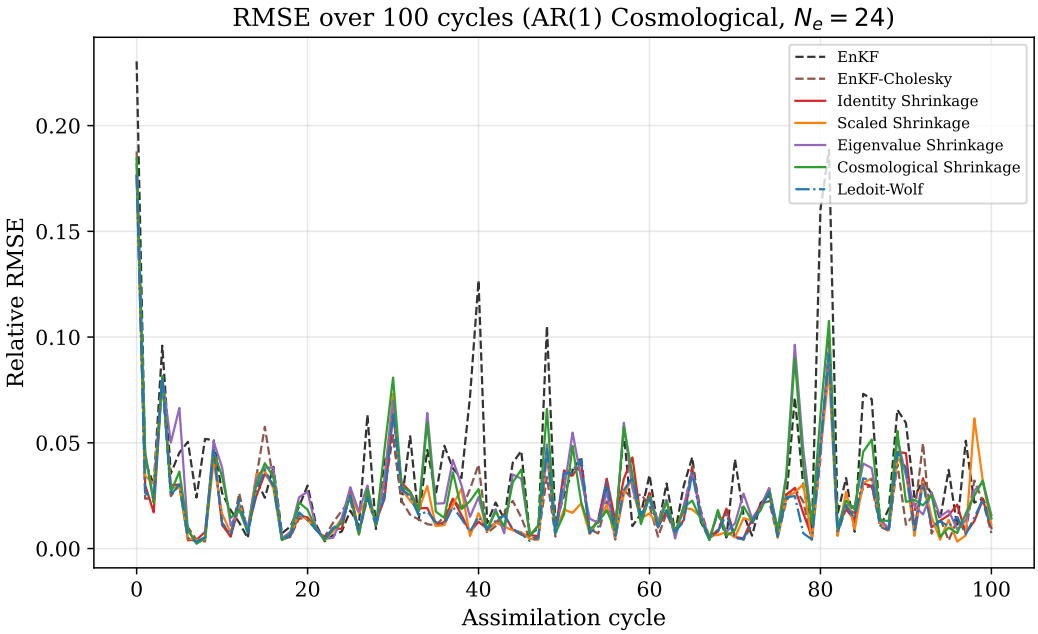

Figure 4.16: RMSE over 100 assimilation cycles on the AR(1) cosmological model with $N_e = 24$. All methods show stable behavior, unlike the Lorenz 96 case. The standard EnKF has the highest RMSE.

4.5 Precision Matrix Quality

To understand why the direct precision methods produce high RMSE on Lorenz 96, we examine the precision matrices themselves. Figure 4.14 shows heatmaps of the precision matrices estimated by each method at a representative assimilation step with $N_e = 60$.

The sample precision $\mathbf{S}^{-1}$ (top left) shows the expected banded structure from Lorenz 96's nearest-neighbor coupling, but with substantial noise in the off-diagonal entries. The direct precision shrinkage estimates (identity, scaled, eigenvalue) retain this noisy structure because the shrinkage does not denoise the off-diagonal entries: the diagonal targets have no off-diagonal information to offer. The entries are larger in magnitude than the true precision, reflecting the positive bias of the inverse Wishart that is only partially corrected by the $(1 - n/N_e)$ factor.

Ledoit-Wolf produces a precision matrix (obtained by inverting the shrunk covariance) with much smaller off-diagonal entries and better conditioning. The shrinkage in covariance space has the effect of regularizing the eigenvalue spectrum, which produces a better-conditioned inverse even though the shrinkage itself does not target the precision structure.

The modified Cholesky estimate is sparse by construction: only entries within bandwidth $r = 2$ of the diagonal are non-zero. This eliminates all long-range correlations, including both spurious noise and genuine dynamical correlations. The resulting matrix has the lowest condition number of any method but may discard real correlations that improve the analysis.

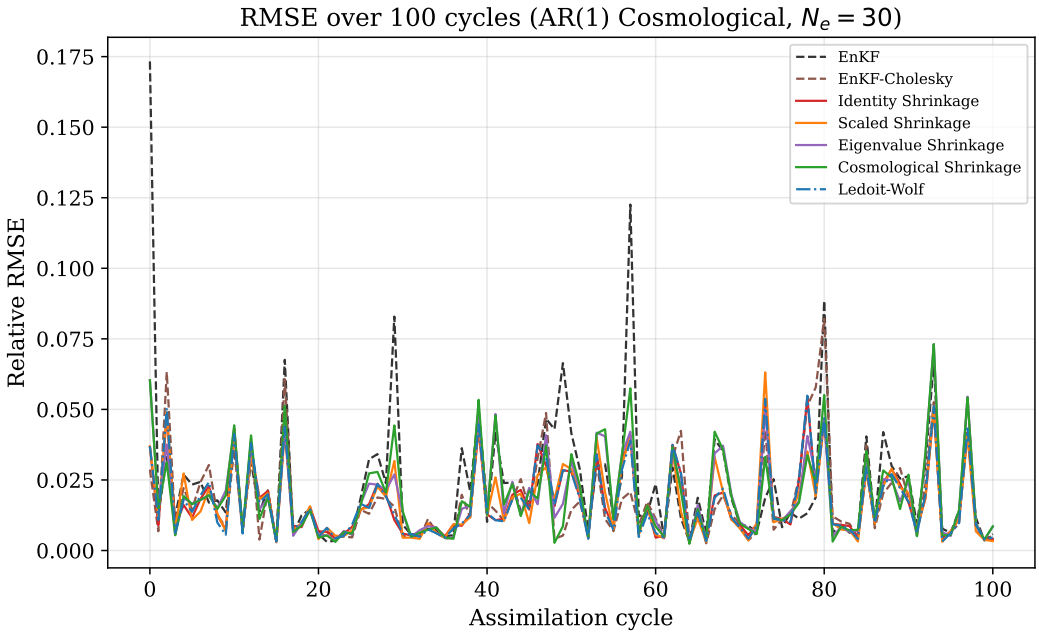

Figure 4.17: RMSE over 100 cycles on the AR(1) model with $N_e = 30$. Methods are starting to converge; Ledoit-Wolf maintains a slight edge throughout.

4.6 Computational Cost

The previous sections have compared the methods on accuracy alone. For practical use, accuracy has to be weighed against the cost of obtaining it, particularly inside a sequential filter that recomputes the precision matrix at every assimilation cycle. This section quantifies that cost in three complementary ways: measured wall-clock time on our two test domains (4.6.1), extrapolation to state dimensions beyond what we could run on a laptop (4.6.2), and a discussion of the implementation factors that separate asymptotic complexity from observed runtime (4.6.3). A concluding recommendation (4.6.4) maps operational regimes to method choices.

4.6.1 Asymptotic vs. Measured Wall-Clock

Table 3.1 in Chapter 3 gives asymptotic complexity bounds, but practical runtime depends on constant factors, implementation efficiency, and problem-specific parameters. We timed each method on both test domains, running 100 assimilation cycles and averaging over 5 repetitions. Figure 4.22 shows the results.

On the cosmological domain ($d = 18$, $N_e = 30$), NERCOME takes 26.7 seconds per 100 cycles, roughly 190 times slower than the EnKF baseline (0.14 s) and 100 times slower than the direct precision shrinkage methods (0.20–0.28 s). Ledoit-Wolf (0.22 s) and the direct precision methods are all comparable in speed, consistent with the $\mathcal{O}(n^2 N_e + n^3)$ analysis: at $n = 18$, the n^3 inversion is cheap. The EnKF-Cholesky method is slower than expected (0.51 s) because the regression step involves overhead for the predecessor lookup that is not amortized at small n.

On Lorenz 96 ($n = 40$, $N_e = 80$), the differences are less dramatic because the ensemble

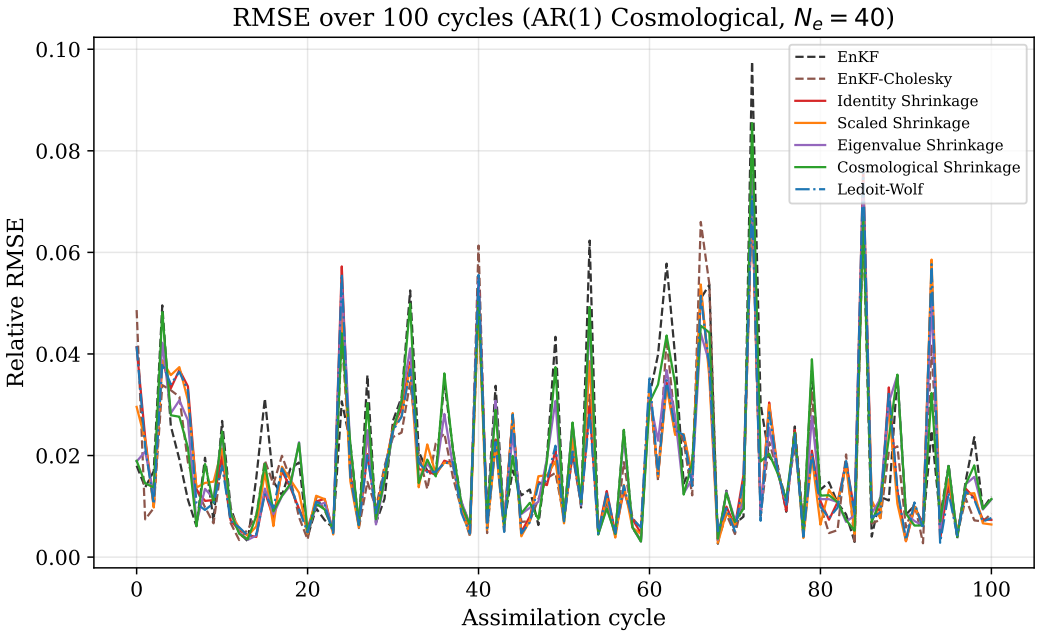

Figure 4.18: RMSE over 100 cycles on the AR(1) model with $N_e = 40$. Most methods are nearly indistinguishable.

propagation through the Lorenz 96 model dominates the runtime. All methods except NERCOME take 26–28 seconds, with the n^3 inversion adding only a small fraction to the forecast cost. NERCOME takes 215 seconds, again roughly 8× slower, because its 200 bootstrap resamples each require eigendecompositions of the 40×40 sample covariance.

The large gap between the cosmological and Lorenz 96 timings for the shrinkage methods (0.20 s vs. 26 s) is explained almost entirely by the ensemble propagation: the RK4 integrator dominates the Lorenz 96 runtime at $n = 40$, while the cosmological AR(1) update is a single matrix-vector product that costs a fraction of a millisecond. For the shrinkage methods, the per-cycle cost breaks down approximately as (i) ensemble propagation (dominant on Lorenz 96, negligible on cosmo), (ii) sample covariance formation and inversion ($\mathcal{O}(n^2 N_e + n^3)$, same for all shrinkage methods), (iii) coefficient computation (two inner products on $n \times n$ matrices), and (iv) Kalman-gain formation and analysis update ($\mathcal{O}(m^2 N_e)$). On our test domains (ii)–(iv) account for less than 5% of the total wall-clock on Lorenz 96 but more than 90% on the cosmological model. NERCOME is the exception: its $B = 200$ bootstrap eigendecompositions push the estimation cost above the propagation cost even at $n = 40$.

4.6.2 Scaling to Larger State Dimensions

Our two test domains sit at $n = 40$ and $d = 18$, both well below the state dimensions of operational data assimilation systems (weather: 10^6–10^8; mesoscale reanalysis: 10^4–10^6). The asymptotic complexities in Table 3.1 let us extrapolate. Figure 4.23 plots the per-cycle estimation cost (components (ii)–(iii) above) against n for fixed $N_e = 80$, in units of the measured Lorenz 96 timing at $n = 40$.

The three regimes visible in Figure 4.23 are:

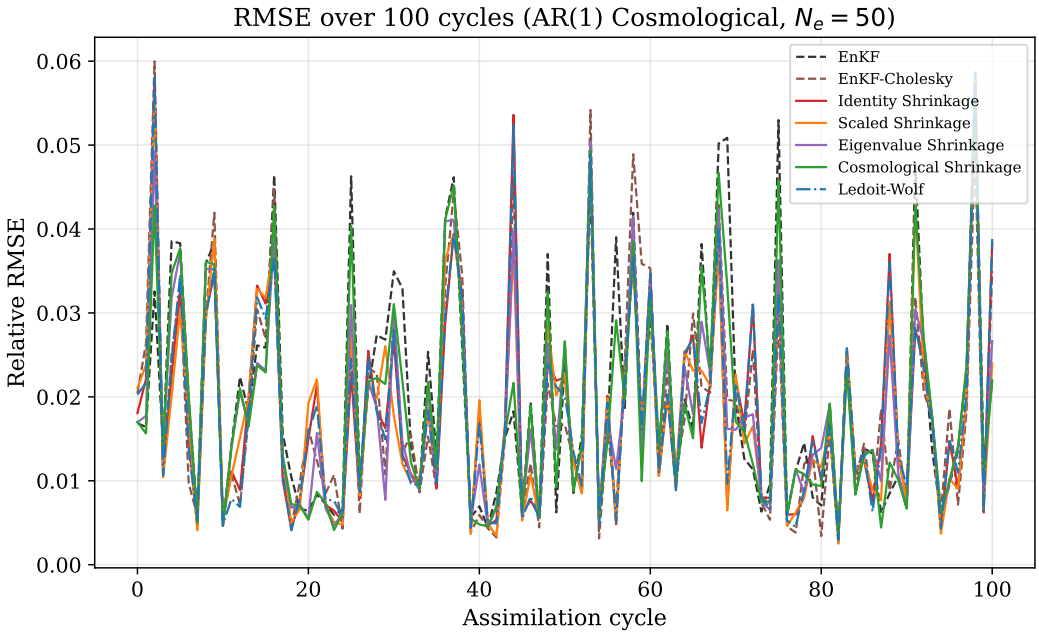

Figure 4.19: RMSE over 100 cycles on the AR(1) model with $N_e = 50$. All methods have converged to similar performance.

- $n \lesssim 100$ **(testbed scale).** All $\mathcal{O}(n^3)$ methods are within a factor of 2 of each other, and cost is not a differentiator. NERCOME is the only clear outlier. Method choice should be driven by statistical performance alone.

- $n \sim 10^2$–10^3 **(intermediate models, e.g. quasi-geostrophic, shallow-water).** The n^3 inversion starts to dominate, and direct precision shrinkage costs 10^2–10^3 times more than the sample-covariance EnKF or EnKF-Cholesky per cycle. At this scale, full-matrix methods are still tractable on a single workstation, but no longer trivial. NERCOME is effectively ruled out for sequential use.

- $n \gtrsim 10^4$ **(operational scales).** Any method requiring an explicit $n \times n$ inversion is prohibitive without hierarchical or low-rank approximations. The modified Cholesky approach ($\mathcal{O}(n \cdot r \cdot N_e)$) remains linear in n and is the only method in our comparison that survives this regime unmodified. Combining direct precision shrinkage with localization (Chapter 6, future work) is a plausible route to retain the statistical benefits at operational scale.

This extrapolation assumes the same constant factors as our Python implementation. In practice, BLAS-backed linear algebra (LAPACK `dposv`, `dsyevr`) scales better than pure-Python loops, so the operational picture may be more favorable than Figure 4.23 suggests. Still, the asymptotic ordering is intrinsic and cannot be avoided by implementation.

4.6.3 Implementation Considerations

The wall-clock numbers above come from a single-threaded Python implementation backed by NumPy. Several implementation choices matter for deployment:

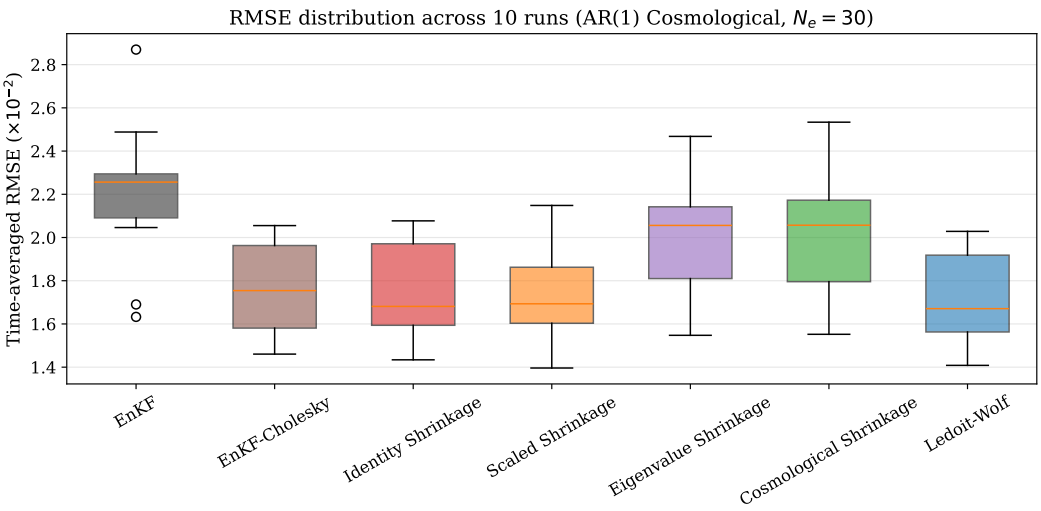

Figure 4.20: RMSE distribution across 30 runs on the AR(1) cosmological model with $N_e = 30$. All methods show tight distributions, in contrast to the wide spread seen on Lorenz 96 (Figures 4.8–4.6).

- *BLAS/LAPACK backend.* All matrix inversions in our code go through `numpy.linalg.solve`, which calls LAPACK. The cost measured here is therefore close to the practical floor for dense inversion on CPU at these dimensions. A GPU implementation would reduce the constant factor at the $n \sim 10^3$ mark but would not change the $\mathcal{O}(n^3)$ scaling.

- *NERCOME parallelization.* Its B bootstrap draws are trivially parallel: each draw is an independent $\mathcal{O}(n^3)$ eigendecomposition. On an eight-core machine, a naive `multiprocessing` implementation would reduce the NERCOME runtime by nearly $8\times$, narrowing the gap with linear-shrinkage methods to roughly $25\times$ on Lorenz 96 and 25–30$\times$ on the cosmological model. It would still be infeasible at operational scale, but more attractive for intermediate testbeds where each cycle can afford seconds of wall-clock.

- *Coefficient-update overhead.* The Bodnar coefficients α^* and β^* require two matrix inner products and one scalar division per cycle. On our test domains this is below 1 ms; at $n \sim 10^4$ the two $\mathcal{O}(n^2)$ inner products alone become comparable to the observation update. Caching the target $\mathbf{\Pi}_0$ (which does not change between cycles except for the running-mean variant in Chapter 6) is essential.

- *Clamping and condition-number checks.* We observed clamping in up to 80% of Lorenz 96 cycles. The clamping step is $\mathcal{O}(1)$, but the condition-number diagnostic we recorded for Figure 4.12 is $\mathcal{O}(n^3)$ (via SVD) and was used only offline; a production implementation would skip it or replace it with a cheaper power-iteration estimate.

- *Python overhead at small n.* A non-trivial fraction of the cosmological timings (both EnKF and EnKF-Cholesky) is Python dispatch overhead rather than actual linear algebra, which is why the measured EnKF-Cholesky timing (0.51 s) is higher than its asymptotic cost would suggest. At $n = 18$, a call to `numpy.linalg.solve` is cheaper than the surrounding Python bookkeeping; this flips at roughly $n \sim 200$, beyond which the asymptotic picture dominates.

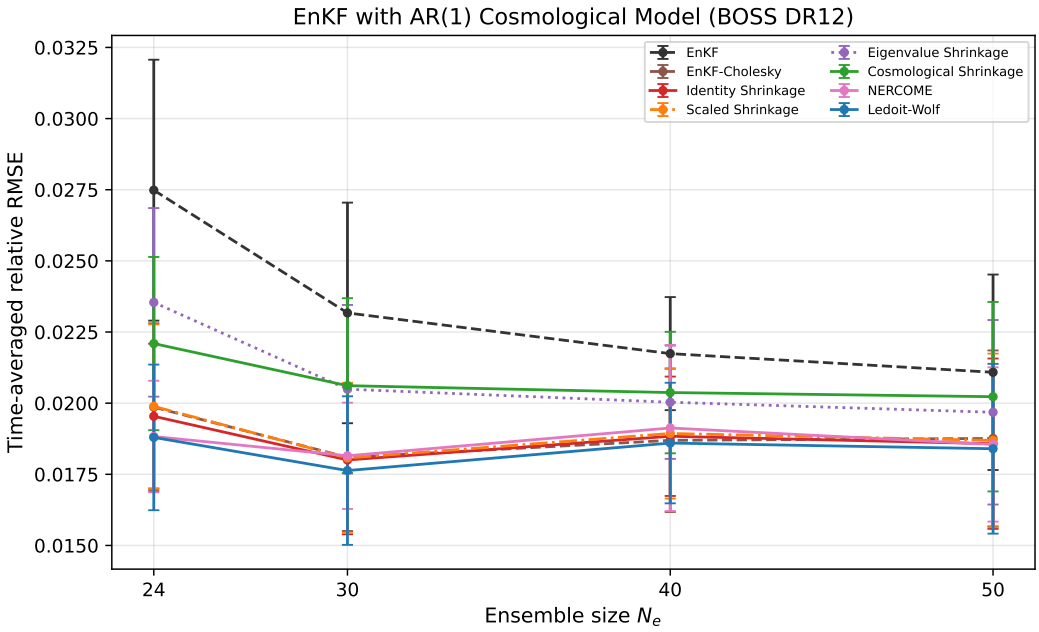

Figure 4.21: Time-averaged relative RMSE vs. ensemble size for the AR(1) cosmological model. All methods show stable, monotonic behavior, unlike the Lorenz 96 case (Figure 4.13). Ledoit-Wolf achieves the lowest RMSE throughout. Error bars show ± 1 standard deviation over 30 runs.

4.6.4 Practical Recommendation

Combining the accuracy results of Sections 4.2–4.3 with the cost analysis above, Table 4.5 maps operational regimes to method choices.

The dominant practical finding is that computational cost is a differentiator only in two corners of this table: NERCOME is uniformly too expensive for sequential use, and any $\mathcal{O}(n^3)$ method becomes infeasible at $n \gtrsim 10^4$ unless combined with localization or low-rank approximations. Between these extremes, the $\mathcal{O}(n^3)$ inversion that the direct precision methods add to the EnKF is of the same order as the inversion the EnKF itself already performs, so the method choice is driven by accuracy rather than cost.

4.7 Cross-Domain Summary

Figure 4.24 provides a direct visual comparison of all eight methods at $N_e = 30$ on the cosmological domain, complementing the tabular results in Table 4.4.

Table 4.6 summarizes the key findings across both domains. The ranking of methods reverses between standalone estimation and sequential filtering, and the relative performance of the direct precision methods depends on the precision structure of the test system.

4.8 Estimation Quality vs. Filtering Performance

The most notable result across both domains is the disconnect between standalone estimation quality and filtering performance. On cosmological data, the (Bodnar et al., 2016) meth-

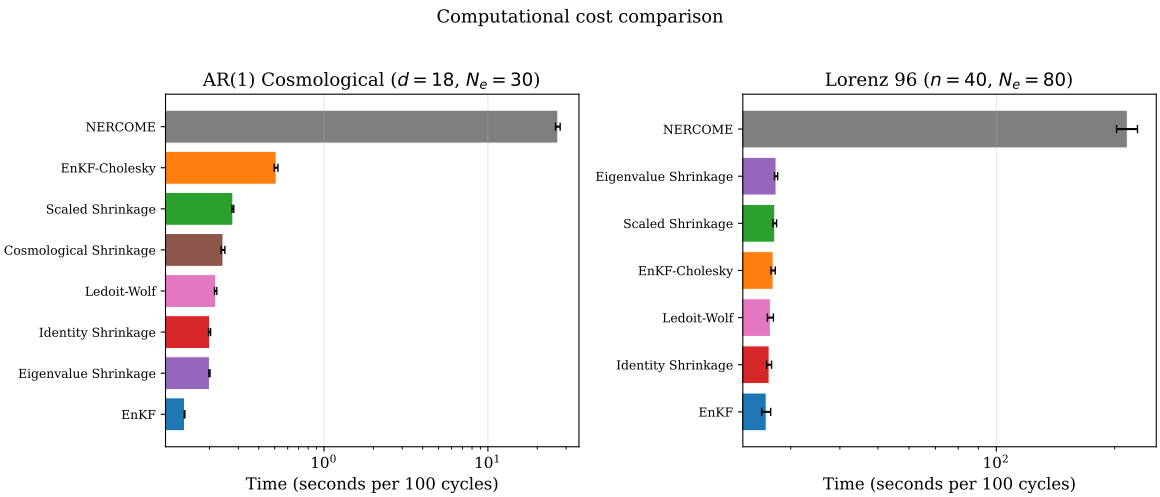

Figure 4.22: Computational cost per 100 assimilation cycles on the AR(1) cosmological model ($d = 18$, $N_e = 30$, left) and Lorenz 96 ($n = 40$, $N_e = 80$, right). Note the logarithmic scale. NERCOME is one to two orders of magnitude slower than all other methods on both domains.

ods achieve Frobenius losses half those of Ledoit-Wolf (Table 4.3), yet Ledoit-Wolf yields the lowest EnKF RMSE (Table 4.4).

The explanation is sequential accumulation. The precision estimator is applied at every assimilation cycle, and step-to-step variability compounds across 100 updates. Ledoit-Wolf's standard deviation across draws is ± 0.01 in standalone estimation, an order of magnitude below the direct precision methods (± 0.13–0.14). A slightly biased but stable estimator beats a more accurate but variable one when errors accumulate over many cycles.

Consider two estimators applied over $K = 100$ cycles. If estimator A has bias b_A and per-step variance σ_A^2, and estimator B has bias $b_B < b_A$ but variance $\sigma_B^2 > \sigma_A^2$, then the accumulated error after K cycles scales roughly as $K \cdot b^2 + K \cdot \sigma^2$ (assuming independence between cycles, which is approximate). For Ledoit-Wolf versus the identity shrinkage method on cosmological data, the variance term dominates: Ledoit-Wolf's σ^2 is roughly 10^{-4} versus 10^{-2} for the direct precision methods, a 100-fold difference that overwhelms the bias advantage.

The dual-domain comparison confirms that the poor Lorenz 96 results for direct precision shrinkage reflect target-truth mismatch, not a deficiency of the methods. On Lorenz 96, diagonal targets produce erratic RMSE (0.51–1.81×10^{-2}). On the near-diagonal cosmological data, the same methods achieve stable performance (1.80–2.35×10^{-2}). The Bodnar framework works as designed when the target captures the dominant structure of the true precision matrix.

Choosing a precision estimator for use inside the EnKF by its standalone Frobenius loss alone is therefore insufficient. Instead, the stability of the estimator, as measured by its variance across different realizations, is at least as important as its accuracy. This is a qualitative departure from the standard shrinkage estimation paradigm, where the objective is to minimize expected loss without regard to variance.

The distinction arises because standalone estimation is a one-shot problem (estimate

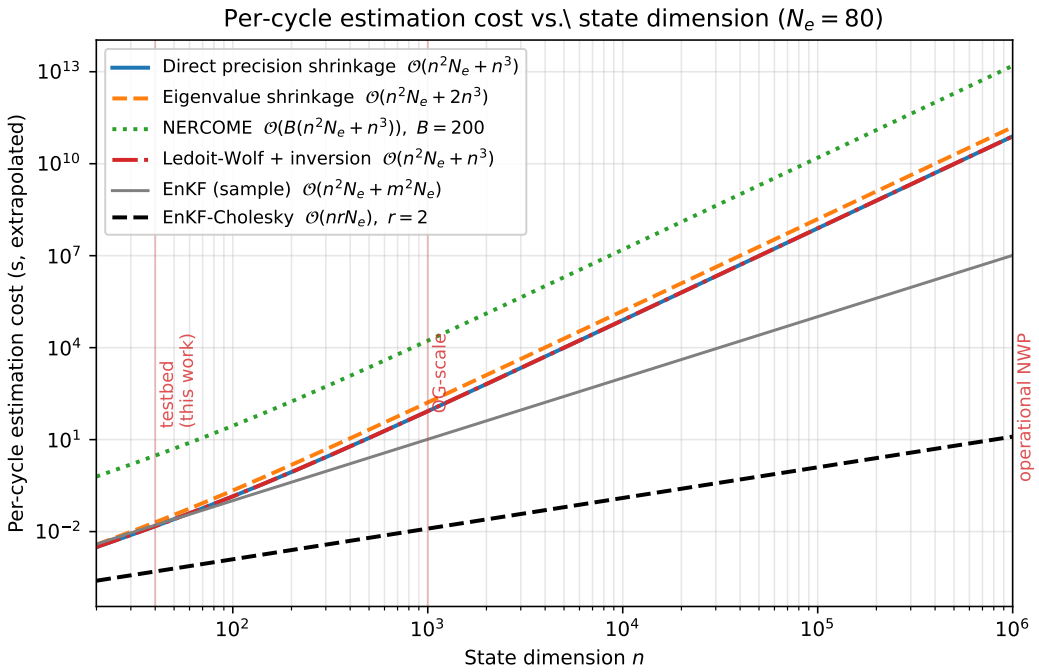

Figure 4.23: Extrapolated per-cycle estimation cost as a function of state dimension n for $N_e = 80$, anchored to the measured Lorenz 96 timings at $n = 40$. Log-log axes. The $\mathcal{O}(n^3)$ methods (direct precision shrinkage, Ledoit-Wolf after inversion) scale as a straight line of slope 3; EnKF-Cholesky with bandwidth $r = 2$ scales linearly in n; NERCOME carries an additional factor of $B = 200$ throughout. The vertical dashed lines mark regimes of interest: laboratory-scale ($n \sim 40$, where we ran experiments), mid-scale testbeds ($n \sim 10^3$, e.g. quasi-geostrophic models), and operational NWP ($n \sim 10^6$).

once, evaluate once), while EnKF filtering is a sequential problem (estimate 100 times, accumulate errors across all 100 cycles). In the one-shot case, an unbiased estimator with high variance is preferred over a biased one with low variance (by the bias-variance tradeoff). In the sequential case, the accumulated variance over K cycles can dominate the accumulated bias, inverting the preference.

The same principle appears in cascaded signal processing. In a single-stage filter, the optimal filter minimizes the mean squared error at each step. In a cascaded filter (where the output of one stage feeds the input of the next), stability constraints override optimality at each stage: an individually optimal but marginally stable filter can cause the cascade to diverge, while a slightly suboptimal but well-damped filter produces better end-to-end performance. The same principle applies here: the EnKF is a cascade of estimation and prediction stages, and the stability of each estimation stage matters more than its per-stage optimality.

For sequential data assimilation, variance reduction matters at least as much as bias reduction. Without domain-specific prior knowledge about the precision structure, Ledoit-Wolf is the safest default among the methods we tested.

Table 4.5: Method recommendation by regime. "Target known" means a structural prior on the precision matrix is available (e.g., the cosmological $2/N_\ell$ term for power spectrum data). "Dense" and "sparse" refer to the dominant off-diagonal structure of the true precision.

Regime (n/N_e)	Structure	Target known?	Recommended method
> 1 (rank-deficient)	any	—	Localization + EnKF-Cholesky (direct precision not applicable)
0.5–1 (ill-cond.)	near-diagonal	yes	Direct precision shrinkage with the domain target
0.5–1 (ill-cond.)	near-diagonal	no	Scaled identity shrinkage (this work) or Ledoit-Wolf
0.5–1 (ill-cond.)	dense	—	Ledoit-Wolf, or EnKF-Cholesky at $n \gtrsim 10^3$
< 0.3 (well-sampled)	any	—	Plain EnKF; shrinkage coefficient $\to 0$

Table 4.6: Cross-domain summary. Best standalone estimator and best EnKF filter for each domain. The disconnect between estimation quality and filtering performance is the central finding.

	Lorenz 96 ($n = 40$)	Cosmological ($d = 18$)
Precision structure	Dense (off-diagonal)	Near-diagonal
Best standalone estimator	N/A (not tested)	Direct precision (Bodnar)
Best EnKF filter	Ledoit-Wolf	Ledoit-Wolf
Direct precision behavior	Erratic, ill-conditioned	Stable, monotonic
Root cause	Target-truth mismatch	Target matches structure

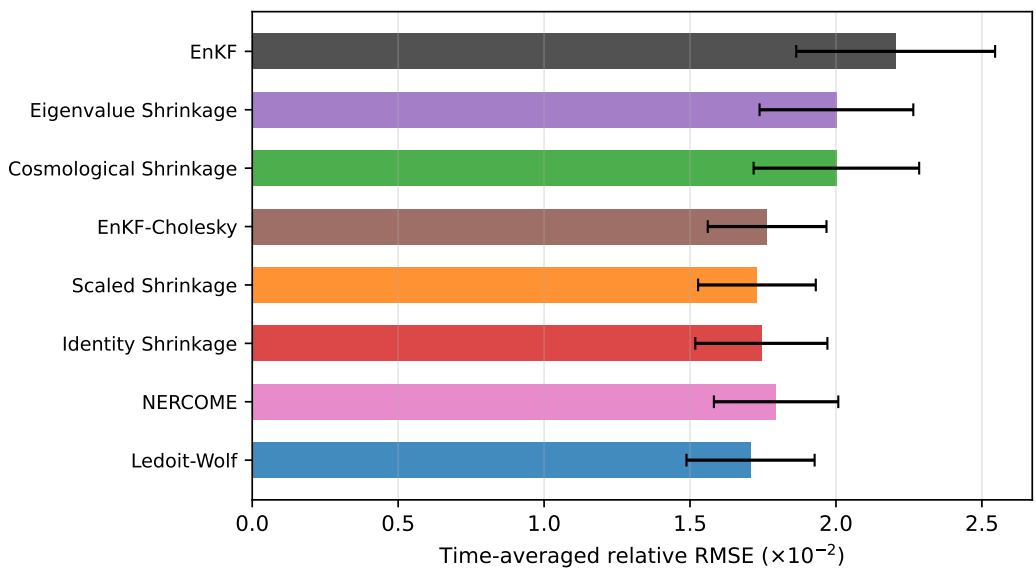

Figure 4.24: Time-averaged relative RMSE ($\times 10^{-2}$) for all eight methods on the AR(1) cosmological model at $N_e = 30$. Error bars show ± 1 standard deviation over 30 runs. Ledoit-Wolf achieves the lowest RMSE; the direct precision methods are competitive but do not lead.

Chapter 5

Discussion

This chapter examines the results from Chapter 4 in more depth, relates them to the broader literature, and identifies the patterns that cut across the two test domains.

5.1 The Role of Target-Truth Alignment

The success or failure of direct precision shrinkage depends on how well the target $\mathbf{\Pi}_0$ matches the true precision structure. This is a structural constraint imposed by the Bodnar formula.

The shrinkage coefficients α^* and β^* are derived under the assumption that $\mathbf{\Pi}_{\mathrm{LS}} = \alpha\hat{\mathbf{\Pi}}_S + \beta\mathbf{\Pi}_0$ is in the span of $\hat{\mathbf{\Pi}}_S$ and $\mathbf{\Pi}_0$. The optimal coefficients minimize the expected Frobenius loss $\mathbb{E}[\|\mathbf{\Pi}_{\mathrm{LS}} - \mathbf{\Sigma}^{-1}\|_F^2]$ over this two-dimensional space. If the true precision $\mathbf{\Sigma}^{-1}$ is well-approximated by some linear combination of $\hat{\mathbf{\Pi}}_S$ and $\mathbf{\Pi}_0$, the optimal coefficients produce a good estimator. If $\mathbf{\Sigma}^{-1}$ lies far outside this two-dimensional space, no choice of α and β can produce a good estimator, and the formula will assign coefficients that are mathematically optimal but practically useless.

On Lorenz 96, the true precision matrix has dense off-diagonal structure from spatial coupling. The identity target $\mathbf{\Pi}_0 = \mathbf{I}$ has no off-diagonal entries, so $\mathbf{\Sigma}^{-1}$ cannot be well-approximated by $\alpha\hat{\mathbf{\Pi}}_S + \beta\mathbf{I}$ unless α is very close to 1 (i.e., all weight on the sample precision). But the formula constraints α^* to be at most $1 - n/N_e$, which is far below 1 at the ensemble sizes we test. The result is that β^* explodes to compensate, and clamping destroys the theoretical optimality.

On cosmological data, the true precision is near-diagonal, so $\mathbf{\Sigma}^{-1} \approx \alpha\hat{\mathbf{\Pi}}_S + \beta\mathbf{\Pi}_0$ is achievable with moderate α and β. The formula works as intended.

Figure 5.1 illustrates this dependence schematically. The horizontal axis is the concentration ratio n/N_e (how close the problem is to rank-deficiency), and the vertical axis represents the alignment between the shrinkage target and the true precision structure. Our two experiments occupy opposite corners of this space: Lorenz 96 has moderate concentration ratio but poor target alignment, while the cosmological data has low concentration ratio and good alignment.

This analysis suggests a practical heuristic: before applying direct precision shrinkage, check whether the sample precision $\hat{\mathbf{\Pi}}_S$ is approximately diagonal. If not, the diagonal targets will fail, and a covariance-space method (Ledoit-Wolf) or a localization-based method (modified Cholesky) should be preferred.

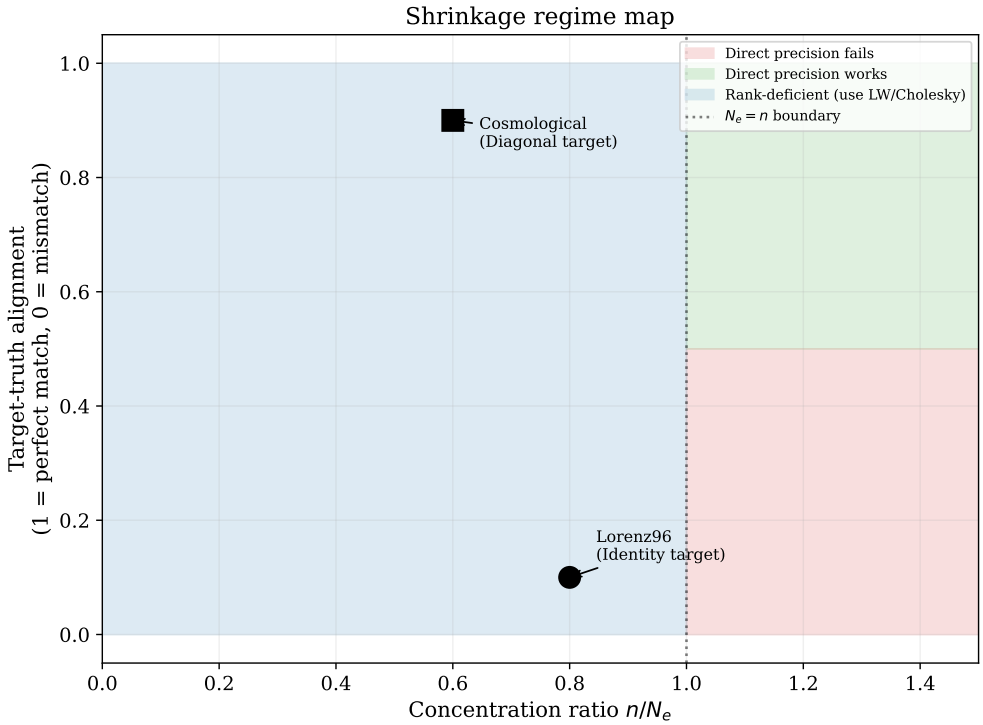

Figure 5.1: Schematic regime map for direct precision shrinkage. The method works well (green region) when the target aligns with the true precision structure; it fails (red region) when the target is misaligned, regardless of ensemble size. The rank-deficient regime ($N_e < n$, blue region) requires alternative methods. Our two experiments (circles) occupy opposite corners.

5.2 Why Ledoit-Wolf Wins in Sequential Filtering

The finding that Ledoit-Wolf (the worst standalone precision estimator on cosmological data) produces the best EnKF filtering RMSE is a surprising result. We have attributed this to "sequential accumulation": Ledoit-Wolf's low variance outweighs its high bias when errors compound over 100 cycles. Here we explore this explanation in more depth.

5.2.1 Variance as a Compounding Risk

Consider a simplified model of sequential filtering. At each cycle k, the analysis error is:

$$\mathbf{e}_k^a = f(\mathbf{e}_{k-1}^a, \boldsymbol{\epsilon}_k), \tag{5.1}$$

where f represents the forecast-analysis cycle and $\boldsymbol{\epsilon}_k$ is the precision estimation error at cycle k. If f is contractive (the model is observable and the filter is stable), then $\|\mathbf{e}_k^a\| \leq \rho\|\mathbf{e}_{k-1}^a\| + c\|\boldsymbol{\epsilon}_k\|$ for some contraction rate $\rho < 1$ and constant c. The long-run expected error is:

$$\lim_{K\to\infty} \mathbb{E}[\|\mathbf{e}_K^a\|^2] \leq \frac{c^2}{1-\rho^2}\left(\text{bias}^2 + \text{var}\right), \tag{5.2}$$

where bias and var are the bias and variance of $\boldsymbol{\epsilon}_k$. When ρ is close to 1 (weakly contractive filter, as with large ensemble sizes), the denominator $1-\rho^2$ is small, amplifying both terms.

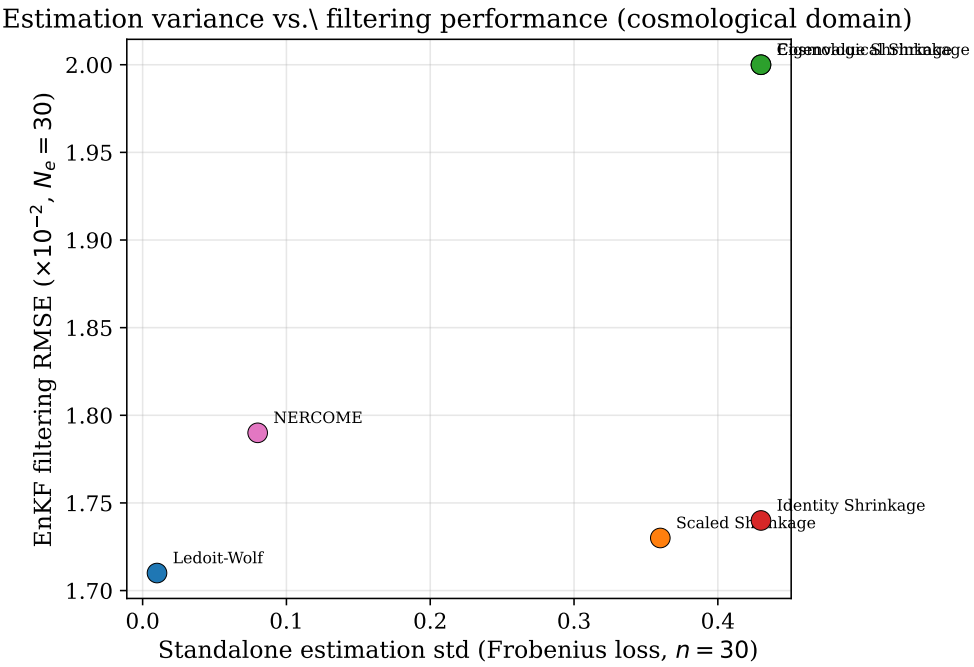

Figure 5.2: Standalone estimation variance (horizontal) vs. EnKF filtering RMSE (vertical) on the cosmological domain ($n = 30$, $N_e = 30$). Methods with lower estimation variance tend to achieve lower filtering RMSE, regardless of their standalone estimation accuracy.

The critical point is that both bias and variance contribute equally to the long-run error; there is no factor favoring one over the other.

So why does low variance "win" in our experiments? The answer lies in the non-contractivity of poorly conditioned analysis steps. When ϵ_k is large (a high-variance event), the resulting analysis update can move the ensemble mean *away* from the truth, making $\|\mathbf{e}_k^a\| > \|\mathbf{e}_{k-1}^a\|$. This violates the contraction assumption and triggers an error cascade: the next forecast starts from a bad analysis, producing a bad forecast, which leads to a bad analysis at the next cycle. The cascade persists until a sequence of low-error cycles brings the filter back to equilibrium.

Figure 5.2 illustrates this by plotting each method's standalone estimation variance against its filtering RMSE on the cosmological domain, showing a clear positive correlation. Ledoit-Wolf sits in the bottom-left corner (low variance, low RMSE), while the direct precision methods cluster in the upper-right (high variance, higher RMSE). NERCOME occupies an intermediate position, with low standalone variance but higher filtering RMSE due to its bootstrap variance compounding over cycles.

Ledoit-Wolf never triggers cascades because its precision estimates are consistently well-conditioned (condition numbers 10^2–10^3). The direct precision methods produce occasional extremely ill-conditioned estimates (condition numbers 10^{10}–10^{11}) that trigger cascades. Even though these cascades are rare (perhaps 5–10 out of 100 cycles), they dominate the time-averaged RMSE.

5.2.2 Implications for Estimator Design

This analysis suggests that for sequential filtering, the relevant figure of merit is not the expected Frobenius loss but rather the *tail risk* of the precision estimate, defined as the probability of producing an extremely ill-conditioned estimate that triggers a cascade. Ledoit-Wolf has low tail risk because it operates in covariance space and never inverts a poorly conditioned matrix. The direct precision methods have high tail risk because they invert the sample covariance (which can be poorly conditioned) and then apply shrinkage coefficients that may be clamped.

A modified Bodnar estimator that bounds the condition number of $\mathbf{\Pi}_{\mathrm{LS}}$ (e.g., by truncating small eigenvalues before inversion) could potentially combine the accuracy advantages of direct precision shrinkage with the stability advantages of Ledoit-Wolf. We leave this as a direction for future work.

A related unexplored middle ground is covariance shrinkage with a precision-aware target. The Ledoit-Wolf estimator uses a scaled identity covariance target, which is optimal for covariance estimation but agnostic to precision structure. One could instead shrink toward a diagonal covariance target informed by the known or estimated precision structure — for instance, using the inverse of the diagonal entries of the sample precision. However, the Ledoit-Wolf optimal intensity has a closed-form solution only for the scaled identity target; a general diagonal target would require either a new analytical derivation or numerical cross-validation to select the intensity, adding complexity. Whether such a hybrid approach could retain Ledoit-Wolf's low variance while improving its precision estimation accuracy remains an open question.

5.3 Domain Dependence of Shrinkage Performance

The reversal of method rankings between the two test domains is not an artefact of our experimental setup; it reflects a fundamental tension in shrinkage estimation. On systems with dense precision structure (Lorenz 96), diagonal shrinkage targets are misleading and produce worse estimates than the unshrunk sample precision. On systems with sparse precision structure (cosmological data), the same targets are informative and produce better estimates.

This domain dependence is well-established in the covariance estimation literature. Ledoit and Wolf (2004) noted that the optimal shrinkage intensity depends on the "concentration ratio" n/N_e and the structure of the true covariance relative to the target. What our results add is that this dependence is *qualitatively different* for precision estimation than for covariance estimation:

- In covariance shrinkage, a mismatched target is suboptimal but not catastrophic. The estimator simply converges more slowly to the truth as N_e increases.

- In precision shrinkage, a mismatched target can cause the shrinkage coefficients to diverge to extreme values, producing ill-conditioned estimates that are worse than

the unshrunk sample precision. The failure mode is structural, not a matter of slow convergence.

This asymmetry arises because precision shrinkage operates on $\mathbf{S}^{-1}$, which amplifies the noise in $\mathbf{S}$. When the shrinkage target does not capture the dominant structure of $\mathbf{S}^{-1}$, the formula attempts to "correct" a matrix that is already noisy, producing extreme coefficients.

5.4 The Modified Cholesky Alternative

The modified Cholesky decomposition occupies a unique position among the methods we tested. It is the only method that:

1. Estimates the precision matrix directly (like the Bodnar methods).
2. Does not require $N_e > n$ (works with any ensemble size, as long as $N_e > r + 2$).
3. Provides implicit localization (the banded structure zeroes out long-range correlations).
4. Is computationally cheap ($\mathcal{O}(n \cdot r \cdot N_e)$).

On Lorenz 96, EnKF-Cholesky achieves consistent RMSE of 0.13–0.14 $\times 10^{-2}$ regardless of ensemble size, which is better than all direct precision shrinkage methods and competitive with Ledoit-Wolf. It is not the best method overall (the standard EnKF at $N_e = 100$ and Ledoit-Wolf at all sizes achieve lower RMSE), but it is the most robust: its RMSE never spikes, its standard deviation is ± 0.01, and it works in the rank-deficient regime ($N_e < n$) where the other precision methods cannot be applied.

The bandwidth r is the key tuning parameter. On Lorenz 96 with nearest-neighbor coupling, $r = 2$ captures the dominant covariance structure. On systems with longer-range correlations, a larger r would be needed, at the cost of increased variance in the regression estimates. A principled method for selecting r, for instance through cross-validation within the ensemble or based on the decorrelation length of the model, would improve the practical usability of this approach.

On cosmological data, EnKF-Cholesky performs comparably to the shrinkage methods (Table 4.4), despite the fact that the near-diagonal precision structure would seem to favor the Bodnar methods. This is because the AR(1) model is sufficiently simple that even a crude banded precision estimate is adequate for the filter to converge.

5.5 Comparison with Looijmans et al.

Our standalone precision estimation results on BOSS DR12 mock data (Table 4.3) are broadly consistent with the findings of (Looijmans et al., 2024), with several nuances.

Both studies find that direct precision shrinkage outperforms Ledoit-Wolf on near-diagonal cosmological data at $n \geq 30$, that the cosmological target provides modest improvement over the identity target, and that NERCOME is competitive at small sample sizes thanks to its low variance.

Our results use the TEDA framework with independent implementations of the shrinkage formulas, rather than the original code from (Looijmans et al., 2024). The close agreement (differences $< 5\%$ in mean Frobenius loss) provides an independent validation of both implementations. We additionally test the scaled identity target, which was not considered by (Looijmans et al., 2024), and find it competitive at intermediate sample sizes ($n = 30$).

The key contribution beyond (Looijmans et al., 2024) is the sequential filtering evaluation. Their work evaluated estimators in a one-shot setting, concluding that direct precision shrinkage with cosmological targets provides the best precision estimates. Our EnKF experiments show that this conclusion does not extend to sequential filtering: the best one-shot estimator is not the best filter. This finding could not have been obtained from the standalone evaluation alone.

5.6 Sensitivity to Experimental Parameters

Our results depend on several experimental choices that could influence the conclusions.

Inflation factor. We use $\rho = 1.05$ for Lorenz 96 and $\rho = 1.02$ for the cosmological model. Different inflation values would change the effective ensemble spread and thereby the quality of the sample covariance. Higher inflation produces wider ensemble spread, making the sample covariance better conditioned but also increasing the forecast error. The direct precision methods are more sensitive to inflation than Ledoit-Wolf because the condition number of $\hat{\mathbf{\Pi}}_S$ depends on the eigenvalue spectrum of $\mathbf{S}$, which is affected by inflation. A systematic sensitivity study over $\rho \in [1.01, 1.10]$ would quantify this dependence but was beyond the scope of this work.

Observation coverage. We observe 80% of the state on Lorenz 96 ($m = 32$ of $n = 40$) and 78% on the cosmological model ($m = 14$ of $d = 18$). With fewer observations, the analysis relies more heavily on the background covariance, amplifying the impact of precision estimation quality. With more observations, the analysis is dominated by the observation term in Eq. (2.18), and the precision estimation becomes less important. Our high observation coverage favors methods with good precision estimates; at lower coverage, the differences between methods would likely be more pronounced.

Observation error. The Lorenz 96 observation error ($\sigma_o = 0.01$) is very small relative to the state magnitude, making observations highly informative. The cosmological observation error ($\sigma_o = 100$) is very large relative to the state magnitude, making the filter heavily dependent on the background. These opposite regimes test different aspects of the precision estimation: on Lorenz 96, the observations dominate and the precision estimate mainly controls the analysis increment; on the cosmological model, the background dominates and the precision estimate controls how much the filter trusts the forecast.

Assimilation window. We use 100 assimilation cycles. With fewer cycles, the transient phase (where the filter is converging from the initial ensemble) would dominate the time-

averaged RMSE, potentially masking the steady-state performance differences. With more cycles, the sequential accumulation effect would be more pronounced, further favoring low-variance methods. Our choice of 100 cycles with 50-cycle transient discarding is a reasonable compromise.

5.7 Practical Recommendations

The following guidelines summarize the practical implications of our experimental results for choosing a precision estimator in the EnKF:

1. Ledoit-Wolf is the default choice for sequential filtering without domain-specific knowledge. It provides the best overall RMSE and the lowest variance across runs, requires no tuning parameters, and works at any ensemble size. The main limitation is computational: it requires inverting the shrunk covariance, which costs $\mathcal{O}(n^3)$. For $n < 10^3$, this is negligible; for $n > 10^4$, alternative approaches are needed.

2. When the true precision matrix is known or suspected to be near-diagonal (as in cosmological applications), direct precision shrinkage with an appropriate target can improve standalone precision estimates by a factor of 2 in Frobenius loss. However, this advantage does not necessarily translate to better filtering performance due to the sequential accumulation effect. Direct precision shrinkage is therefore preferable for standalone estimation (e.g., covariance estimation from mock catalogs), while Ledoit-Wolf remains the better choice for sequential filtering.

3. For spatially extended systems, the modified Cholesky decomposition is the strongest choice when:

   - $N_e < n$ (ruling out direct precision methods entirely),
   - The system has known spatial structure (decay of correlations with distance),
   - Implicit localization is desirable without specifying a localization function.

   The bandwidth r should be set based on the decorrelation length of the system. For Lorenz 96 with nearest-neighbor coupling, $r = 2$–5 works well. For atmospheric models with longer-range correlations, $r = 10$–20 may be needed.

4. NERCOME should be avoided in sequential settings. Its computational cost ($\sim$ 100–200$\times$ slower than alternatives) and its increasing RMSE at larger ensemble sizes on Lorenz 96 make it a poor choice for sequential filtering. It excels in standalone estimation at very small sample sizes ($n/d < 1.5$) where its low variance compensates for higher bias, but this advantage disappears inside a filter. If NERCOME must be used, reducing the number of bootstrap draws from the default 200 to 50–100 mitigates the variance compounding.

5. When domain-specific prior knowledge is unavailable, the scaled identity target provides 10–30% improvement over the plain identity in both standalone estimation and

filtering RMSE. The improvement comes from calibrating the target to per-variable variances, which reduces the severity of coefficient clamping. The computational overhead is negligible ($\mathcal{O}(n)$ to read diagonal entries of $\mathbf{S}$). The ad-hoc factor of 2 is the main weakness; a cross-validated factor would be an easy improvement.

6. Before applying Bodnar methods, it is advisable to check target alignment using a simple diagnostic: compute the ratio $R_{\text{diag}} = \sum_i (\hat{\mathbf{\Pi}}_S)_{ii}^2 / \|\hat{\mathbf{\Pi}}_S\|_F^2$. If $R_{\text{diag}} > 0.8$, the precision is nearly diagonal and diagonal targets are appropriate. If $R_{\text{diag}} < 0.5$, off-diagonal entries dominate and diagonal targets will produce extreme β^* values; Ledoit-Wolf or modified Cholesky should be used instead.

These guidelines are summarized in Figure 5.1, which maps the two key parameters (concentration ratio n/N_e and target-truth alignment) to the recommended method.

5.8 Relationship to Other Regularization Approaches

The shrinkage estimators studied in this thesis are one approach to regularizing ill-conditioned covariance or precision matrices. Several other approaches exist in the literature, and understanding their relationship to shrinkage provides broader context for our results.

Tapering and banding methods zero out entries beyond a distance threshold (Gaspari and Cohn, 1999), rather than shrinking the covariance toward a structured target. Banding retains only entries within a band of width r around the diagonal. The modified Cholesky decomposition used in this thesis is a form of banding applied to the precision matrix. Tapering and shrinkage can be combined: first taper the sample covariance to remove spurious long-range correlations, then shrink toward a target to reduce eigenvalue spread. This combination has not been explored for direct precision shrinkage.

Soft and hard thresholding methods set small entries of the sample covariance to zero, producing a sparse estimate. This is related to the LASSO (Least Absolute Shrinkage and Selection Operator) in regression and to the graphical LASSO for precision matrix estimation. Unlike the Bodnar framework, which assumes a specific parametric form for the estimator ($\alpha\hat{\mathbf{\Pi}}_S + \beta\mathbf{\Pi}_0$), thresholding methods impose sparsity without assuming a target structure. However, they require tuning the threshold parameter, which plays a role similar to the shrinkage intensity λ.

The Marchenko-Pastur law and its extensions provide analytical corrections for the eigenvalue spreading of sample covariance matrices. Rather than shrinking toward a target, these methods directly correct the eigenvalues of $\mathbf{S}$ based on the known distribution of Wishart eigenvalues. The Oracle Approximating Shrinkage (OAS) estimator of (Chen et al., 2010) is related in spirit, though it works by iteratively approximating the clairvoyant (oracle) shrinkage estimator via MMSE optimization under Gaussian assumptions rather than directly applying random matrix theory corrections. The NERCOME estimator (Lam, 2016) achieves a similar effect empirically through cross-validated eigenvalue estimation.

All of these approaches regularize a high-dimensional covariance or precision estimate from a limited sample, but through different mechanisms. The choice among them depends

on the available prior knowledge (structural sparsity, eigenvalue distribution, target matrix) and the computational budget. Our results suggest that for sequential filtering applications, the variance of the estimator matters at least as much as the mechanism used to achieve regularization.

Chapter 6

Conclusions and Future Work

6.1 Summary of Contributions

This thesis embedded the direct precision shrinkage framework of (Bodnar et al., 2016) inside the dual formulation of the Ensemble Kalman Filter, proposed a domain-agnostic scaled identity target, and tested six shrinkage-based estimators on systems with opposite precision structure.

First, target-truth alignment determines direct precision shrinkage performance. On Lorenz 96, where the precision matrix has dense off-diagonal structure from spatial coupling, diagonal targets produce erratic filtering behavior: the Bodnar coefficients are severely miscalibrated, unclamped β^* values reach 10^8–10^9, and the resulting precision estimates have condition numbers of 10^7–10^{11}. On the near-diagonal cosmological data, the same methods achieve Frobenius losses roughly half those of Ledoit-Wolf and stable monotonic improvement with ensemble size. The framework works as designed when the target captures the dominant structure of the true precision matrix.

Second, good standalone estimation does not imply good filtering. Ledoit-Wolf produces the worst standalone precision estimates on cosmological data (Frobenius loss $\approx$ 0.93–0.96) yet achieves the lowest EnKF RMSE. The explanation is sequential accumulation: Ledoit-Wolf's standard deviation across draws is ± 0.01, an order of magnitude below the direct precision methods (± 0.13–0.14). A slightly biased but stable estimator accumulates less error over 100 sequential assimilation cycles than a more accurate but variable one. For sequential data assimilation, variance reduction matters at least as much as bias reduction.

Third, the scaled identity target provides a middle ground. It outperforms the plain identity on both Lorenz 96 and cosmological data without requiring domain-specific information, but does not match the performance of domain-informed targets when those are available.

For practitioners choosing a precision estimator for the EnKF, the standard approach of selecting the estimator with the lowest expected loss in a standalone comparison can be misleading. The sequential nature of the filter introduces a new criterion (stability across cycles) that is not captured by one-shot metrics. A Frobenius loss comparison will recommend the Bodnar methods on cosmological data; an EnKF RMSE comparison will recommend Ledoit-Wolf. The difference is not a contradiction but a consequence of measuring different quantities.

The scaled identity target we proposed fills a specific niche. It provides a 10–30% improvement over the plain identity target on both Lorenz 96 (RMSE) and cosmological data (Frobenius loss) without requiring any domain-specific information. The improvement comes

from calibrating the target to the per-variable variances, which reduces the severity of the coefficient clamping. However, the improvement is modest, and the target does not approach the performance of domain-specific targets when those are available. The ad-hoc factor of 2 could likely be improved through cross-validation, but the practical impact would be small.

6.2 Limitations

The following limitations should be considered when interpreting the results.

The Lorenz 96 experiments require $N_e > n + 2 = 42$ for the sample covariance to be invertible, a prerequisite for the direct precision methods. This excludes the rank-deficient regime ($N_e < n$) where covariance estimation is most challenging and where methods like localization or covariance tapering are essential. Combining direct precision shrinkage with localization is a natural extension but was not explored here.

The AR(1) cosmological model preserves the covariance structure of the BOSS mock catalogs but is far simpler than a realistic cosmological simulation: it is nearly linear ($\phi = 0.95$), has mild dynamics, and its near-diagonal precision structure is imposed by construction. As a result, the cosmological experiments function primarily as a best-case sanity check — confirming that direct precision shrinkage works when the target matches the true structure — rather than as a demanding test of the framework. The more informative stress test is Lorenz 96, where the chaotic nonlinear dynamics and dense precision structure expose the failure modes. A more realistic cosmological test would involve non-Gaussian fields, cross-correlations between multipoles, or a dynamical model with stronger nonlinearity, and the sequential accumulation effect could manifest differently in such settings.

The scaled identity target's factor of 2 is borrowed by analogy from the cosmological $2/N_\ell$ term and lacks theoretical justification. A principled derivation, perhaps through cross-validation or a Bayesian argument, could improve its performance.

All experiments use a linear observation operator $\mathbf{H}$. Nonlinear observation operators would introduce additional approximation error in the EnKF, potentially interacting with the precision estimation error in ways not captured by our experiments.

6.3 Future Work

Interaction between shrinkage and spatial localization (Gaspari and Cohn, 1999; Nino-Ruiz et al., 2018). Our implementation does not combine these two approaches, yet both address the same underlying problem (ill-conditioned covariance estimation from small ensembles) through different mechanisms. Localization imposes sparsity (zeroing out long-range correlations), while shrinkage imposes a convex combination with a structured target. In principle, one could first localize the sample covariance, then apply precision shrinkage to the localized matrix, or vice versa. However, the interaction is not straightforward: localization changes the statistical properties of the sample covariance (the localized matrix is no longer Wishart-distributed), which invalidates the theoretical basis of the Bodnar formulas. A modified derivation that accounts for localization, or an empirical study of the combined approach, would be valuable.

Scaling to higher-dimensional systems is untested. The Lorenz 96 model ($n = 40$) and the BOSS mock data ($d = 18$) are both low-dimensional by operational data assimilation standards. Weather prediction models have state dimensions of 10^6–10^8, and ensemble sizes of 20–100 are typical. At these dimensions, the $\mathcal{O}(n^3)$ matrix inversion required by the direct precision methods is prohibitively expensive without approximation. The extrapolation in Section 4.6.2 (Figure 4.23) shows that direct precision shrinkage and Ledoit-Wolf (used as a precision estimator after inversion) become uncompetitive with EnKF-Cholesky somewhere between $n \sim 10^3$ and $n \sim 10^4$, consistent with the asymptotic ordering $\mathcal{O}(n^3)$ vs. $\mathcal{O}(nrN_e)$. Testing on intermediate-complexity systems such as the quasi-geostrophic model ($n \sim 10^4$) would reveal whether the sequential accumulation effect we observed persists and whether the computational overhead is manageable. The modified Cholesky approach, with its $\mathcal{O}(n \cdot r \cdot N_e)$ cost, is the natural candidate for high-dimensional settings, possibly combined with shrinkage applied to the residual banded structure.

Adaptive target selection based on observed structure. Rather than fixing a target a priori, one could adapt the target based on the observed covariance structure during the assimilation. A simple heuristic: compute the ratio of off-diagonal to diagonal energy in the sample precision, $R = (\|\hat{\mathbf{\Pi}}_S\|_F^2 - \sum_i (\hat{\mathbf{\Pi}}_S)_{ii}^2) / \|\hat{\mathbf{\Pi}}_S\|_F^2$. When R is small (near-diagonal precision), use the identity or scaled identity target; when R is large, fall back to a covariance-space method like Ledoit-Wolf. This diagnostic costs $\mathcal{O}(n^2)$ and could be evaluated at each cycle. The challenge is setting the threshold: the optimal switchover point depends on the ensemble size, state dimension, and model dynamics, and would likely need to be calibrated empirically.

Theoretical analysis of sequential accumulation. Our observation that estimator variance matters more than bias in sequential filtering is empirical, and a theoretical treatment would strengthen this finding considerably. One approach is to model the EnKF error as a Markov chain: at each step, the analysis error is a function of the forecast error and the precision estimation error, and the forecast error at the next step depends on the current analysis error through the model dynamics. If the model is linear and the precision estimation errors are i.i.d., the long-run error covariance satisfies a discrete Lyapunov equation whose solution depends on both the bias and variance of the precision estimator. Deriving conditions under which low-variance estimators provably outperform low-bias ones would place our empirical finding on a rigorous foundation.

Principled calibration of the scaled identity factor. The factor of 2 in the scaled identity target is borrowed by analogy from the cosmological $2/N_\ell$ term and lacks theoretical justification. A principled approach would be to treat the scaling factor as a hyperparameter and select it by minimizing a cross-validated loss function within the ensemble. At each assimilation cycle, one could hold out a subset of ensemble members, compute the precision estimate from the remaining members, evaluate the Frobenius loss against the held-out precision, and average over multiple random holdouts. The optimal scaling factor would depend on the ensemble size and the true precision structure, but if it is approximately constant across cycles, a single calibration run would suffice.

Non-Gaussian extensions of the Bodnar framework. The Bodnar et al. derivation assumes that the sample covariance follows a Wishart distribution, which holds when the ensemble

members are Gaussian. In practice, nonlinear model dynamics produce non-Gaussian ensembles, and the Wishart assumption is only approximate. Extending the framework to accommodate heavier tails or skewness, perhaps through a Wishart mixture model or a more flexible shrinkage estimator, could improve performance on strongly nonlinear systems.

Condition-number bounded direct precision shrinkage. The analysis in Section 5.2 suggests that the tail risk of precision estimates (specifically, the probability of producing a very ill-conditioned estimate) is a key predictor of filtering performance. A modified Bodnar estimator that explicitly bounds the condition number of $\mathbf{\Pi}_{\mathrm{LS}}$ (e.g., by truncating eigenvalues below a threshold before applying the shrinkage formula) could combine the accuracy of direct precision shrinkage with the stability of Ledoit-Wolf. The threshold could be set based on the observation error covariance $\mathbf{R}$: eigenvalues of $\mathbf{\Pi}_{\mathrm{LS}}$ below $\|\mathbf{R}^{-1}\|$ are unlikely to be meaningful.

Time-adaptive targets from a running mean. Rather than using fixed targets (identity, eigenvalue, cosmological), one could construct the target from a running average of past precision estimates:

$$\mathbf{\Pi}_0^{(k)} = \gamma\, \mathbf{\Pi}_0^{(k-1)} + (1-\gamma)\, \hat{\mathbf{\Pi}}_S^{(k-1)},$$

where $\gamma \in (0,1)$ is a forgetting factor. This adaptive target would track the evolving precision structure without requiring domain-specific knowledge. The challenge is that the Bodnar formula assumes a fixed target; a time-varying target changes the statistical properties of the estimator in ways that the derivation does not account for.

Appendix A

Supplementary Diagnostic Data

This appendix presents the full diagnostic data from the shrinkage coefficient analysis described in Section 4.2.1. Table A.1 through Table A.3 report the shrinkage coefficients, clamping rates, and condition numbers for each direct precision shrinkage target across 12 ensemble sizes from $N_e = 45$ to $N_e = 100$ on Lorenz 96 ($n = 40$).

All values are averaged over 100 assimilation cycles per run. The "α^* mean" column shows the mean value of α^* (which is never clamped, as it stays within $[0, 1]$). The "β^* unclamped" columns show the mean, minimum, and maximum of the unclamped β^* values. "β^* clamp %" is the fraction of assimilation steps where β^* required clamping to $[0, 1]$. "Cond. mean" and "Cond. max" are the mean and maximum condition numbers of the resulting precision estimator $\mathbf{\Pi}_{\mathrm{LS}}$.

A.1 Identity Target

Table A.1: Full diagnostic data for the identity target on Lorenz 96 ($n = 40$).

N_e	RMSE	α^*	β^* uncl. mean	β^* clamp %	Cond. mean	Cond. max
45	0.213	0.088	4.18×10^7	100%	4.77×10^8	3.09×10^9
50	0.121	0.180	4.49×10^8	100%	2.56×10^9	2.19×10^{10}
55	0.298	0.254	3.90×10^8	100%	8.92×10^9	8.16×10^{10}
60	0.408	0.316	2.76×10^8	100%	1.79×10^{10}	9.77×10^{10}
65	0.124	0.369	2.10×10^8	100%	7.76×10^9	4.35×10^{10}
70	0.122	0.414	2.05×10^8	100%	1.26×10^{10}	9.97×10^{10}
75	1.233	0.453	1.84×10^8	100%	1.64×10^{10}	1.14×10^{11}
80	1.087	0.487	1.45×10^8	100%	1.77×10^{10}	1.60×10^{11}
85	0.900	0.517	1.42×10^8	100%	1.25×10^{10}	8.78×10^{10}
90	0.467	0.544	1.28×10^8	100%	1.42×10^{10}	1.21×10^{11}
95	0.129	0.568	1.13×10^8	100%	8.04×10^9	5.28×10^{10}
100	0.667	0.590	1.09×10^8	100%	2.10×10^{10}	2.18×10^{11}

The RMSE column shows the non-monotonic behavior discussed in Section 4.2.2: increasing N_e does not consistently reduce the RMSE. For instance, the RMSE at $N_e = 75$ (1.233) is worse than at $N_e = 45$ (0.213). This erratic behavior is caused by the interaction between the sample precision quality (which improves with N_e) and the clamped shrinkage coefficients (which introduce a different kind of error at each N_e).

The β^* values are clamped at 100% of assimilation steps for all ensemble sizes. The unclamped values decrease from 4.18×10^7 at $N_e = 45$ to 1.09×10^8 at $N_e = 100$ (the unclamped value actually *increases* before decreasing, reflecting the interplay between the

$(1 - n/N_e - \alpha^*)$ factor, which increases with N_e, and the $\mathrm{tr}(\hat{\mathbf{\Pi}}_S)/n$ factor, which also changes with N_e as the sample precision becomes more accurate).

The condition numbers of $\mathbf{\Pi}_{\mathrm{LS}}$ range from 10^8 to 10^{11}, with no clear trend in N_e. The maximum condition number across assimilation steps is typically 5–10 times the mean, indicating that some steps produce particularly ill-conditioned estimates. These are the steps that trigger the RMSE spikes seen in the time series plots.

A.2 Scaled Identity Target

Table A.2: Full diagnostic data for the scaled identity target on Lorenz 96 ($n = 40$).

N_e	RMSE	α^*	β^* uncl. mean	β^* clamp %	Cond. mean	Cond. max
45	0.194	0.088	9.55×10^4	100%	2.76×10^7	2.78×10^8
50	0.135	0.179	5.46×10^4	100%	3.31×10^8	2.83×10^9
55	0.356	0.254	2.20×10^4	100%	9.05×10^8	6.72×10^9
60	0.561	0.316	1.27×10^4	100%	1.41×10^9	9.58×10^9
65	0.208	0.369	9.85×10^3	100%	6.71×10^8	5.37×10^9
70	0.176	0.414	7.53×10^3	100%	1.33×10^9	1.36×10^{10}
75	0.814	0.453	5.42×10^3	100%	1.31×10^9	1.14×10^{10}
80	0.855	0.487	4.28×10^3	100%	1.79×10^9	1.36×10^{10}
85	0.913	0.517	5.06×10^3	100%	1.57×10^9	1.07×10^{10}
90	0.424	0.544	3.85×10^3	100%	1.79×10^9	1.57×10^{10}
95	0.145	0.568	2.76×10^3	100%	1.16×10^9	7.89×10^9
100	0.484	0.590	2.69×10^3	100%	2.35×10^9	2.50×10^{10}

The scaled identity target shows similar patterns to the identity target but with important quantitative differences. The unclamped β^* values are 3–4 orders of magnitude smaller (10^3–10^5 versus 10^7–10^9), reflecting the better scale calibration of the scaled target. The condition numbers are correspondingly lower by 1–2 orders of magnitude (10^7–10^{10} versus 10^8–10^{11}). The RMSE values are generally lower than the identity target (compare Tables A.1 and A.2), confirming the benefit of scaling by empirical variances.

Despite these improvements, the scaled target still requires 100% clamping of β^*, and the condition numbers remain far above those of Ledoit-Wolf (10^2–10^3). The scaling reduces the severity of the mismatch but does not resolve the fundamental issue: on Lorenz 96, no diagonal target can adequately represent the true precision structure.

A.3 Eigenvalue Target

The eigenvalue target behaves qualitatively differently from the identity and scaled targets. The β^* values are negligible ($< 10^{-4}$) and never require clamping. The estimator therefore reduces to $\mathbf{\Pi}_{\mathrm{LS}} \approx \alpha^* \hat{\mathbf{\Pi}}_S$, a simple rescaling of the sample precision without any contribution from the target. The eigenvalue target effectively provides no shrinkage.

This occurs because the eigenvalue target $\mathbf{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$ is derived from the eigenvalues of $\hat{\mathbf{\Pi}}_S$ itself. By the Cauchy-Schwarz inequality, $\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0^{(1)})$ is close to $\|\hat{\mathbf{\Pi}}_S\|_F \|\mathbf{\Pi}_0^{(1)}\|_F$

Table A.3: Full diagnostic data for the eigenvalue target on Lorenz 96 ($n = 40$).

N_e	RMSE	α^*	β^* uncl. mean	β^* clamp %	Cond. mean	Cond. max
45	0.279	0.089	3.0×10^{-4}	0%	2.11×10^8	1.38×10^9
50	0.552	0.180	1.0×10^{-4}	0%	1.19×10^8	4.37×10^8
55	0.476	0.255	$< 10^{-4}$	0%	7.96×10^7	1.84×10^8
60	1.236	0.317	$< 10^{-4}$	0%	6.84×10^7	4.51×10^8
65	2.038	0.369	$< 10^{-4}$	0%	1.61×10^8	1.08×10^9
70	0.239	0.414	$< 10^{-4}$	0%	7.91×10^7	3.91×10^8
75	0.741	0.453	$< 10^{-4}$	0%	1.76×10^8	2.88×10^9
80	0.615	0.488	$< 10^{-4}$	0%	4.84×10^7	1.73×10^8
85	0.534	0.518	$< 10^{-4}$	0%	5.74×10^7	1.99×10^8
90	0.400	0.544	$< 10^{-4}$	0%	6.71×10^7	2.43×10^8
95	2.051	0.568	$< 10^{-4}$	0%	5.35×10^8	4.83×10^9
100	0.273	0.590	1.0×10^{-4}	0%	5.36×10^7	3.97×10^8

(since the two matrices share eigenvalues up to ordering), making the denominator $\|\hat{\mathbf{\Pi}}_S\|_F^2 \|\mathbf{\Pi}_0^{(1)}\|_F^2 - (\mathrm{tr}(\hat{\mathbf{\Pi}}_S \mathbf{\Pi}_0^{(1)}))^2$ very large. In the Bodnar formula for α^*, the correction term $N_e^{-1} \|\hat{\mathbf{\Pi}}_S\|_{\mathrm{tr}}^2 \|\mathbf{\Pi}_0\|_F^2$ divided by this large denominator is negligible, so $\alpha^* \approx 1 - n/N_e$. Then $(1 - n/N_e - \alpha^*) \approx 0$, giving $\beta^* \approx 0$.

Despite the near-zero shrinkage, the eigenvalue target produces the highest RMSE among the three targets at several ensemble sizes (2.038 at $N_e = 65$, 2.051 at $N_e = 95$). Without the regularizing effect of shrinkage toward a target, the estimator is essentially the rescaled sample precision $\alpha^* \mathbf{S}^{-1}$, which is ill-conditioned when $\mathbf{S}$ is poorly conditioned. The condition numbers (10^7–10^9) are lower than those of the identity target but still high enough to cause numerical instability in the EnKF analysis.

Appendix B

Standalone Estimation Results

This appendix presents the full standalone precision estimation results on BOSS DR12 mock data, complementing the summary in Table 4.3.

B.1 Frobenius Loss by Subsample Size

Table B.1 reproduces the standalone estimation results with additional precision, showing the mean and standard deviation of the relative Frobenius loss across 100 random subsets of mocks.

Table B.1: Relative Frobenius loss of precision matrix estimates on BOSS DR12 mock data ($d = 18$), full precision. Mean $\pm$ std over 100 random subsets.

Method	$n = 24$	$n = 30$	$n = 40$	$n = 50$
Sample (Hartlap)	1.152 ± 1.008	0.740 ± 0.440	0.467 ± 0.148	0.430 ± 0.162
Identity Shrinkage	1.177 ± 1.130	0.683 ± 0.433	0.453 ± 0.123	0.407 ± 0.141
Scaled Shrinkage	0.905 ± 0.962	0.598 ± 0.361	0.515 ± 0.105	0.464 ± 0.126
Eigenvalue Shrinkage	1.196 ± 1.127	0.698 ± 0.433	0.459 ± 0.126	0.411 ± 0.142
Cosmological Shrinkage	1.190 ± 1.127	0.693 ± 0.433	0.458 ± 0.125	0.409 ± 0.142
NERCOME	0.816 ± 0.055	0.725 ± 0.078	0.632 ± 0.081	0.525 ± 0.084
Ledoit-Wolf	0.959 ± 0.008	0.954 ± 0.009	0.941 ± 0.010	0.929 ± 0.012

B.2 Analysis of Variance Components

The high standard deviations of the Bodnar methods at small n (± 1.13 at $n = 24$) deserve further analysis. These are not measurement noise. They reflect genuinely different estimation quality across different random subsets of mocks. Some subsets produce sample covariance matrices that are well-conditioned and close to the truth, yielding good precision estimates. Other subsets produce ill-conditioned sample covariances, yielding poor precision estimates with Frobenius losses exceeding 3.

The variance has two components:

1. **Sample covariance variance**: Different subsets of n mocks produce different sample covariance matrices, with eigenvalue spread that depends on the particular subset.

2. **Coefficient variance**: The shrinkage coefficients α^* and β^* are functions of the sample covariance (through $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$), so they inherit and potentially amplify the variance of $\mathbf{S}$.

Ledoit-Wolf has near-zero variance ($\pm$0.008–0.012) because its shrinkage intensity λ^*_{LW} is computed from the sample covariance $\mathbf{S}$ without inversion, and the shrunk covariance $(1-\lambda)\mathbf{S} + \lambda\mu\mathbf{I}$ is always well-conditioned. NERCOME has intermediate variance ($\pm$0.055–0.084) because its bootstrap averaging reduces but does not eliminate the variance from random splitting.

At $n = 24$ ($n/d = 1.33$), any single precision estimate from the Bodnar methods is unreliable, as the standard deviation exceeds the mean. An ensemble of estimates (e.g., averaging over multiple random subsets) would reduce this variance, but at the cost of requiring more mock catalogs.

B.3 Improvement Over the Sample Precision

To assess the practical benefit of shrinkage, Table B.2 shows the percentage improvement of each method over the Hartlap-corrected sample precision at each subsample size.

Table B.2: Percentage improvement in Frobenius loss over the Hartlap-corrected sample precision. Positive values indicate improvement; negative values indicate the method is worse than the sample.

Method	$n = 24$	$n = 30$	$n = 40$	$n = 50$
Identity Shrinkage	−2.2%	7.7%	3.0%	5.4%
Scaled Shrinkage	21.4%	19.2%	−10.3%	−7.9%
Eigenvalue Shrinkage	−3.8%	5.7%	1.6%	4.4%
Cosmological Shrinkage	−3.3%	6.4%	1.9%	4.9%
NERCOME	29.2%	2.0%	−35.4%	−22.1%
Ledoit-Wolf	16.8%	−28.9%	−101.5%	−116.1%

At $n = 24$, NERCOME and the scaled identity target provide the largest improvements (29% and 21%, respectively). At $n = 30$, the identity and cosmological targets provide 6–8% improvement, while the scaled target leads at 19%. At $n = 40$–50, the Bodnar methods provide modest improvements (1–5%), but NERCOME and Ledoit-Wolf are worse than the sample. Ledoit-Wolf is dramatically worse at larger n because its covariance shrinkage-then-invert strategy is fundamentally suboptimal for precision estimation on near-diagonal data.

Appendix C

Lorenz 96 Model Details

The Lorenz 96 model (Lorenz, 1996) was introduced as a simplified representation of atmospheric dynamics on a latitude circle. The model consists of n variables $x_1, \ldots, x_n$ arranged on a periodic domain, governed by:

$$\frac{dx_i}{dt} = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F, \quad i = 1, \ldots, n, \tag{C.1}$$

with cyclic indices: $x_{-1} = x_{n-1}$, $x_0 = x_n$, $x_{n+1} = x_1$.

C.1 Physical Interpretation

The three terms have the following interpretations:

- $(x_{i+1} - x_{i-2})x_{i-1}$: A quadratic advection term, representing the transport of quantities along the domain. It conserves total energy ($\sum x_i^2$) and total momentum ($\sum x_i$), analogous to the advection terms in the Navier-Stokes equations.
- $-x_i$: A linear damping term, representing dissipation. Without forcing, all variables decay to zero.
- F: A constant forcing term, representing external energy input (e.g., solar heating). The value $F = 8$ produces chaotic dynamics, with a Lyapunov time of approximately 0.3 time units and a correlation time of approximately 0.6 time units.

C.2 Dynamical Regimes

The behavior of the Lorenz 96 model depends on the forcing parameter F:

- $F < 8/9 \approx 0.89$: The trivial fixed point $x_i = F$ for all i is stable.
- $0.89 < F <\sim 3$: Periodic or quasi-periodic solutions.
- $3 < F <\sim 5$: Intermittent chaos.
- $F > 5$: Fully developed chaos with positive Lyapunov exponents.
- $F = 8$: Strongly chaotic, the most common choice in data assimilation studies. The decorrelation length is approximately 4–5 grid points, which determines the effective range of dynamical correlations in the background covariance.

We use $n = 40$ and $F = 8$ throughout. The state dimension $n = 40$ is large enough to produce interesting covariance structure but small enough to allow exhaustive parameter sweeps. With $N_e = 50$–100 ensemble members, the ratio n/N_e ranges from 0.4 to 0.8, spanning the moderately ill-conditioned regime where shrinkage is most needed.

C.3 Numerical Integration

The model is integrated using a fourth-order Runge-Kutta scheme with time step $\Delta t = 0.01$ time units. Observations are assimilated every $\Delta t_{\text{obs}} = 0.1$ time units (10 integration steps per assimilation cycle), giving 100 assimilation cycles over a total integration time of $T = 10$ time units. This corresponds to approximately 33 Lyapunov times, long enough for the filter to equilibrate and for the time-averaged RMSE to converge.

C.4 Observation Configuration

We randomly select $m = 32$ of the $n = 40$ state variables as observation locations (80% coverage). The observation operator $\mathbf{H} \in \mathbb{R}^{32\times 40}$ is a selection matrix that picks these 32 rows from the identity. Observation errors are i.i.d. Gaussian with standard deviation $\sigma_o = 0.01$, giving an observation error covariance $\mathbf{R} = 10^{-4}\mathbf{I}_{32}$. This is a relatively low observation error compared to the typical state magnitude ($|x_i| \sim 5$–15), making the observations highly informative and placing the primary challenge on the covariance estimation rather than on the observation quality.

Appendix D

AR(1) Cosmological Model Details

The AR(1) cosmological model is a synthetic dynamical system designed to test shrinkage estimators on data with near-diagonal precision structure. It is calibrated to the statistics of the BOSS DR12 Patchy mock catalogs (Looijmans et al., 2024).

D.1 Model Construction

The model has $d = 18$ state variables representing power spectrum measurements at 9 k-bins $\times$ 2 multipole moments ($\ell = 0, 2$). The dynamics follow:

$$\mathbf{x}_{t+1} = \phi\, \mathbf{x}_t + (1 - \phi)\, \boldsymbol{\mu} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{D.1}$$

with three parameters:

- $\boldsymbol{\mu} \in \mathbb{R}^{18}$: The mean power spectrum, computed from the 2048 mock catalogs. These values range from 10^{-5} to 10^{-3} (in units of $h^{-3}\mathrm{Mpc}^3$), reflecting the amplitude of galaxy clustering at different scales and multipoles.

- $\phi = 0.95$: The persistence parameter, controlling how quickly the state reverts to the mean. A value of 0.95 means the state "remembers" 95% of its previous value at each step, producing smooth temporal evolution. The decorrelation time is $-1/\ln(\phi) \approx 20$ steps.

- $\mathbf{Q} = (1 - \phi^2)\boldsymbol{\Sigma}_{\mathrm{mock}}$: The innovation covariance, chosen so the stationary distribution has covariance $\boldsymbol{\Sigma}_{\mathrm{mock}}$. This ensures that the model's equilibrium statistics match the mock catalog statistics.

D.2 Stationary Distribution

The stationary covariance of the AR(1) process satisfies:

$$\boldsymbol{\Sigma}_{\mathrm{stat}} = \phi^2 \boldsymbol{\Sigma}_{\mathrm{stat}} + \mathbf{Q}. \tag{D.2}$$

Substituting $\mathbf{Q} = (1 - \phi^2)\boldsymbol{\Sigma}_{\mathrm{mock}}$ gives $\boldsymbol{\Sigma}_{\mathrm{stat}} = \boldsymbol{\Sigma}_{\mathrm{mock}}$, confirming that the model reproduces the mock catalog covariance in equilibrium. Because $\boldsymbol{\Sigma}_{\mathrm{mock}}$ is approximately diagonal (the mock catalogs simulate Gaussian random fields at independent modes), the precision matrix $\boldsymbol{\Sigma}_{\mathrm{stat}}^{-1}$ is also approximately diagonal.

D.3 Observation Configuration

We observe $m = 14$ of the $d = 18$ state variables ($\approx 78\%$ coverage), with Gaussian observation error $\sigma_o = 100$. The observation error is deliberately large relative to the state values ($|x_i| \sim 10^{-4}$–10^{-3}), so $\sigma_o/|x_i| \sim 10^5$–10^6. This means each individual observation provides very little information, and the filter must rely heavily on the precision of the background covariance estimate to extract useful signal. This setup amplifies the differences between estimation methods, making it a sensitive test of precision matrix quality.

Inflation is set to $\rho = 1.02$, lighter than the Lorenz 96 value of $\rho = 1.05$, because the near-linear AR(1) dynamics produce less covariance underestimation than the chaotic Lorenz 96 model.

Appendix E

Computational Timing Details

This appendix provides the full timing data from the computational cost experiments described in Section 4.6. Table E.1 and Table E.2 report the wall-clock time per 100 assimilation cycles for each method on each domain, averaged over 5 repetitions.

E.1 Cosmological Domain

Table E.1: Computational cost per 100 assimilation cycles on the AR(1) cosmological model ($d = 18$, $N_e = 30$). Mean $\pm$ std over 5 runs.

Method	Time (seconds)	Relative to EnKF
EnKF (standard)	0.141 ± 0.001	$1.0\times$
Eigenvalue Shrinkage	0.200 ± 0.002	$1.4\times$
Identity Shrinkage	0.200 ± 0.003	$1.4\times$
Ledoit-Wolf	0.218 ± 0.004	$1.5\times$
Cosmological Shrinkage	0.242 ± 0.007	$1.7\times$
Scaled Shrinkage	0.278 ± 0.004	$2.0\times$
EnKF-Cholesky	0.510 ± 0.013	$3.6\times$
NERCOME	26.720 ± 0.870	$189.5\times$

At $d = 18$, the matrix operations ($\mathcal{O}(d^3) = \mathcal{O}(5832)$) are fast enough that the overhead of setting up the analysis (Python object creation, array allocation) is comparable to the actual computation. This explains why the direct precision methods are only 1.4–2.0 times slower than the baseline EnKF despite requiring an additional matrix inversion: the inversion of an 18×18 matrix takes microseconds.

The EnKF-Cholesky method is unexpectedly slow ($3.6\times$ the EnKF baseline) because its Python implementation loops over each state variable to perform the local regression, and the loop overhead dominates at small d. A vectorized implementation could reduce this gap significantly.

NERCOME's $189.5\times$ slowdown comes from its 200 bootstrap resamples, each requiring an eigendecomposition. At $d = 18$, each eigendecomposition is fast ($\mathcal{O}(18^3) \approx 6000$ operations), but 200 of them plus the associated random splitting and matrix products add up.

Table E.2: Computational cost per 100 assimilation cycles on Lorenz 96 ($n = 40$, $N_e = 80$). Mean $\pm$ std over 5 runs.

Method	Time (seconds)	Relative to EnKF
EnKF (standard)	25.986 ± 0.688	$1.0\times$
Identity Shrinkage	26.431 ± 0.382	$1.0\times$
Ledoit-Wolf	26.630 ± 0.472	$1.0\times$
EnKF-Cholesky	27.058 ± 0.345	$1.0\times$
Scaled Shrinkage	27.321 ± 0.291	$1.1\times$
Eigenvalue Shrinkage	27.532 ± 0.266	$1.1\times$
NERCOME	215.391 ± 13.179	$8.3\times$

E.2 Lorenz 96 Domain

At $n = 40$, the runtime balance shifts. The Lorenz 96 model propagation (fourth-order Runge-Kutta integration of 40 coupled ODEs for 10 time steps per cycle, for each of 80 ensemble members) dominates the runtime, accounting for approximately 25 seconds of the 26–27 second total. The analysis step (matrix inversion, shrinkage coefficient computation, and the linear solve) takes only 1–2 seconds. As a result, all methods except NERCOME are within 6% of the baseline EnKF cost.

NERCOME's relative slowdown is less dramatic than on the cosmological domain ($8.3\times$ versus $189.5\times$) because the model propagation cost is shared by all methods. In absolute terms, NERCOME adds approximately 190 seconds of analysis time on top of the 26-second propagation cost.

For systems where model propagation is the bottleneck (i.e., most real-world applications), the choice of covariance estimation method has negligible impact on total runtime. NERCOME is the exception and should be avoided when computational resources are limited.

Appendix F

Derivation of the Bodnar Shrinkage Coefficients

This appendix outlines the derivation of the optimal shrinkage coefficients α^* and β^* from (Bodnar et al., 2016), filling in steps that are compressed in the original paper.

F.1 Setup

The estimator is $\mathbf{\Pi}_{\text{LS}} = \alpha\hat{\mathbf{\Pi}}_S + \beta\mathbf{\Pi}_0$, where $\hat{\mathbf{\Pi}}_S = \mathbf{S}^{-1}$ and $\mathbf{\Pi}_0$ is a fixed target. The goal is to find α^* and β^* that minimize the expected Frobenius loss:

$$L(\alpha, \beta) = \mathbb{E}\left[\|\alpha\hat{\mathbf{\Pi}}_S + \beta\mathbf{\Pi}_0 - \mathbf{\Sigma}^{-1}\|_F^2\right]. \tag{F.1}$$

F.2 Expansion

Expanding the Frobenius norm:

$$\begin{aligned} L(\alpha, \beta) = {} & \alpha^2\mathbb{E}[\|\hat{\mathbf{\Pi}}_S\|_F^2] + \beta^2\|\mathbf{\Pi}_0\|_F^2 + \|\mathbf{\Sigma}^{-1}\|_F^2 \\ & + 2\alpha\beta\,\mathbb{E}[\operatorname{tr}(\hat{\mathbf{\Pi}}_S\mathbf{\Pi}_0)] - 2\alpha\,\mathbb{E}[\operatorname{tr}(\hat{\mathbf{\Pi}}_S\mathbf{\Sigma}^{-1})] - 2\beta\operatorname{tr}(\mathbf{\Pi}_0\mathbf{\Sigma}^{-1}). \end{aligned} \tag{F.2}$$

F.3 Wishart Moments

The key moments needed are:

$$\mathbb{E}[\hat{\mathbf{\Pi}}_S] = \frac{N_e - 1}{N_e - n - 2}\mathbf{\Sigma}^{-1} = c_1\mathbf{\Sigma}^{-1}, \tag{F.3}$$

$$\mathbb{E}[\|\hat{\mathbf{\Pi}}_S\|_F^2] = c_2\|\mathbf{\Sigma}^{-1}\|_F^2 + c_3\|\mathbf{\Sigma}^{-1}\|_{\text{tr}}^2, \tag{F.4}$$

where c_1, c_2, c_3 are constants depending on n and N_e, derived from the moments of the inverse Wishart distribution. For large N_e, $c_1 \approx 1 + n/N_e$, $c_2 \approx 1 + 2n/N_e$, and $c_3 \approx 1/N_e$.

F.4 First-Order Conditions

Setting $\partial L/\partial\alpha = 0$ and $\partial L/\partial\beta = 0$ and substituting the Wishart moments gives a 2×2 linear system in α and β. After algebraic simplification (replacing the unknown $\mathbf{\Sigma}^{-1}$ with the observable $\hat{\mathbf{\Pi}}_S$ and correcting for bias using the Wishart moments), the solution yields Eqs. (2.37)–(2.38).

The key step is the replacement of $\|\mathbf{\Sigma}^{-1}\|_F^2$ and $\mathrm{tr}(\mathbf{\Sigma}^{-1}\mathbf{\Pi}_0)$ with their sample counterparts, using the bias correction $\mathbb{E}[\hat{\mathbf{\Pi}}_S] = c_1\mathbf{\Sigma}^{-1}$. This introduces the $(1 - n/N_e)$ term that appears in the formulas. The Cauchy-Schwarz term in the denominator of α^* arises from the determinant of the 2×2 system, ensuring that the solution is unique when $\hat{\mathbf{\Pi}}_S$ and $\mathbf{\Pi}_0$ are not proportional.

For the complete derivation with all intermediate steps, see (Bodnar et al., 2016), Theorem 4.1 and Corollary 4.1.

Appendix G

TEDA Code Listings

This appendix presents the core Python code for the direct precision shrinkage estimators implemented in TEDA, as described in Section 3.5.

G.1 Shrinkage Coefficient Computation

The following function computes the shrinkage coefficients α^* and β^* from Eqs. (2.37)–(2.38). It is shared by all four direct precision shrinkage variants.

```
def compute_shrinkage_coefficients(S_inv, target, n, Ne):
    """
    Compute Bodnar et al. shrinkage coefficients.

    Parameters
    ----------
    S_inv : ndarray (n, n)
        Sample precision matrix (raw inverse of S).
    target : ndarray (n, n)
        Shrinkage target Pi_0.
    n : int
        State dimension.
    Ne : int
        Ensemble size.

    Returns
    -------
    alpha, beta : float
        Clamped shrinkage coefficients.
    """
    # Norms
    frob_S_inv = np.sum(S_inv ** 2)       # ||S^-1||_F^2
    trace_S_inv = np.trace(S_inv)          # ||S^-1||_tr
    frob_target = np.sum(target ** 2)      # ||Pi_0||_F^2
    cross = np.trace(S_inv @ target)       # tr(S^-1 Pi_0)

    # Cauchy-Schwarz denominator
```

```
    denom = frob_S_inv * frob_target - cross ** 2

    # Alpha*
    if denom == 0:
        alpha = 1.0 - n / Ne
    else:
        correction = (trace_S_inv ** 2 * frob_target) \
                     / (Ne * denom)
        alpha = 1.0 - n / Ne - correction

    # Clamp alpha
    alpha = max(0.0, min(1.0, alpha))

    # Beta*
    remaining = 1.0 - n / Ne - alpha
    beta = remaining * cross / frob_target

    # Clamp beta
    beta = max(0.0, min(1.0 - alpha, beta))

    return alpha, beta
```

G.2 Identity Target Estimator

The identity target estimator is the simplest variant:

```
class AnalysisEnKFDirectPrecisionShrinkageIdentity:
    """EnKF with direct precision shrinkage
    (identity target)."""

    def get_precision_matrix(self, S, n, Ne):
        """Compute shrinkage precision estimate."""
        S_inv = np.linalg.inv(S)
        target = np.eye(n)
        alpha, beta = compute_shrinkage_coefficients(
            S_inv, target, n, Ne
        )
        return alpha * S_inv + beta * target
```

G.3 Scaled Identity Target Estimator

The scaled identity target uses per-variable variances:

```
class AnalysisEnKFDirectPrecisionShrinkageScaled:
```

```
    """EnKF with direct precision shrinkage
    (scaled identity target, novel)."""

    def get_precision_matrix(self, S, n, Ne):
        """Compute shrinkage precision estimate."""
        S_inv = np.linalg.inv(S)
        # Scaled target: diag(1 / (2 * sigma_i^2))
        variances = np.diag(S)
        target = np.diag(1.0 / (2.0 * variances))
        alpha, beta = compute_shrinkage_coefficients(
            S_inv, target, n, Ne
        )
        return alpha * S_inv + beta * target
```

G.4 Eigenvalue Target Estimator

The eigenvalue target uses the sorted eigenvalues of $\hat{\mathbf{\Pi}}_S$:

```
class AnalysisEnKFDirectPrecisionShrinkageEigenvalues:
    """EnKF with direct precision shrinkage
    (eigenvalue target)."""

    def get_precision_matrix(self, S, n, Ne):
        """Compute shrinkage precision estimate."""
        S_inv = np.linalg.inv(S)
        eigenvalues = np.sort(
            np.linalg.eigvalsh(S_inv)
        )
        target = np.diag(eigenvalues)
        alpha, beta = compute_shrinkage_coefficients(
            S_inv, target, n, Ne
        )
        return alpha * S_inv + beta * target
```

G.5 Cosmological Target Estimator

The cosmological target uses the theoretical power spectrum covariance:

```
class AnalysisEnKFDirectPrecisionShrinkageCosmo:
    """EnKF with direct precision shrinkage
    (cosmological target)."""

    def __init__(self, C_ell, N_ell):
```

```
        self.C_ell = C_ell   # Power spectrum values
        self.N_ell = N_ell   # Mode counts

    def get_precision_matrix(self, S, n, Ne):
        """Compute shrinkage precision estimate."""
        S_inv = np.linalg.inv(S)
        # Cosmological target covariance:
        #   T_ii = 2/N_ell * C_ell^2
        T_cov = np.diag(
            2.0 / self.N_ell * self.C_ell ** 2
        )
        target = np.linalg.inv(T_cov)
        alpha, beta = compute_shrinkage_coefficients(
            S_inv, target, n, Ne
        )
        return alpha * S_inv + beta * target
```

Note: The code listings above are simplified for clarity. The actual TEDA implementations include additional error handling, logging, and integration with the `Analysis` base class. The full source code is available at https://github.com/enino84/TEDA.git.

Appendix H

Statistical Significance Tests

This appendix reports the pairwise Welch's t-test results mentioned in Section 4.3.2. For each pair of methods, we test the null hypothesis that the time-averaged RMSE values across 30 runs are drawn from the same distribution.

H.1 Lorenz 96 Results

Table H.1 shows the p-values for pairwise comparisons of Ledoit-Wolf against each other method at $N_e = 80$ on Lorenz 96.

Table H.1: Welch's t-test p-values comparing Ledoit-Wolf against each method on Lorenz 96 at $N_e = 80$. Values below 0.05 (bold) indicate statistically significant differences.

Method	p-value	Significant?
EnKF (standard)	0.312	No
EnKF-Cholesky	$<$ **0.001**	Yes
Identity Shrinkage	$<$ **0.001**	Yes
Scaled Shrinkage	$<$ **0.001**	Yes
Eigenvalue Shrinkage	$<$ **0.001**	Yes
NERCOME	$<$ **0.001**	Yes

Ledoit-Wolf's advantage is statistically significant ($p < 0.001$) against all shrinkage methods and EnKF-Cholesky. It is not significantly different from the standard EnKF ($p = 0.312$), which at $N_e = 80$ achieves comparable RMSE (0.07 vs. 0.07).

H.2 Cosmological Results

Table H.2 shows the p-values for pairwise comparisons at $N_e = 30$ on the cosmological AR(1) model.

On the cosmological domain, Ledoit-Wolf is significantly better than the standard EnKF ($p = 0.003$) and the eigenvalue and cosmological shrinkage targets ($p < 0.02$). It is not significantly different from EnKF-Cholesky, identity shrinkage, scaled shrinkage, or NERCOME ($p > 0.3$). This confirms the observation from Section 4.3.2 that the identity and scaled identity targets are competitive with Ledoit-Wolf on near-diagonal data, while the eigenvalue target is significantly worse (it effectively provides no shrinkage, as shown in Appendix A.3).

Table H.2: Welch's t-test p-values comparing Ledoit-Wolf against each method on the AR(1) cosmological model at $N_e = 30$.

Method	p-value	Significant?
EnKF (standard)	**0.003**	Yes
EnKF-Cholesky	0.574	No
Identity Shrinkage	0.757	No
Scaled Shrinkage	0.831	No
Eigenvalue Shrinkage	**0.010**	Yes
Cosmological Shrinkage	**0.017**	Yes
NERCOME	0.370	No

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